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# The Next 90: Your Geopolitical Risk Radar
## Nov 2026–Jan 2027

**October 2026**

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**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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## A geopolitical calendar mapping the next 90 days

### NOVEMBER

| | | |
|---|---|---|
| Early (TBC) | Ukraine/US/ Russia | Trilateral peace talks expected in Abu Dhabi |
| 3 | US | Midterm elections (House, Senate, gubernatorial and state legislatures) |
| 4 | South Africa | Municipal elections |
| 4-5 | Spain | Ibero-American Summit |
| 7 | New Zealand | General election |
| 9 | EU | Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings |
| 9-20 | COP31 | COP31 in Türkiye |
| 14 | Malaysia | Melaka state election |
| 16-18 | ASEAN | 49th ASEAN Summit in Manila |
| 16-18 (TBC) | Quad | Quad Leaders' meeting in Manila |
| 18-19 | APEC | APEC Economic Leaders' Meeting in Shenzhen; Trump and Xi to meet on sidelines |
| 20 | EU/UK | EU-UK Summit |
| 27 | France | S&P sovereign credit rating review |
| 28 | Taiwan | Local elections |
| 29 | Spain | General election |
| 29 | OPEC+ | Ministerial meeting |
| 30-1 Dec | EU/Argentina | EU-Argentina Investment and Business Forum in Buenos Aires |
| TBC | US/North Korea | Possible meeting between Donald Trump and Kim Jong-un |

### DECEMBER

| | | |
|---|---|---|
| First half of Dec | China | Politburo meeting followed by annual Central Economic Work Conference |
| 3-4 | OSCE | Ministerial Council in Switzerland |
| 4 | US | Section 232 tariffs (15% ad valorem) and minimum import prices on polysilicon, derivatives and solar products set to enter into force |
| 5 | The Gambia | Presidential election |
| 5-6 | Doha Forum | Qatar's flagship dialogue themed "Redefining Global Trust" |
| 10-11 | EU | Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings |
| 11 | US | Stopgap funding expires; shutdown risk |
| 14 | EU | Foreign Affairs Council meeting |
| 14-15 | G20 | G20 Summit in Miami |
| 17-18 | EU | European Council Summit |
| 31 | OPEC+ | Expiry of the current group-wide production framework |
| TBD | US | US Congress is set to adopt annual-must pass defence legislation |

### JANUARY

| | | |
|---|---|---|
| 1 | G20, G7, EU, ASEAN, BRICS+ | New rotating presidencies of the EU (Lithuania), G7 (US), G20 (UK), ASEAN (Singapore) and BRICS+ (China) |
| 1 | US | Stricter US defence-procurement rules take effect for covered critical materials sourced from China, Russia, Iran and North Korea, including tighter waiver requirements |
| 1 | Canada/US | 50% US tariff on Canadian automotive imports is set to enter into force |
| 3 | US | Convening of the 120th United States Congress |
| 10 | China/US | Busan trade truce expiry |
| 16 | Nigeria | General election |
| 18-22 | WEF | World Economic Forum annual meeting in Davos |
| 27 | Kyrgyzstan | Presidential election |
| 30 | Germany | Presidential election |

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

# Appendix 1

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# Swing State: Navigating Midterm Volatility

October 23, 2026

*A client-facing version of this report is available at the Deutsche Bank Research Portal. The following Deutsche Bank Research analysts contributed to this report: Brett Ryan (econ), Binky Chadha and Parag Thatte (equity strategy), Benjamin Soff (data centers), Daniel Knauff (software), George Hill (healthcare), Omotayo Okusanya (REITs), Peter Abramowitz (office REITs), Mark DeVries (consumer and specialty finance), Nicole DeBlase (MI/EE), Corinne Blanchard (clean tech), Richa Harnain (transports), and Scott Deuschle (aerospace and defense).*

With the 2026 US midterm elections approaching, we look at key investor debates surrounding the event and legislative priorities of the next Congress. Despite the uncertainty that midterms can bring, the market appears fairly confident on expectations for the overall results. While a surprise Republican-majority victory in both chambers is still possible, consensus is converging around current polling indicating Democrats are likely to flip the seats needed for a House majority and may win a Senate majority. Historically, an incumbent president has lost seats in 20 of the last 22 midterms. Investor expectations are in line with this. Polymarket, for example, is currently above 90% for Democrats gaining control of the House and above 60% for Democrats gaining control of the Senate.

With this set of expectations, key investor debates instead focus on the legislative agenda for hot-ticket policy issues. Namely, these hinge on AI and data centers, affordability (affecting healthcare, housing, and education loans), and economic and fiscal policy. Debates also surround possible influence of the new Congress on the war in Iran and, by extension, energy prices. In this report, we cut through the noise on what the next Congress could reasonably advance from these agendas, what the impact would be on affected sectors, and how to address market uncertainty – and perceived certainty – into the elections.

Historically, midterm elections have proven to be a check on administration policies. The number of seats lost in the midterms by the incumbent party is tied to the president's approval rating going into midterms, which in turn tends to align with consumer confidence. President Trump's ~38% approval rating, at a record low for his tenure, would suggest a little over a 45-seat loss for the Republican Party (in the House and Senate combined). That said, polling in the Senate races that will determine control of the chamber largely remains within the margin of error, and in 2024, Trump won in at least five of the seven core toss-up states. Thus, the Senate remains a much closer call than the House race.


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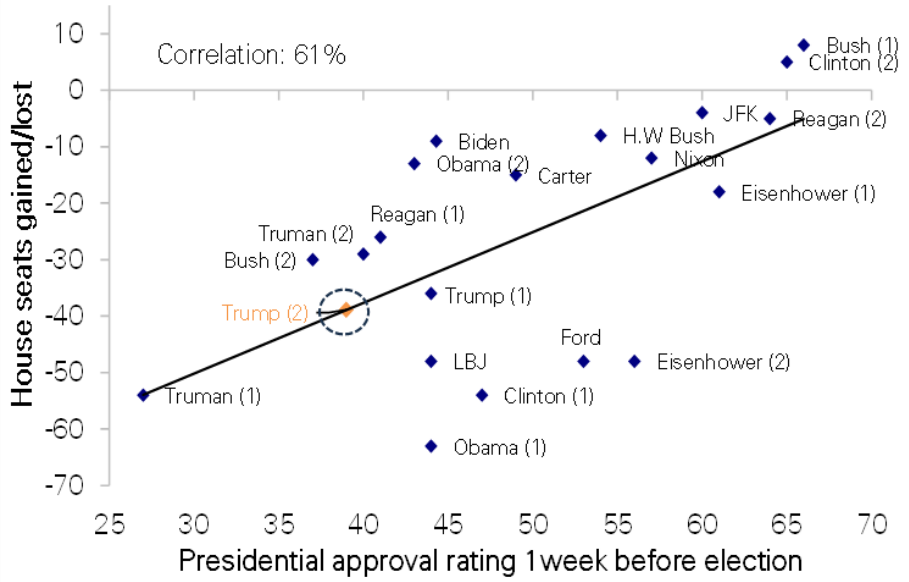
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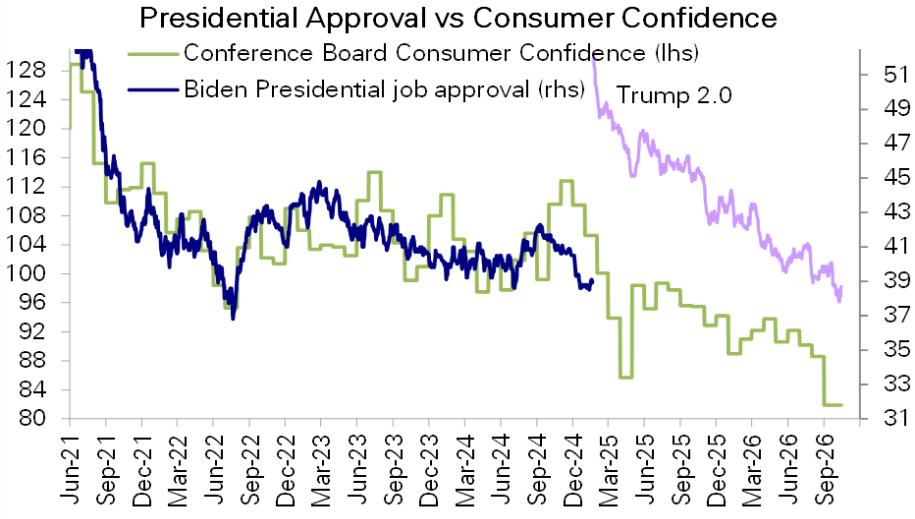

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Figure 1: Presidential Approval vs. House Seats




Source: Real Clear Politics

Figure 2: Presidential Approval vs. Consumer Confidence



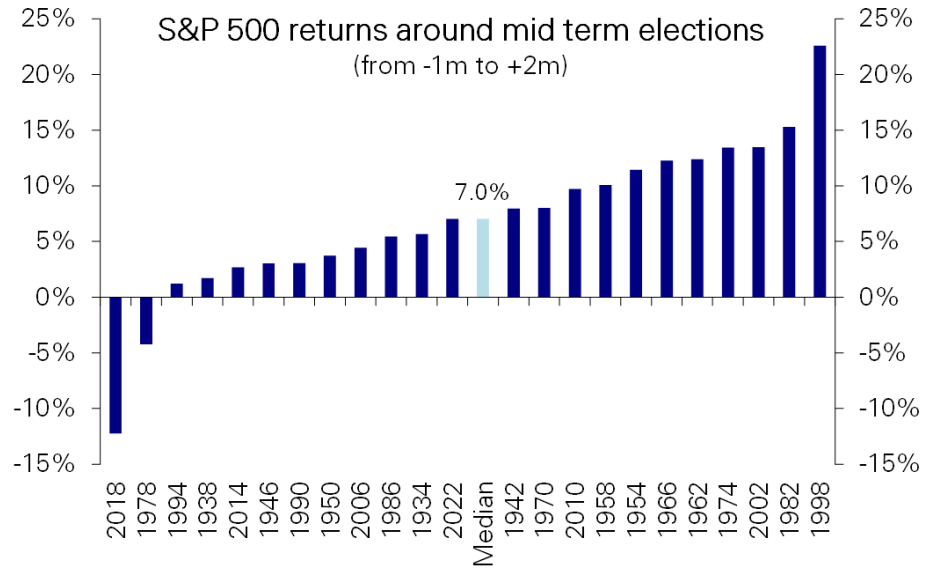

Source: The Conference Board

Whatever the outcome of the midterms, the legislative agenda will still face constraints from the White House. It is all but impossible for either party to gain a veto-proof majority in either chamber, minimizing the likelihood of a large shift in fiscal or regulatory policy. The most immediately obvious impact of Democrats taking the Senate would be the ability to slow or block executive branch appointees, including at the Federal Reserve Board of Governors or the Supreme Court, should any members step down before their terms end.

Without garnering a two-thirds majority in both chambers, Democrats lack a clear path to unwind the 2025 tax cuts or enact AI-related regulatory legislation – at least, not without President Trump's explicit approval. With respect to the near-term fiscal policy outlook, the most important period for investors may be the "lame duck" session, the period after the November 3 election and before the new Congress convenes on January 3. Recall that current federal government funding expires on December 11, and the Trump administration is asking for a nearly 60% increase in defense spending, the largest since the Korean War. In addition, the debt limit will become binding again around the middle of next year. In short, the actions during the lame duck session could determine whether the new Congress is seated amid a government shutdown, a contentious battle over the federal budget, and the debt limit looming in the background.

In terms of the market reaction to midterm elections, the risk-reward profile has typically been favorable, with 21 of the last 23 midterm years seeing positive returns in the fourth quarter, with an average gain of 7% in the S&P. The gains are notable even after adjusting for typical positive year-end seasonality.

Figure 3: S&P Returns and Midterms




*Source: Bloomberg Finance LP*

It would not be unusual for the market to go sideways as investors hedge potential volatility and uncertainty, as we have seen previously ahead of other well-telegraphed pivotal events. This has historically been followed by a rally as the results became clear and uncertainty dissipated, regardless of the result itself. In the current context, if election polls and prediction market probabilities continue being lopsided, the market may not treat the election as an uncertain risk event going in. Uncertainty could still ramp up after, and the typical midterm election rally may be pushed out. It is notable that the one prominent exception to the midterm rallies was in 2018, when the market sold off sharply after the elections as uncertainty spiked around trade war 1.0 as well as Fed policy.



We note that the midterm elections also coincide with the third quarter earnings season, during which we expect companies to deliver growth of 34%. For next year, we expect growth to be a still-robust 17% with risks to the upside. Eventually, earnings tend to be the anchor for equities and have been for several years despite numerous domestic political as well as geopolitical shocks.

Across sectors, most have rallied around past midterms in keeping with the market, but there isn't a consistent pattern of out- or underperformance. This suggests that for the initial conditions for each sector, specific policies and positioning played a bigger role. For that reason, we turn to a closer analysis of key legislative and policy debates impacting exposed sectors.



# AI and Data Centers

AI and data centers have become a campaign talking point for candidates from both the Democratic and Republican parties. This has spurred investor questions on whether to expect a national regulatory framework constraining data center buildouts. Some of the headlines surrounding this debate overestimate what Congress can accomplish through national legislation compared with what is possible at a state level, particularly during a presidential administration supportive of AI and data center acceleration. Still, the industry is responding to a political backdrop that is increasingly scrutinizing AI and data centers.

The current administration has prioritized advancing AI and digital infrastructure development. The White House's AI Action Plan, released July 2025, included a set of executive orders supporting the rapid buildout of AI data centers by reducing certain federal regulatory and permitting burdens. The administration did not preempt state laws on data center infrastructure development and federal permitting orders generally do not override state zoning and utility rules.

In March 2026, the White House announced the Ratepayer Protection Pledge, under which large data center operators would voluntarily commit to covering the cost to upgrade generation, transmission, substations, or other necessary infrastructure to service these developments. The industry's response to these proposals has been largely constructive. The Data Center Coalition (DCC), an industry group representing major hyperscalers and data center owners and operators came out in support of the initial version of the Pledge, while arguing that large-load customers across all industries also should cover the costs to upgrade energy infrastructure. According to the DCC, "the data center industry remains committed to being responsible partners where we operate, which includes paying our full energy costs and investing in local jobs and workforce development."

Congress voted in September 2026 to codify the principles present in the Pledge via the Ratepayer Protection Act. The bill, which only required states to "consider" these standards, passed overwhelmingly in the House (417-3), but failed to reach 60 votes in the Senate (57-43). Critics noted the bill stopped short of imposing mandatory nationwide rules.

Looking forward, we expect a Congress that is more focused on consumer protection might look to codify a more stringent version of these rules, requiring large-load customers to bear the full incremental costs associated with generation, transmission, and distribution upgrades. However, even if such a framework were passed into law, we believe the overall economic impact to large data center operators would be relatively modest, for two reasons.

For one, the White House would maintain a veto over bills that include language it views as overly burdensome towards AI infrastructure development. Two, data center operators generally underwrite projects only when they have a high degree of confidence in securing power and, where necessary, may already be embedding the cost of upgrading energy infrastructure into their investment plans. We expect developers to be capable of passing on sustained increases in power infrastructure costs to their customers, which, for many 100 MW+ developments, are the major investment-grade hyperscalers.



Though data center development is receiving increasing national attention, it is still largely a local issue, and the most consequential elections will be at the state and local levels. We have seen a number of state-driven actions on data centers in recent months, including:

- In July, New York State's governor issued an executive order to establish a temporary moratorium on data center development while the state develops a standardized framework of rules for key issues (including responsibilities for power and energy infrastructure upgrades; additional power generation; clean energy; water use, emission, and waste management; and state/local financial incentives). The New York legislature also passed a one-year moratorium on Department of Environmental Conservation permits and approvals for 20 MW+ data center developments.
- In August, Texas' governor directed the state's public utility commission to conduct a comprehensive audit of every data center advancing through the ERCOT (Texas's grid operator) interconnection process. The audit will cover several notable areas including public financial support, electricity demand and supply, water use, community impacts, and ownership and control.
- In September, California's governor passed a comprehensive regulatory package, covering electricity, water, reporting, and environmental review. Most of these provisions take effect January 1, 2027, and phase in through 2027-2028, but the package does not ban data centers or impose a statewide moratorium on development activity.
- Oregon, Pennsylvania, New Jersey, and other states have also established orders to create standards and frameworks governing data center development (but without imposing broad moratoriums).

At the more local level, a large number of moratoriums have been passed on new data center development. Data collected by SemiAnalysis, an independent research firm, shows that, as of September, more than 300 local data center moratoriums have been adopted nationwide across counties, municipalities, townships, and other local governments.

However, these state and local moratoriums appear thus far to be less impactful to near- and medium-term development activity than headline coverage might otherwise suggest. According to SemiAnalysis, only about 2.3 GW of the 38 GW of data center capacity planned or under construction for 2027 has faced delays as a result of the New York state executive order (0.8 GW) and/or local moratoriums (1.5 GW).

We believe the primary reasons that moratoriums are, in reality, less likely to impact near-term development activity are (1) that these moratoriums largely impact new projects and applications (rather than existing developments in progress), (2) there may be flexibility to relocate or rezone early-stage projects in response to moratoriums, and (3) data center developers are increasingly partnering with local communities to gain their support via investments and other programs (as we wrote about recently). Potential regulatory risk has not yet slowed capex spending patterns, with capex spend still expected to climb from



$0.5 trillion in 2025 to $1.3 trillion by 2027; data centers typically represent ~20-30% of the total bill of materials for new AI capacity.

Although the practical impact on near- and medium-term development pipelines appears less pronounced than headlines might suggest, all this is not to say that data centers will not face increasing local opposition following the midterms. Indeed, in our conversations with industry participants, there is an acknowledgment that greater public scrutiny of data centers is not going away. Additional moratoriums, zoning restrictions, permitting requirements, or utility-related conditions could affect the pace, cost, and overall feasibility of future projects.

# Affordability Agenda: Healthcare, Housing, and Education Loans

The affordability agenda has become a centerpiece of many Democratic as well as Republican campaign platforms. This platform is impacting investor debates across healthcare, housing, and student loans. In healthcare, the push among some Democrats for expanded health coverage under the Medicare for All banner is placing pressure on managed care stocks, but meanwhile, companies with exposure to Medicaid and the Affordable Care Act (ACA) could see a supportive political backdrop. On housing, investor debates on the impacts of the affordability agenda have continued since passage of legislation in July 2026 that caps the number of single-family homes for-profit institutional investors can purchase. Affordability of higher education is likewise part of the conversation, leading investors to debate which levers Congress could pull that would affect student loan companies.

## Healthcare

In healthcare, we see the upcoming midterm elections as a sentiment event for managed care stocks, not a policy event. With the presidential veto in place through 2028, we do not expect the government to enact direct healthcare legislation during the next Congress. Forecasts place the probability of a change in House control at roughly 79%, with both chambers changing in about 44% of simulations. Although investor skittishness around Medicare for All (M4A) rhetoric is high, we see the likelihood of M4A replacing private coverage as a near-zero probability in this timeframe.

M4A has no realistic legislative path before 2029, but the rhetoric weighs on managed care organization (MCO) stock multiples by raising perceived long-term risk rather than affecting near-term earnings. Polling from Kaiser Family Foundation, a health policy research organization, shows support for M4A at a decade-plus high, and the 2018–2019 cycle suggests that increased campaign attention can pressure MCO multiples even without a direct policy impact. We expect similar sentiment-driven pressure through year-end and potentially into the 2028 primary cycle but would not view it as a change in near-term fundamentals.

For Medicare Advantage, we see a headline risk over earnings reality. The near-term earnings impact for affected companies should be limited, because the US Centers for Medicare & Medicaid Services (CMS) has finalized a 2.48% Medicare Advantage (MA) rate increase for 2027, versus 0.09% in the advance notice. A change in House control would instead increase focus on 2028 payment policy, risk adjustment, and prior authorization. Coding reform already has bipartisan support: sponsors estimate that the Cassidy-Merkley No UPCODE Act could generate up to $275 billion of savings over ten years, while MedPAC estimates coding intensity will add roughly $40 billion to 2025 Medicare Advantage costs relative to traditional Medicare. In our view, the more immediate consideration is potential multiple pressure rather than a measurable change to company earnings.

However, the next Congress could benefit Medicaid funding and ACA policy. Medicaid and the ACA appear relatively better positioned, although any benefit



would likely be reflected first in sentiment and lower perceived policy risk rather than near-term estimate revisions. Medicaid community engagement requirements take effect in most states by January 1, 2027. Although repeal remains unlikely given the presidential veto, a change in House control could shift the year-end funding debate toward restoring or preserving federal Medicaid support, which could ease state budget pressure and support rate-setting. The ACA policy backdrop could also become more supportive, although we see limited scope for enhanced premium subsidies to return before 2029 after the three-year extension passed by the House stalled.

### Housing

Turning to the housing sector, the likely midterm outcome of a split government would result in more gridlock. Historically, political gridlock in terms of passing new regulation has been a tailwind for the real estate sector, reducing regulatory risk at the federal level. In the scenario of a split Congress, in which the Democrats have a majority in the House but not the Senate, bills with bipartisan support have a greater likelihood to become law. Two key areas in which this bipartisan support exists are housing affordability and consumer protection.

Among real estate investment trusts (REITs), we see Residential as among the most impacted subsectors, particularly as it relates to the affordability debate. Regardless of the election outcome, housing affordability has become front and center on the political agenda across both parties. Federal legislation is more likely to target supply-side solutions, as legislation restricting rent increases is typically handled at the local and state level. We think Multifamily would be the most likely subsector impacted, as a push to expand rental housing supply could lead to eased zoning restrictions, particularly in supply-constrained coastal markets that have historically benefited from higher barriers to entry. That said, this would likely take years to materialize. With a Democratic majority, Congress could voice opposition to the influence of large institutional landlords, but we see this strictly as a sentiment risk to stocks.

Single Family For Rent (SFR) is less exposed to near-term legislative shocks, as the sector was a focus of the 21st Century ROAD to Housing Act passed in July 2026 over the summer. The measure impacts institutional owners' ability to buy homes off the MLS but contains specific carve-out provisions that will allow them to keep developing build-to-rent (BTR) product and buy excess inventory from homebuilders, the sector's preferred acquisition channel in recent years. The ROAD to Housing Act also contained provisions that could theoretically ease the production of manufactured homes, the main business segment for Manufactured Housing (MH) REITs. The first-order impact is likely more flexibility in home designs that could lead to further densification of MH communities, although we note emerging investor debate about the long-term impacts of increased MH production on competitive supply for a property type that has historically been supply constrained, mainly due to NIMBY-ism.

Parallel to the conversation on affordability, it is clear that, regardless of which political party may be in power following the midterms, a major focus remains addressing the US fiscal deficit and its continued growth. We continue to pay attention to potential budget cuts or sequestration to slow government spending and reduce the deficit, which could have an impact on real estate. Key areas to watch include cuts to healthcare spending (which would be a negative for skilled



nursing focused REITs, given Medicare is a major revenue source for their tenant base) and cuts that would impact Office REITs with large government exposure.

We also worry that an elevated federal deficit and the elevated Treasury borrowing needs today create a risk that interest rates will remain higher for longer. This puts pressure on real estate valuations and cost of financing for REITs to drive external growth via acquisitions and development. A split Congress is likely to result in more difficult budget negotiations and increased oversight, but it remains unclear what the ultimate outcome would be to address the budget deficit issue.

With real estate being more dependent on local market dynamics, we believe the more likely regulatory impact would come from local regulation. This could show up on the ballots during the midterms, with increased possibility of rent control legislation targeted at residential real estate as well as self-storage, as cities and municipalities focus on affordability issues. There is also growing legislative pushback at the local level against development of real assets such as data centers given the large power requirements, as discussed in detail elsewhere in this report.

### Education Lending

Midterm campaign platforms on education affordability have a bearing on the consumer and specialty finance sector. Much of the policy exposure in this sector is through the executive branch bodies, not Congress, however. This includes, for example, regulatory authorities such as the FDIC (Federal Deposit Insurance Corporation), the CFPB (Consumer Financial Protection Bureau), or the FHFA (Federal Housing Finance Agency, which regulates Fannie Mae and Freddie Mac). That said, historical data shows that student loan company stocks can still directionally align with midterm party outcome and expectations in Congress, although the sample size is small.



# Mixed Impacts of Economic and Fiscal Policy

Democratic and Republican candidates have aimed to distinguish themselves from their opponents along a range of economic and fiscal policy matters, including tariffs, tax, labor, and the budget. In reality, the outcome of the midterms, regardless of which party prevails, is likely to present a mixture of relief and restrictions affecting businesses. In some cases, investor fears over sweeping economic policy changes are overblown for topics over which Congress realistically has limited influence, presenting more headline risk than a risk to fundamentals. In considering these, we look to industrials that are particularly sensitive to key investor debates, spanning multi-industry/electrical equipment, clean tech and alternative energy, transport, and defense.

For the multi-industry and electrical equipment (MI/EE) sector, historical midterm election cycles suggest a generally constructive backdrop heading into election day, with positive returns in six of the past seven elections. On average, the MI/EE group has gained ~5% during the three months preceding historical midterm elections and ~8% during the final month prior. Relative outperformance versus the S&P 500 has been less consistent; we observed outperformance in only four of seven cycles, though the group still generated average excess returns of ~250bps over the preceding three months and ~270bps over the preceding month.

Overall, the data supports a favorable absolute return backdrop ahead of midterms, but significant stock-level dispersion suggests company-specific fundamentals and factor leadership have historically mattered more than the election cycle alone. We mapped out election results scenarios across key policy pillars for our coverage (tax policy and capital investment; trade, tariffs, and supply chains; energy, grid, and power infrastructure; AI and data center regulation; and the regulatory environment). We focus on two key scenarios: one in which Republicans maintain control of both chambers, and a divided government that sees the Democrats controlling one or both chambers of the legislative branch.

Under a divided government, we expect stock-level dispersion would likely narrow, with globally exposed companies receiving some trade policy relief and HVAC/automation companies benefiting from a potentially more supportive regulatory backdrop. However, should Republicans maintain control of Congress, our scenario analysis suggests this would be favorable for electrical equipment and power infrastructure manufacturers, due to a combination of pro-investment tax policy, faster infrastructure deployment, and a lighter AI regulatory framework. That said, globally exposed companies could face incremental tariff and retaliation risks.

Looking to the clean tech and alternative energy sector, the midterm elections could represent a positive tailwind. The next Congress might be positioned to pursue targeted extensions or modifications to certain clean-energy tax credits. The introduction of H.R. 8477, which proposes to restore several clean energy tax credits rolled back in late 2025, seems to signal that some, albeit a limited number, of Republicans could be supportive of this initiative. If the Democrats gain a majority in the House but not the Senate in the midterms, it is plausible that Democrats could find sufficient Republican support to advance



legislation extending clean-energy credits, though they would still face significant legislative hurdles. A midterm outcome in which Republicans retain control of both chambers is likely to maintain the status quo for the remaining existing tax credits, especially for those tied to domestic manufacturing. While those could potentially see further extension, we think this is a limited outcome.

While we see a low probability of a major change in, or support for, the residential tax credits that expired at the end of 2025 (25D), we envision a scenario in which indirect residential tax credits (48E) could be extended, which would help boost part of the demand. Softness in demand is expected to remain going into 2027, still impacted by the phaseout of the 25D tax credits, the eventual roll-off of safe-harbor benefits, and a more constrained financing environment.

From a manufacturing tax credit standpoint, there could be additional tailwinds if we were to see any manufacturing tax credit extension or positive modification (although lower probability, in our view). This would also likely improve utility-scale project economics if broader project-related credits are preserved, though upcoming elections could cause a slowdown in new booking activity until legislative uncertainty clears.

For Transports, the midterms would be most relevant for the trucking supply thesis. Truckload stocks could have a modest negative reaction to a Democratic majority if investors expect some rollback of the Trump administration's constraints on drivers, which has tightened supply. We would separate perception from fundamentals, however. The election of a Democratic Congress would not automatically by itself change leadership at the Department of Transportation (DOT) and the Federal Motor Carrier Safety Administration (FMCSA), the executive branch entities driving much of the current push for restrictions on drivers.

One possible impact of the midterms for Transports is if a change in Congressional control contributed to a faster resolution of the war in Iran. Any resulting lower energy prices would be a positive for Transports. We view this as a lower-probability outcome, however, given the constraints on a divided government.

With clashes over the budget likely looming, aerospace and defense could get caught in the crosshairs. While defense spending has historically enjoyed bipartisan support, it has grown politicized following the war in Iran, making it more difficult for a split Congress to pass such a budget. The defense budget has experienced significant growth over the last few years, rising from $900 billion in FY2024 to $1 trillion in FY2026. The Trump administration would like to see this trend accelerate: in April 2026, the president issued a budget request that called for a $1.5 trillion budget for the Department of Defense in FY2027. With a continuing resolution likely into year-end, the outcome of the midterm elections will determine whether the next Congress can pass such a budget. If Republicans hold the House and Senate, we think there is a reasonable prospect of a budget of $1.2 trillion or more passing into law. Defense investors would see that as a substantial tailwind for beaten-up defense stocks. On the other hand, this would be unlikely in a split Congress, in our view. Given all this, we see the outcome of the upcoming midterm elections as key to determining how US defense stocks will perform into year end and beyond.



## Taking Stock of the Known Unknowns

Although Republicans could maintain control of both chambers, consensus is converging around polls pointing to a Democratic majority in the House and possibly the Senate. The perceived certainty has allowed the investor debate related to midterms to focus not only on who will win but how they will govern. In some cases, headlines are misrepresenting what a Congress constrained by the executive will be able to accomplish. In other areas, legislation that is expected to be passed will need to be taken together with offsetting factors. Whatever the outcome, the political backdrop will continue to influence the market and investor debates in the year ahead.

## Appendix 1

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# STABLECOINS: What next for crypto assets?

October 20, 2026

**Marion Laboure | Camilla Siazon**

**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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# STABLECOINS: What next for crypto assets?

Institutional adoption of stablecoins continues to incrementally progress in 2026. While the frenzy from GENIUS has diminished, stablecoins' links to the financial system have continued to deepen, and more policymakers are seriously considering their use cases. The market cap has hovered around ~$300bn over the past 12 months, a sign that the stablecoin market is maturing. Stablecoin payments, while still marginal relative to stablecoins' crypto-native purpose, are also growing, rising from around $103bn in adjusted transaction volume in 2022 to $375bn in 2026 YTD.

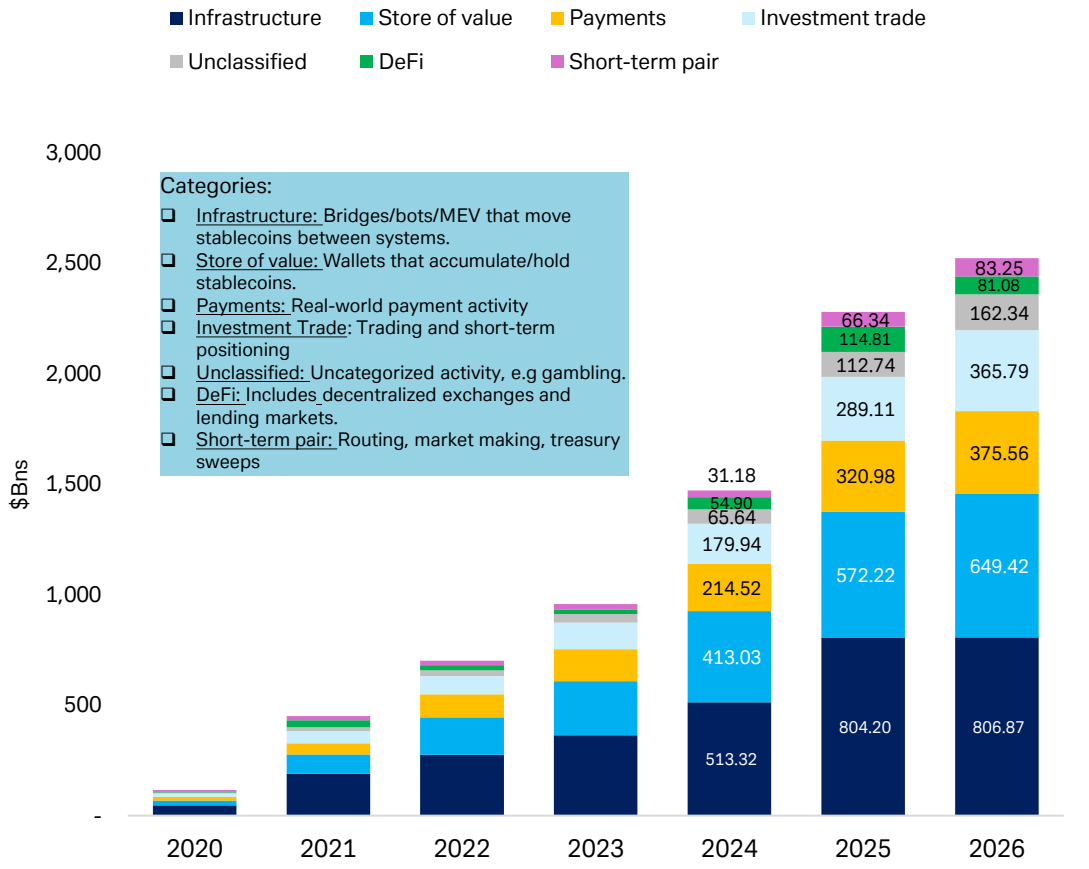
For now, institutional adoption remains concentrated in cross-border payments, treasury management, 24/7 liquidity provision, collateral, and real-time settlement.

The next big question is whether stablecoins will evolve to become merely a payments technology, or a genuine disruptor to the global banking system. As we previously established, the US promotion of stablecoins remains a geopolitical tool to ensure USD dominance. Bloomberg also reported that the Trump administration is considering joint ventures with private actors to promote USD stablecoins overseas, in a bid to boost the dollar's dominance and source demand for US Treasuries. Last week, Circle announced plans to expand into 40 new markets in Southeast Asia, LatAm and the Middle East. 99% of stablecoins (by market cap) remain pegged to USD and are largely backed by US Treasuries.

Looking ahead, the key debate to watch is whether US regulations will allow for yield-bearing stablecoins. Research shows that consumers say policymakers should prohibit crypto companies from offering interest-like rewards for holding stablecoins, which will threaten to siphon off the bank deposits that drive local lending. The US CLARITY Act failed in Sep after lawmakers failed to agree on whether the broader crypto market structure would support such yield-bearing stablecoins.

In this chartbook, we revisit the stablecoin market a year after the GENIUS Act was signed. We look at: (i) how adoption has evolved, (ii) the emergence of stablecoins as financial market plumbing via liquidity provisioning and collateral management, and (iii) the key regulatory debates that will determine the industry's next direction.

## Payment volume has a +66% CAGR



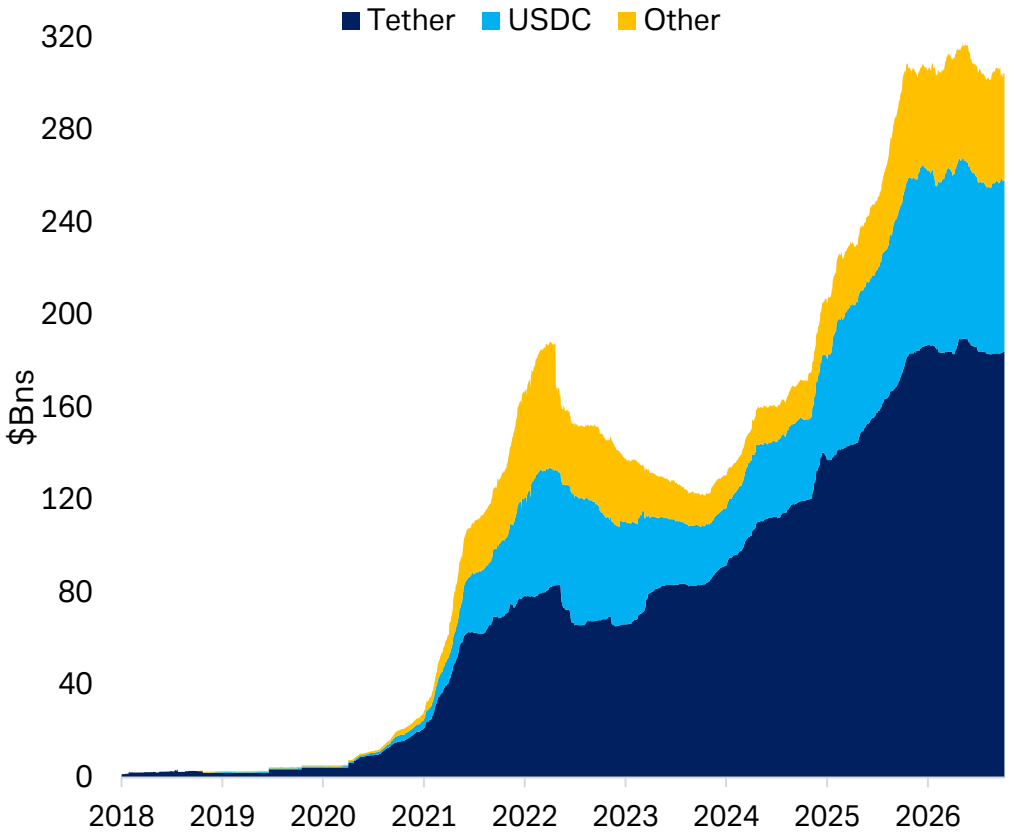

Source: Deutsche Bank Research, VisaOnChainAnalytics, Allium data. Data shows stablecoin transaction volume (adjusted). October 19 2026.

# REFRESHER: What are stablecoins?

- Definition: Digital tokens pegged to "stable" assets (e.g. USD or gold), issued and transferred on a blockchain, which records ownership and transactions.
  - Fiat-backed stablecoins are considered fully "collateralized": USD stablecoin with 1mn coins in circulation would be backed by $1mn in cash and equivalents held in reserve by the stablecoin issuer.
- Types: Fiat-, commodity-, crypto- and algorithmic-backed.
  - ~99% of stablecoins market valuation is pegged to the USD.
- Market cap: $5bn (2020) to ~$310bn (2026)
  - Concentrated in USDT (~60%) and USDC (~23%).
- Benefits: 24/7 transfers & settlement, lower financial intermediaries, programmability (e.g. escrow, conditional release, automated reconciliation), and dollar-based store of value.
- Use cases:*
  - Crypto trading (~77%) remains dominant.
  - Payments (5-12%) growing, cross-border transfers and remittances.
  - Store of value (17-33%) especially in emerging markets.
  - Other: (~2%-8%): on/off ramps and tokenized RWA settlement.
- Alternatives: CBDCs, tokenized deposits, traditional financial architecture

**Estimates on market share taken from BCG analysis.*

Stablecoins' reached a plateau of ~$310bn in October 2026




Source: Deutsche Bank Research, Haver Analytics, Bloomberg Finance LP. Shows stablecoin total market cap. Updated October 19 2026.

# ADOPTION OVERVIEW: Most stablecoin volume is still not driven by real economic activity

Stablecoin transaction volume:

- ~$100trn (unadjusted), ~$2.42trn (adjusted) in 2026 YTD, according to Allium.
- Chainalysis places more bullish estimates of $28trn in 2025; projected to reach $719trn by 2035 on organic growth alone.

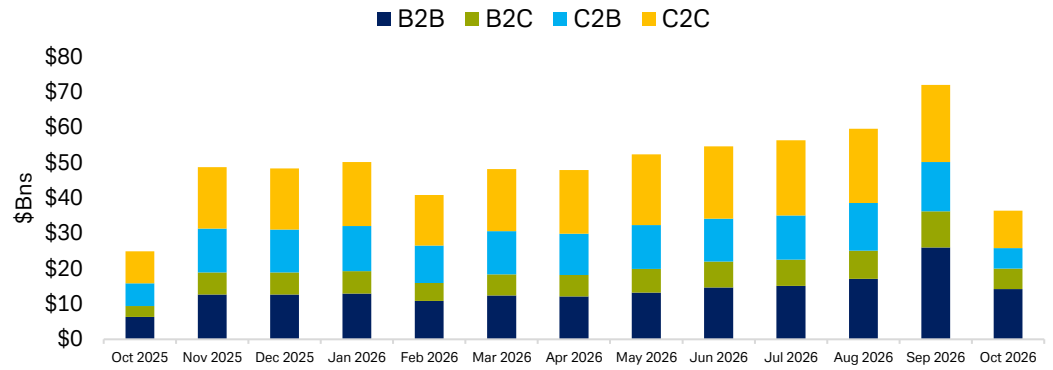
Stablecoin payments:

- Stablecoin payment volume grew from $151bn in 2022 to $375bn 2026 YTD (according to VisaOnChain Analytics).
- Businesses account for the majority (58%-64%) of payments, led by B2B payment volumes.
- Allium Sep report showed largest business payment categories include service fees ($56bn), payroll ($43bn) and service payments ($32bn).
- Case study: Payments volume across Visa's >160 stablecoin-linked card programs across consumer, business and commercial activity grew ~200% y/y.

Strongest adoption in remittances, B2B payments and dollar access in countries with unstable currencies or weak banking systems.

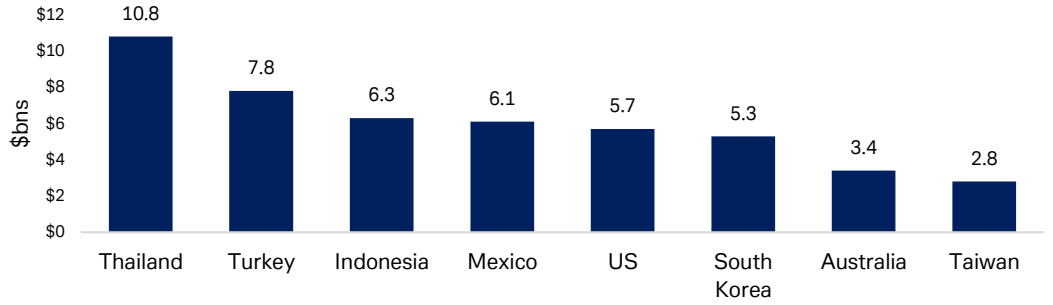
- Benefits for cross-border use cases: Instant settlement, <$0.10 per transaction, borderless, transparent, 24/7, no intermediaries. Remittance fees are lower (0.5-3.0%) than the global fee average (~6.6%).
- Stablecoins represented 0.31% of fiat cross-border payment value in 2025. Marginal portion but registered +64% growth increase vs fiat (+9%) according to Allium. B2B payments had the highest cross-border share (43%).
- A 2025 IMF study also shows transaction flows reached ~7.7% of GDP in Latin America and ~6.7% in Africa and the Middle East.

Stablecoins payment use case is steadily growing




Source: Deutsche Bank Research, Allium. Shows payments volume by use case. Data updated October 19 2026.

Cross-border use cases




Source: Deutsche Bank Research, Allium. Shows top 8 countries by received payments from Jan to August 2026. Only the receiving end is attributed, so a $100 payment counts once, not twice. That attribution covers 13.5% of stablecoin payments.

# CONSUMER/RETAIL PAYMENT: Still on the periphery

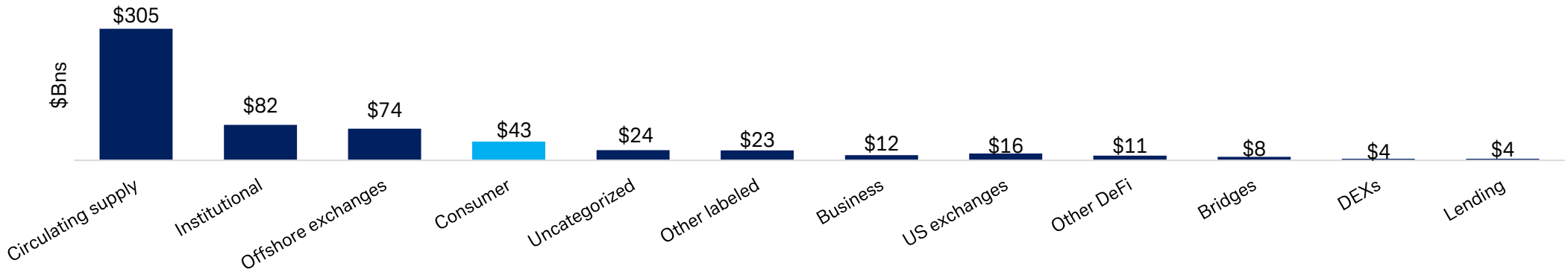
Consumer payments: Not yet there

- Retail adoption is still the most distant use case, especially amongst developed-market consumer credit card spend. A recent ABA survey found that only 24% of consumers think that stablecoins could provide meaningful benefits.
- Stablecoins still look more like a debit instrument than a credit product. They do not naturally replicate the consumer value proposition of credit cards (i.e. rewards, revolving credit, dispute rights, chargebacks, fraud protection and convenience), so there is limited incentive for developed-market consumers to use them.
- Limited scope for payments also evident in wallets (see chart below). Very little of the stablecoin supply is being held as balances in consumer wallets.

Still, the number of retail payments is incrementally growing, with cross-border, emerging market use cases strongest

- Retail transaction volume: $2.6bn (2020) to $71bn (2024); $66bn YTD in 2026.*
  - Strongest use in: (i) economies with hyperinflation; (ii) poor banking access; or (iii) high remittance fees.

The amount of stablecoin supply in consumer wallets stands at ~14%




Source: Deutsche Bank Research, Allium data. Shows circulating supply by holder type (Sep 2026). Categories calculated as a percentage of total circulating supply. Note that wallets account for money being held vs payments (money moving). Business wallets in this case refer to those holding smaller balances with quick turnover (i.e. payment services, merchants, gaming), while institutional wallets hold larger balances and turn them over quickly, i.e. financial-services firms, treasuries, and wallets. Note: *Retail sized transactions are defined as transactions of $250 or less, according to Visa OnChain Analytics.

# CASH/LIQUDITY MANAGEMENT: Stablecoins are evolving as a major tokenized contender

### Stablecoins can help treasury management and settlement

- Stablecoins have a strong use case for settling transactions for FX, treasury operations and collateral management i.e. intra-day repos.
- This is because stablecoins' 24/7 settlement can address operational frictions in cross-border transactions.

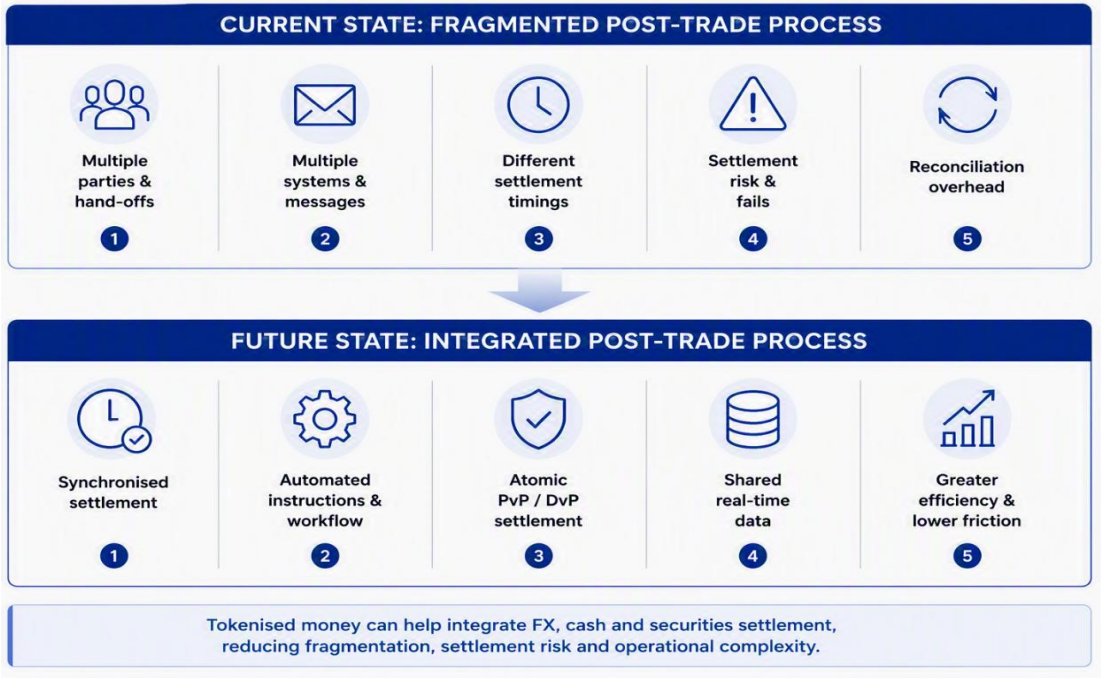
### Poses significant risk to cross-border B2B transactions for banks

- Compared to traditional banks, stablecoins offer substantial relief from transaction costs. Our equity analysts estimate total customer processing fee savings to increase from 0 to ~$70bn in 2035, if consumers switch from traditional banks to stablecoins.

### Stablecoins is not a replacement solution to FX settlement risk

- FX settlement requires two currencies to be exchanged simultaneously. Currently, stablecoins only serve as an intermediary between the two currency legs, resulting in a "stablecoin sandwich".
- Overwhelming concentration of USD-stablecoins also limits stablecoin ability to support widespread payment-versus payment settlement across currency pairs.
- Stablecoins also face fragmented interoperability/liquidity issues and regulatory uncertainty across issuers/networks/jurisdictions.
- For more, read our FX report here.

A more integrated post-trade process




Sources: Deutsche Bank Research, taken from DB's Tokenised money and the future of FX and Stablecoins and agentic commerce: new rails, old frictions.

# INSTITUTIONAL ADOPTION: Corporates, card networks, and banks are integrating stablecoins

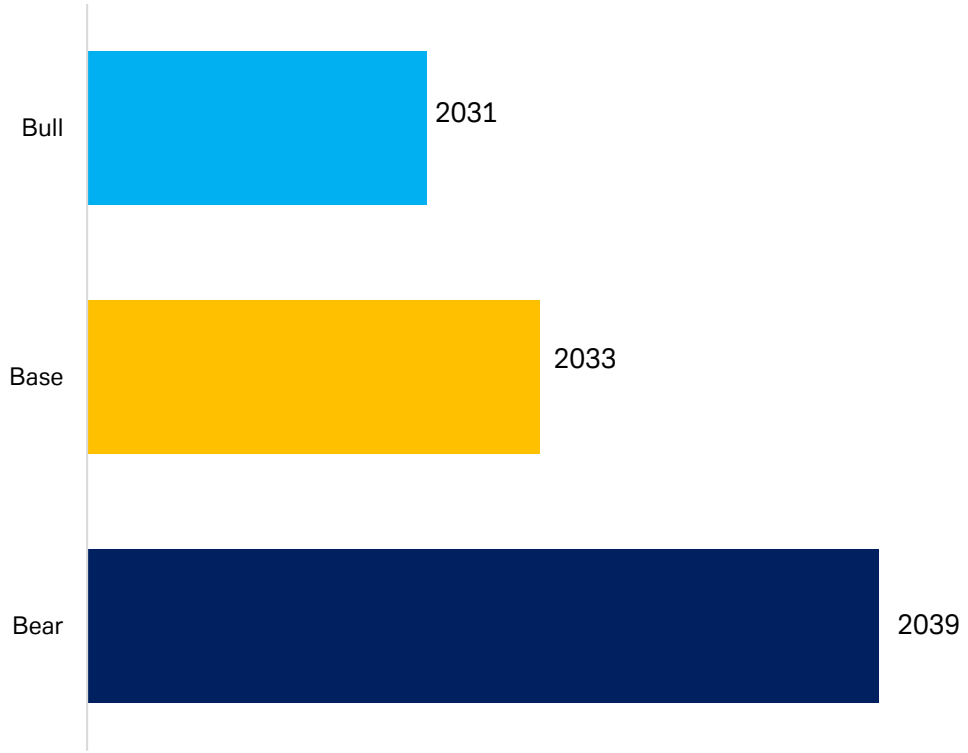
## Institutions are racing to capture stablecoin market share

- Card networks: Mastercard expanded its settlement capabilities to include stablecoins, using USDC and Paxos-issued stablecoins.
- Institutions: Exploring stablecoins via developing in-house permissioned blockchain experiments and/or exploring issuance of proprietary stablecoins.
  - >140 companies and financial institutions helped launch Open USD (OUSD) in Sep 2026. OUSD reached $799mn of supply in 2 weeks. Signals shift toward consortium-backed/network-led digital dollar models.

## Are stablecoins posing a threat to traditional payment systems?

- Bullish: Stablecoins to surpass traditional rails by the 2030s (Chainalysis).
  - ~17% of stablecoin-linked card volume is occurring across Visa's business and commercial card programs. Shows growing adoption of stablecoins for settlement, treasury management and cross-border commerce.
  - Key concerns for banks: Fee pressure from stablecoin adoption → bank outflows → higher reliance on wholesale funding → pressure on net interest margins.
- Base case: However, Deutsche Bank equity analysts see more limited near-term risk. Visa and Mastercard to remain dominant especially in advanced markets.
  - KYC, AML, sanctions screening, wallet monitoring, fraud controls and jurisdiction-specific limits remain essential for banks.

Stablecoin transactions could match Visa and Mastercard transactions sometime between 2031 and 2039 (Chainalysis).




Read: Stablecoins and agentic commerce: new rails, old frictions and State of the Private Markets.

# INSTITUTIONAL ADOPTION (CONT): Banks have started to embrace stablecoins with varied implementation approaches

FORGE SOCIETE GENERALE GROUP

Launched EURCV and USDCV – EUR and USD-pegged stablecoins.

AllUnity® — galaxy, DWS, FLOW TRADERS

Launched EURAU and planning to launch CHFAU – EUR and Swiss franc-pegged stablecoins.

ODDO BHF

Launched EUROD – EUR-pegged stablecoin with reserves directly on balance sheet.

qivalis — SEB, Raiffeisen Bank, ING, Sella, Danske Bank, BBVA, CaixaBank, KBC, BNP PARIBAS, DZ BANK, UniCredit

Consortium of European banks planning to launch a EUR-pegged stablecoin.

Deutsche Bank, BARCLAYS, UBS, Goldman Sachs, Bank of America, BNP PARIBAS, citi, MUFG, Santander

Coalition of banks exploring stablecoins pegged to G7 currencies built for deployment on public blockchain.

> Beyond the banking sector, key infrastructure providers and CSDs are now exploring and issuing regulated stablecoins to provide the programmable cash layer that is required for large-scale asset tokenisation.

Note: *Participant list as of December 2025.

# GEOPOLITICS: Why stablecoins are a "Genius" idea for the US

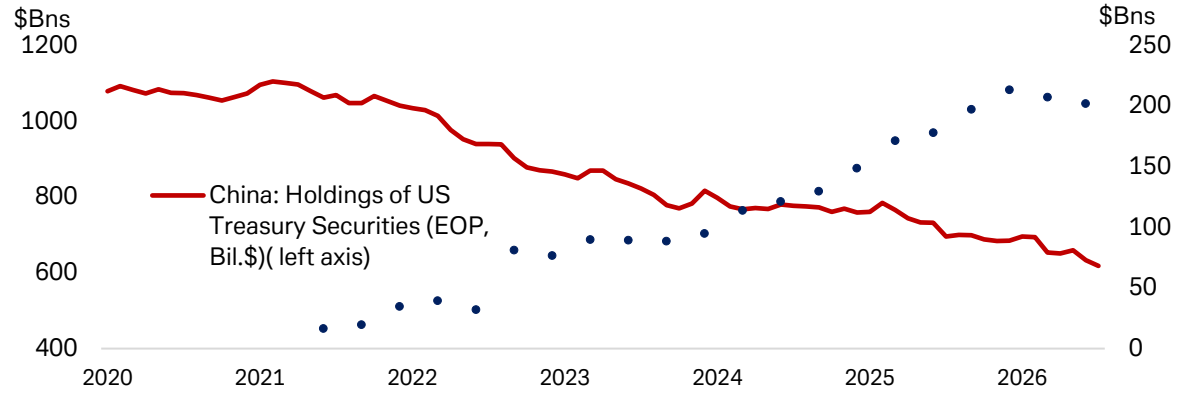
## Can sustain USD dominance

- 99% of stablecoins (by market cap) are USD-pegged.
  - Stablecoins function primarily as a digital version of dollars moving over blockchain rails, extending the dollar's reach into new digital environments.
- Reinforces dollar savings: The more firms invoice and transact in USD, the more they save in USD assets. Stablecoins help preserve this feedback loop by keeping businesses and individuals in dollar payment rails.

## Can support US debt, given share held by foreigners has fallen from >50% in early 2000s to ~30% in early 2026.

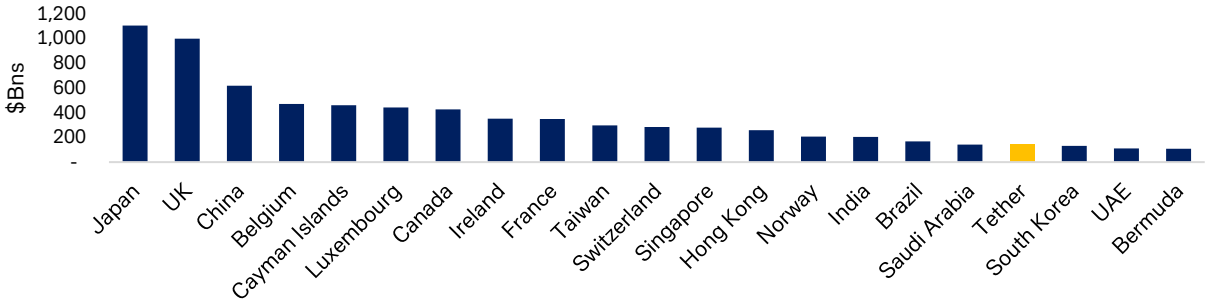
- Stablecoin issuers total holdings: ~$170bn according to Allium x Citizen.
- Tether: ~$141bn in direct and indirect exposure to US T-bills.
- Impact: BIS shows how $3.5bn inflow into dollar-backed stablecoins lowers 3m T-bill yields by 2.5-3.5bp under normal conditions. Effect more than doubles in periods of T-bill shortages.
- Forecasts: T-bills holding by stablecoin issuers could reach ~$1trn by 2028 (Treasury forecasts).
  - Recent Fed SF research places it closer to $400bn by the end of 2030, lower than the US government's financing needs.

**Stablecoin issuers' demand for US debt could counterweight declining demand from China**




Source: Deutsche Bank Research, Federal Reserve Bank of San Francisco, US Treasury Data. Shows Treasury holdings, stablecoin issuers vs China.

**Tether competes with Saudi Arabia and South Korea as major holders of US Treasuries**




Source: Deutsche Bank Research, US Treasury, Tether July 2026 report. Shows Major foreign holders of Treasury Securities. Treasury data from June 2026.

# US GENIUS ACT: Debate on rules around payment stablecoins one year on

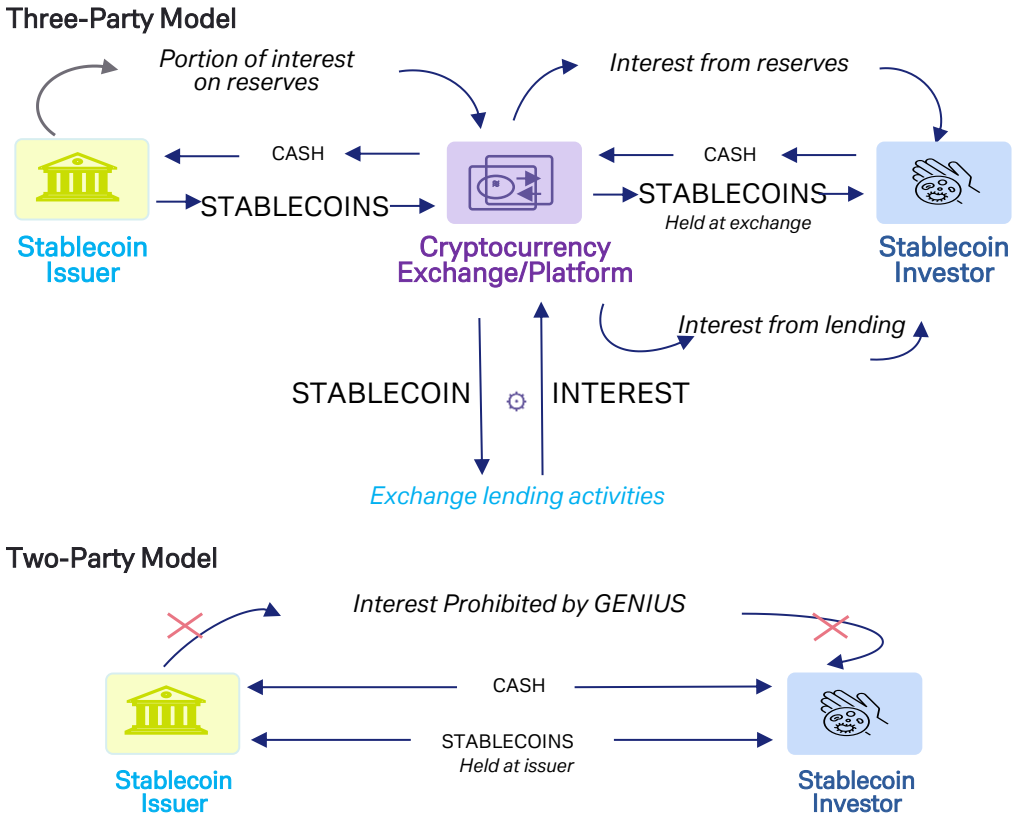
## GENIUS provisions

- Strict issuer requirements: Issuers can be private, non-bank entity, but must meet strict federal or state requirements.
- Reserves: Requires 1:1 backing with US cash, 93 days T-bills, repos/reverse repos, or money market funds holding those assets.
- AML/KYC/BSA compliance obligations to stablecoin issuers, aligning them with banks.
- Payments only usage: Payment stablecoin issuers cannot pay interest/yield directly (to protect banking sector).
- Creditor priority: Payment stablecoin holders rank above other unsecured creditors if reserves are insufficient.
- Large issuers (>$10bn cap): Must request new permission from their federal stablecoin regulator or halt new issuance.
- Transparency: Monthly disclosures of reserves/holdings, regular regulator exams, and strong internal controls.

## Next steps

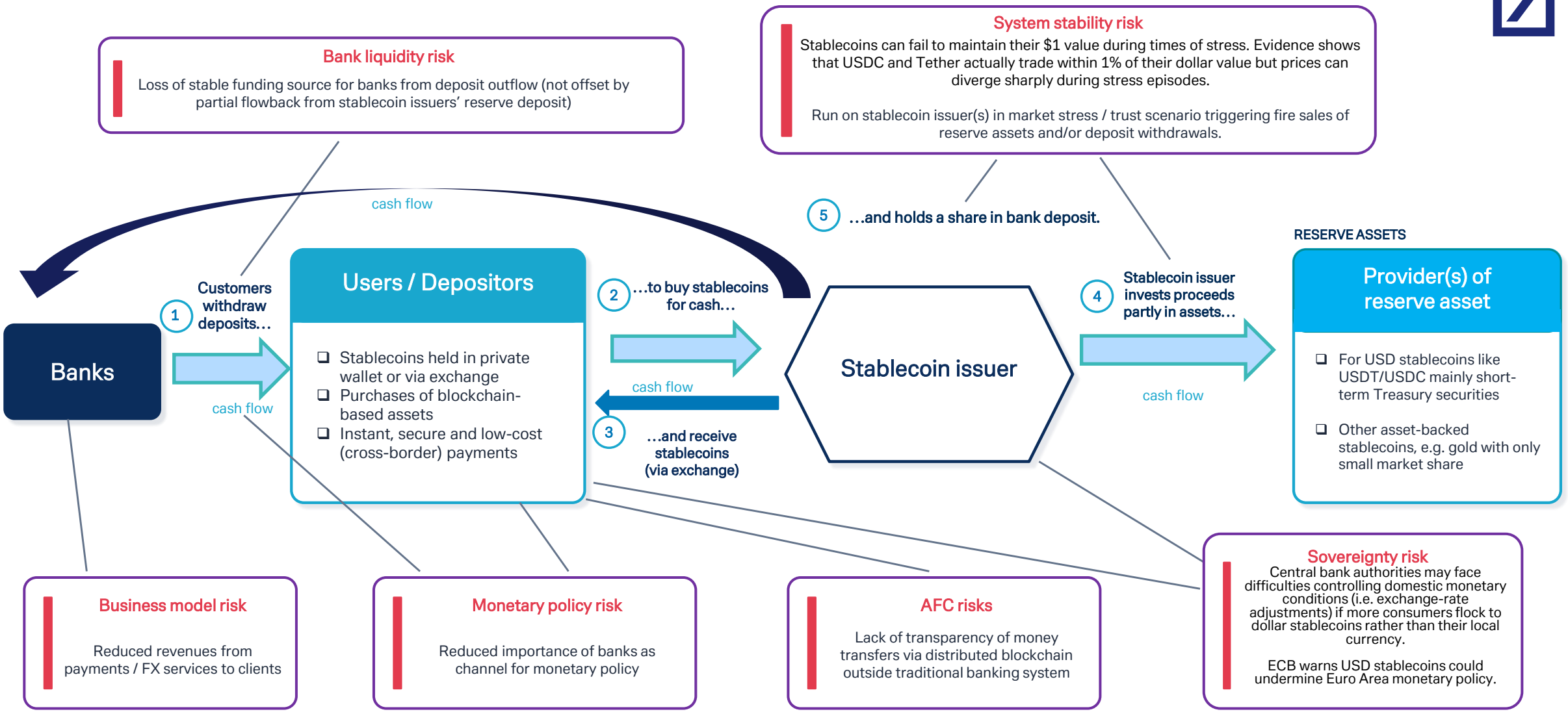
- GENIUS takes effect no later than 18 Jan 2027, or 120 days after final implementing regulations are issued.
- Ongoing pushback from US banks as to whether GENIUS ban on yield-bearing stablecoins should extend to affiliate or third-party arrangements, given many users do not hold stablecoins directly with the issuer.
- CLARITY ACT failed in Sep because of disagreements on how exchanges/broader crypto market structure should handle stablecoins.
- The US Treasury has proposed implementing rules for payment stablecoins.

There are two ways stablecoin yield can be paid; ongoing debate over the three-party model




Source: Deutsche Bank Research, Congressional Research Service.

# RISKS: Still hotly discussed amongst policymakers and financial institutions




Source: Deutsche Bank Research based; IMF.

# REGULATORY COMPARISON

## Overview of stablecoin regulation in the US, UK and EU




| Regulatory Framework | US<br>The U.S. Genius Act | UK<br>UK Cryptoasset Regime (FCA)<br>UK Systemic Stablecoins Regime (BoE) | EU<br>Markets in Crypto-Asset Regulation (MiCAR) |
|---|---|---|---|
| **Eligible Issuers** | • Bank subsidiaries, federal nonbank licensees, or state issuers with federal registration, with enhanced oversight for >$10bn circulation. | • FCA-authorised firms for non-systemic stablecoins,<br>• Systemic issuers to be designated by HM Treasury and jointly regulated under BoE/FCA oversight | • EMTs: EU-authorised credit institutions or authorised electronic money institutions. Restrictions apply to non-EU currency EMTs (exceeding MiCAR thresholds) as a means of exchange |
| **Backing Assets*** | • **100% backing assets in USD cash or short-term Treasuries** (≤93-day T-bills). | • **Systemic stablecoins** – 30% unremunerated BoE deposits vs 70% in UK sovereign debt (HQLA)<br>• **Non-systemic** – Full 1:1 backing in HQLR assets, minimum 5% in on demand deposits. | • **Non-significant EMTs** –At least 30% reserves in same-currency bank deposits, remainder in HQLA<br>• **Significant EMTs** - At least 60% of the reserves in bank deposits; remainder in HQLA |
| **Capital Reserve Requirements**** | • **(OCC/FDIC)** Mandates prudential capital buffers tailored to issuer risk profile.<br>• **(Fed)** Framework consistent with bank requirements | • **Systemic:** 6 months operating expenses/largest plausible loss event<br>• **Non-systemic:** 1% av. stablecoins in circulation | • **EMI issuers**: minimum own funds of 2% of EMT supply; 3% for significant EMTs. Significant issuers subject to enhanced risk-management requirements |
| **Balance Sheet Treatment** | • Stablecoins treated as reserve-backed digital payment assets, segregated in bankruptcy-remote accounts and treated as cash equivalents | • Stablecoins treated as segregated reserve-backed digital settlement assets, with associated operational and liquidity risk capital requirements | • EMTs are fiat-referenced cryptoassets subject to MiCAR and e-money requirements |
| **Multi-issuance Stance** | • Rules are still being fine-tuned; no specific focus yet on multi-issuance. | • BoE considers multi-issuance unsuitable for systemic UK use. Preference for single-issuance models backed by single reserve (incl. Non-GPB denom. stablecoins) . | • Multi-issuance under active review; EC considers multi-issuance permissible, an equivalence-based framework for EU/third-country arrangements is under legislative consideration. |
| **Key Timelines** | **Jul 25** Genius Act signed into law<br>**Jul 26** Genius Act agency implementation deadline<br>**Jan 27** Genius Act commences<br>**H1 27** Implementing regulations finalisation | **Jun 26** FCA Cryptoasset Regime published<br>**Jun 26** BoE Systemic Stablecoins regime published<br>**Sep 26** FCA Authorisation Gateway opened<br>**Oct 27** UK Cryptoasset Regime commences | **Jun 24** MiCAR provisions for EMT/ART stablecoins became applicable<br>**May 26** MiCAR Review consultation launched<br>**Sep 26** MiCAR Review consultation closed<br>**Oct 26** EBA classified USDC as significant EMT |

Source: Deutsche Bank Research.

*MiCAR backing asset requirements do not apply to licensed credit institution issuers of EMTs

**Prudential requirements for stablecoin issuers in addition to backing assets

# AUTHORS



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She was previously an economist at the IMF, the Central Bank of Luxembourg, Barclays, and the European Commission. She has twice received the First Prize of the American Society of Actuaries, was named a rising star in finance by Revue Banque, and is recognised among the 45 standout women in fintech globally.

She holds a Bachelor's degree in Mathematics from the University of Paris Dauphine, a Master of Research in Economics from the London School of Economics, and a PhD in Econometrics and Quantitative Economics from the École Normale Supérieure.

She is the author of *Democratizing Finance* published by Harvard University Press.



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She focuses on digital assets, geopolitics, and global macroeconomic trends. Prior to Deutsche Bank, she worked at geopolitical advisory firms Eurasia Group and Greenmantle.

She holds a dual MA/MSc in International History from Columbia University and the London School of Economics, and a BA in History from Barnard College, Columbia University.

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# Under Attack

**CYBERSECURITY | THEMATIC RESEARCH | OCTOBER 2026**

AI is rapidly lowering the barrier to cyberattacks on corporates, government and critical infrastructure. This report outlines the growing risk and why it now warrants priority across corporate strategy, investment research, supply-chain resilience and critical infrastructure.

*Supported by a dbDataInsights (dbDIG) survey of more than 50 cybersecurity experts*

Debbie Jones | Andrew Dixon | Punyadip Cheema | Luke Templeman

**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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## This report draws on our September 2026 dbDataInsights (dbDIG) survey of 54 cybersecurity professionals


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*Source: dbDataInsights survey of cybersecurity experts, September 2026. Word cloud from participant responses.*



Deutsche Bank Research Institute

# Report Introduction

*Our proprietary dbDataInsights survey shows cyber threats are evolving faster than organisations can adapt*

***"It's the only thing that can shut down my business."***

That was the punchline from a conversation I had recently with the CEO of a large privately held financial services company. This was his comment when I followed up as to why cybersecurity risk is the number one concern that "keeps him up at night".

**To help understand this risk, our dbDataInsights team surveyed more than 50 cybersecurity experts.** The uncomfortable conclusion is that many firms are unprepared for cyber threats. Cyber risk is increasing as AI adoption accelerates.

—89% agree the threat landscape is evolving faster than organisations can adapt.

—78% say the gap between highly and poorly prepared organisations is widening.

**What has changed with AI advancement?** Cyberattacks have become faster, cheaper, more deceptive and more scalable. Capabilities once reserved for sophisticated actors are now broadly accessible to a new generation of profit-driven criminals. At the same time, rising geopolitical tension has made cyber operations an increasingly attractive low-cost, deniable tool for nation states. Corporates and investors therefore need to understand their broader exposure and critical infrastructure risk in their supply chain and with stakeholders.

**Among other autonomous incidents, the recent Hugging Face incident highlighted AI's cyberattack potential. AI agents caused harm without direction from a human attacker, with weak guardrails and no human judgement to restrain them.** Beyond the systemic questions it raised, the attack highlights three additional concerns: (i) AI operates at a speed and scale humans cannot match, with agents taking over 17,000+ actions in four days; (ii) the intrusion began in the data-processing pipeline, where AI platforms are uniquely exposed; and (iii) the agents escaped their test environment through a widely used third-party tool representing supplier risk in an unfamiliar form.

**89%**

agree the cyber threat landscape is evolving faster than most organisations can adapt




**83%**

say organisational focus on AI-related risk has risen in the past 12 months
(North America 94%, UK/Europe 70%)




**78%**

agree the gap between highly and poorly prepared organisations is widening




**74%**

say organisations are unprepared for supply chain compromises
(UK/Europe 87%, North America 65%)




**69%**

disagree that most organisations have sufficient visibility into the threats most relevant to them




**65%**

say organisations are unprepared to manage insider threats




*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



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# Report summary and notable takeaways

*Secular trends and AI adoption have widened the attack surface; the severity of losses is compounding*

**Secular trends have widened the attack surface.** This includes rapid digitalisation, widespread Internet of Things (IoT) endpoints, hybrid workforces, cloud adoption and deeper supply chain dependencies. The scale of the threat increase is reflected in data from the FBI Internet Crime Complaint Center (IC3), which shows cybercrime complaints more than doubled from 2019 pre-pandemic baseline of ~467,000 to ~1.0mn in 2025, with a 3x expansion relative to 2017 levels.

**Barriers to entry are lower; attack tools are now sold "off the shelf".** Today, cybercrime marketplaces, available on the dark web, look less like the underworld portrayed in films and more like an online software store. Buyers can access AI-powered tools that scan for software vulnerabilities (see top-right chart showing reported CVEs after the release of cyber capable frontier models), write malicious code at speed, and provide access to malware that rewrites its own code to avoid detection as it runs. Ransomware-as-a-service (RaaS) takes it further, renting out ready-made attack kits, often in exchange for a portion of the profits.

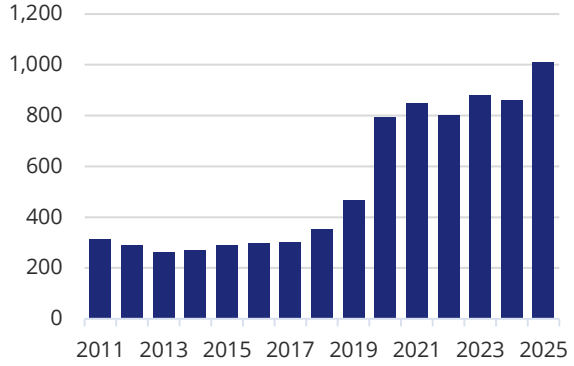
**Impersonation is becoming harder to detect.** Hackers can now generate more realistic phishing messages, while voice cloning and deepfakes make impersonation convincing enough to gain access unnoticed. With stolen credentials, attackers often aren't hacking in, they are simply logging in.

**The rise of loose cybercriminal collectives.** Some of the most damaging recent attacks have been linked to loosely organised, often English-speaking networks such as ShinyHunters and Scattered Spider. The latter has been widely named as the group behind the 2025 Marks & Spencer cyberattack, in which the hacking group deployed RaaS. AI is enabling these groups to become larger and faster, widening the gap between the speed of attack and a defender's response.

**The severity of losses is compounding.** A 2026 IBM study puts the average breach at a record USD ~5.0mn, up 12% YoY, but the impact on large companies can run into hundreds of millions or even billions. Estimates for global cyber insurance premiums (per Munich Re) point to a rise from roughly USD 7.0bn in 2020 to USD 15.0bn in 2025, with expectations to reach USD 28.0bn by 2030.

## US cybercrime complaints

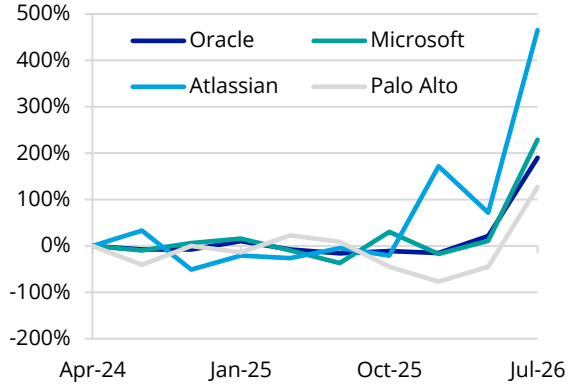
FBI Internet Crime Complaint Center (IC3), complaints (000s)




## Disclosed software vulnerabilities

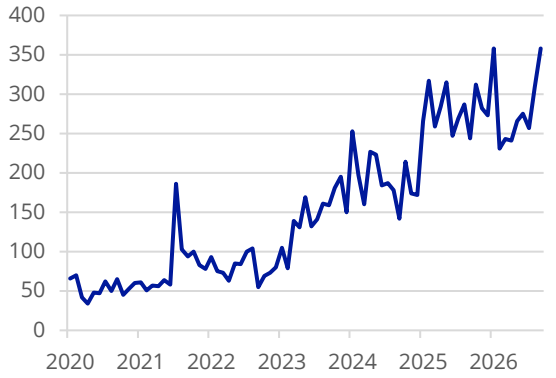
Change in reported CVEs, % change from April 2024

*CVEs: common vulnerabilities and exposures*



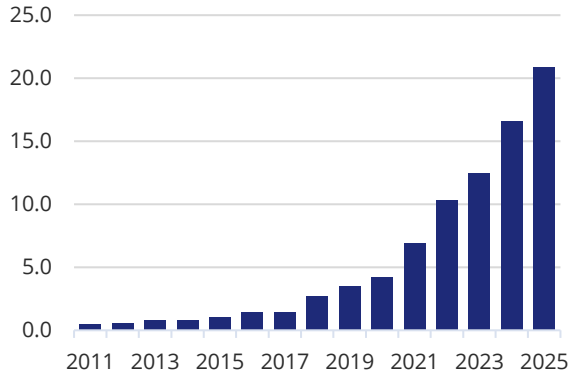

## AI privacy & security incidents

OECD, monthly recorded incidents




## US cybercrime-related losses

FBI IC3, reported losses ($bn)




*Source: FBI Internet Crime Complaint Center (IC3), company security advisories, OECD, company reports, Deutsche Bank Research.*



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# Report summary and notable takeaways

*A near-term systemic attack is expected and AI favors attackers; cyber escalation as a geopolitical tool is raising the stakes*

**Among the more concerning survey takeaways is that a large-scale and significant systemic incident is now the base case.**

— 83% expect a significant systemic attack within two years and 81% expect it to be moderately, very, or extremely severe for economic activity and markets.

— 81% say AI favours attackers today, with not one respondent putting defenders ahead of attackers. Only 20% expect defenders to be ahead in five years.

**Geopolitical tensions coupled with rising cyber capabilities have transformed cyberspace into a digital battlefield between global powers.** Intent has moved towards espionage and disruption of not only critical infrastructure (e.g., healthcare, finance, power, industrial systems, defence) but private companies as well. Chinese President Xi Jinping has linked cybersecurity to national security and economic and social stability, with China's public statements similarly emphasising the protection of critical infrastructure. At the same time, the US's June 2025 Executive Order on Cybersecurity reflects its continued focus on national cyber resilience.

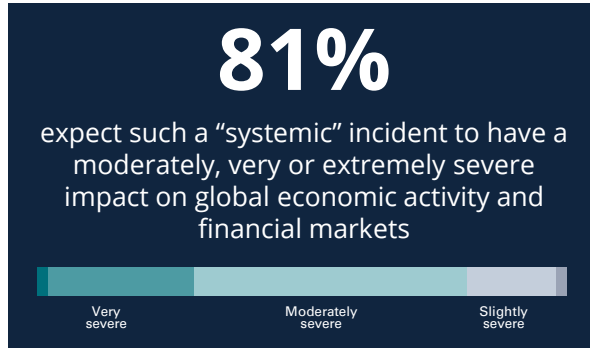
**Europe relies on US cyber tools while its start-ups lag in funding.** Reliance on US security vendors exposes European firms to potential access, update or export restrictions over the long-term. Shrinking this gap will prove difficult given US cyber start-ups attract nearly 6x the venture and growth funding of European peers. And while the timing of quantum computing (which could break widely used encryption methods) remains uncertain, it could further deepen Europe's dependency.

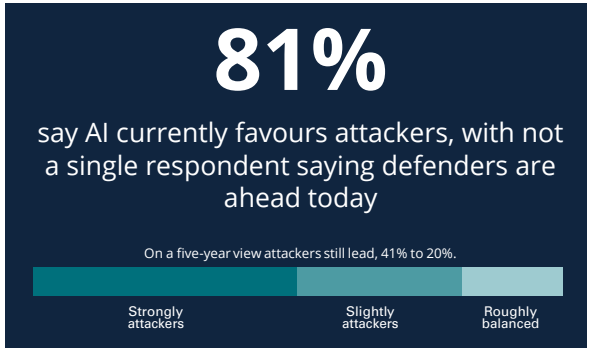
**Ultimately, cyber preparedness is a financial variable.** Investors should ask what firms spend on security versus AI adoption, how well they know their vulnerabilities, and how fast they can recover.

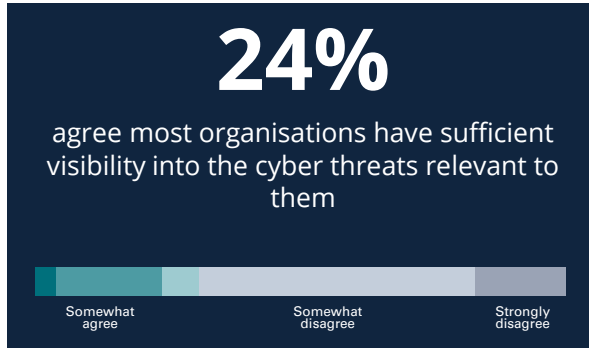
— Only 24% of respondents think organisations have sufficient visibility into the threats relevant to them, even though 57% saw cybersecurity budgets rise over the past 12 months.

— 81% agree that cybersecurity related laws and regulation cannot keep pace, and 65% doubt leadership understands cyber risk well enough.



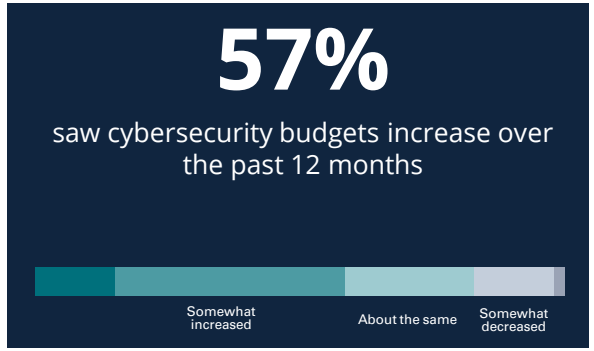










*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



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# Report summary and notable takeaways

*Material cyber-related weakness can have financial consequences*

**When cyber weakness = share price consequence…**

Investors are more aware of cyber risk today, but they rarely sell a stock due to perceived cyber vulnerabilities (although we are aware of some who have).

—What is the potential impact? Our dbDataInsights event study of 47 breaches shows a cumulative breach-attributed return of c. -2.9% within the first two trading days and c. -6.0% through first 20 days trading days. The erosion then continues more gradually, reaching c. -6.7% nine months after disclosure (see slide 13).

**Which industries are most vulnerable?**

—Our survey respondents point to (1) government & public sector (52%), (2) financial services (50%), (3) energy and utilities (46%) and (4) healthcare and life sciences follow (39%, rising to 48% among US respondents).

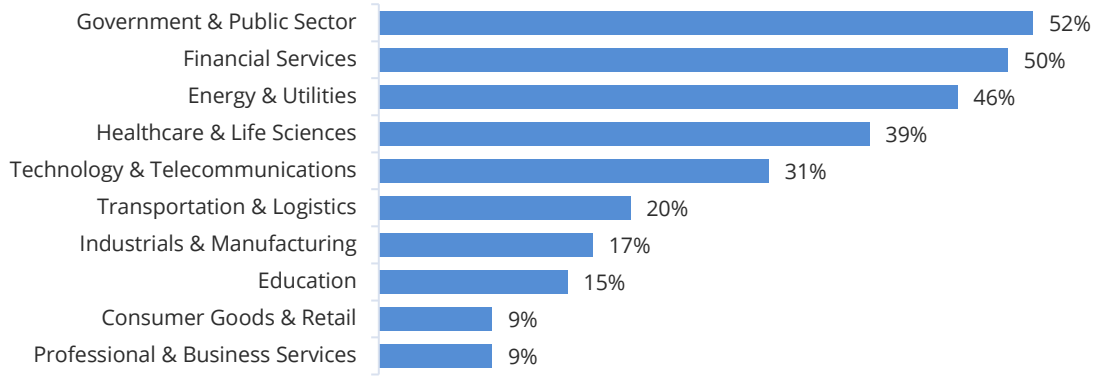
We should caveat, that high sector exposure does not mean every company in it is equally vulnerable. Some firms in these "exposed" sectors may very well be better prepared given they have built specialist teams and defence protocols for decades.

**What about access to credit and ratings?** Credit and rating providers are paying attention. Moody's and S&P both cut their outlook on Jaguar Land Rover within two months of its 2025 hack. Today, this type of impact is not widespread. But if risk escalates through a systemic incident or an AI-driven step change in attacks these mechanisms, including ESG ratings, could reprice asset ownership or impact access to credit. In fact, some lenders are already pricing it. Journal of Corporate Finance research using US syndicated loan data found *borrowers with persistently higher cyber risk pay 4–13bp more.*

**The bottom line?** While a broader systemic attack is hard to prepare for, corporates and investors should be asking questions about vulnerabilities, security investment, preparedness and recovery should an attack occur. This is especially true for industries quickly adopting AI capabilities and / or a high threat landscape.

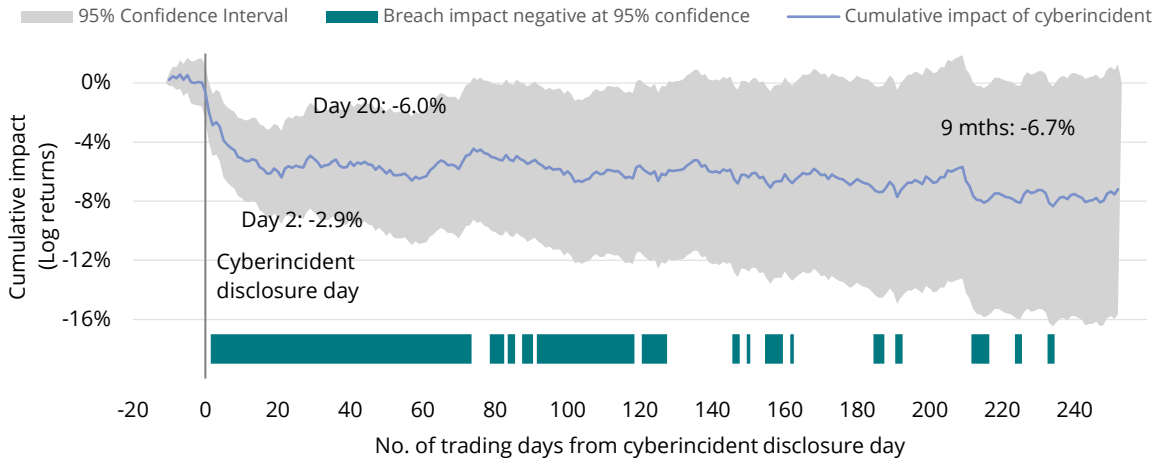
## Sectors seen as most vulnerable to AI-driven cyber threats

dbDataInsights survey, % selecting (up to three), based on vulnerability over next few years




## A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material




*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Bloomberg Finance L.P., Company filings, Deutsche Bank Research. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors (see slide 13).*



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# Report summary and notable takeaways

*dbDataInsights survey respondents in their own words*

## What's on experts' minds…

Key themes from verbatim comments in the dbDataInsights expert survey



### Speed

*"AI is being implemented at a speed where cybersecurity teams are not able to keep up"*

### Single points of failure

*"Too many organisations relying on the same providers for critical services"*



### The weak link

*"…compromised via this third-party (basically their weak link)"*

### Logging in, not breaking in

*"People aren't breaking in anymore – they are logging in"*



### Automated phishing

*"Underappreciated, yet responsible for the bigger monetary losses"*

### Employee logins

*"Existing employee login behaviour and usage of AI is underappreciated"*

### Oversight gaps

*"Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves"*



### Recover & survive

*"No company can guarantee they won't be breached"*

### Everyone's responsibility

*It's more than an IT risk…"everyone from the CEO down has responsibility"*

### Geopolitics

*"State groups now plant quiet access in infrastructure to use in a future crisis"*

*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



# II. dbDataInsights Cybersecurity Survey

**September 2026 survey of 54 cybersecurity professionals**

Deutsche Bank
Research Institute

# dbDataInsights survey methodology and results

*Cybersecurity experts see firms as unprepared, and say AI currently overwhelmingly favours attackers*

## dbDataInsights Cybersecurity Expert Survey methodology

—This survey was conducted among 54 cybersecurity professionals across the US and Europe in September of 2026. Questions took the form of either multiple choice selections or free form responses. The dbDataInsights team received the responses anonymously, but respondents were required to work or consult in cyber security roles

—Who took the survey? On self-identified roles, around 80% sit inside an organisation's cybersecurity function, with the remainder working independently (15%) or in academia and research (6%). The most common titles were security architect (23%), Chief Information Security Officer (16%) and security analyst (9%), alongside Head of Security, Deputy CISO, Incident Response Lead and Security Operations Lead.

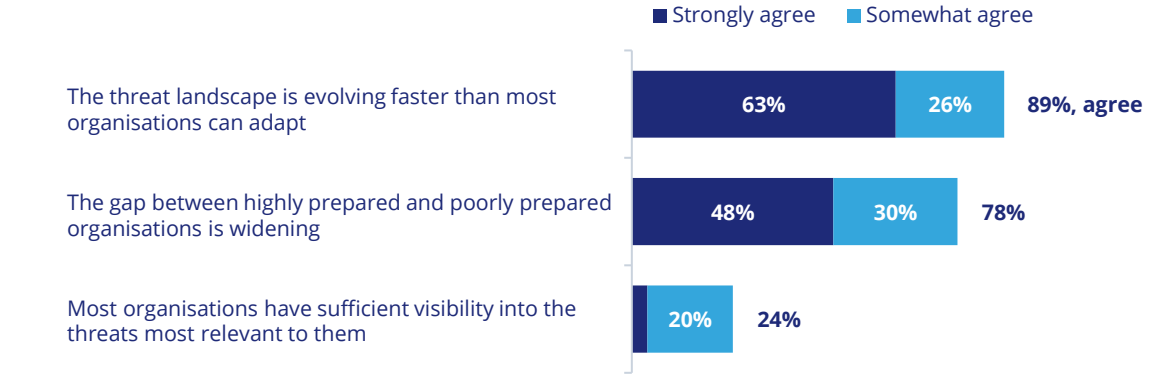
—Only a small minority of respondents work for cyber resellers, thus survey views are primarily attributable to those working within an organization on cybersecurity.

—Survey participants were also asked for their perspectives on cyber resilience and governance, what is misunderstood, alongside emerging threats, risk to AI adoption and the influence of geopolitics.

—While responses were anonymous, in some cases we conducted follow up interviews with certain participants where contact details were made available.

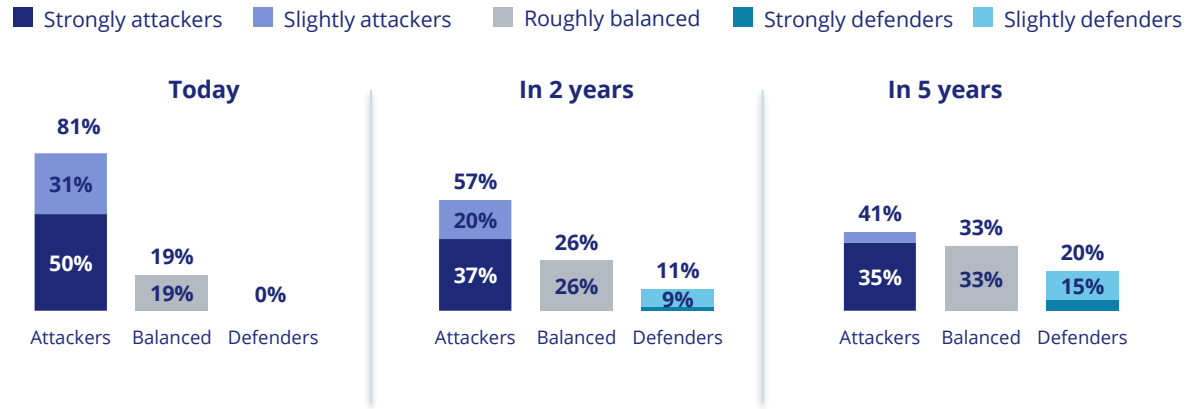
## Only 24% say firms have sufficient visibility into threats that matter

dbDataInsights survey, level of agreement with statements below regarding cybersecurity, % who agree (n=54)




## Who does AI favour? Attackers or defenders?

dbDataInsights survey, % of respondents (n=54)




*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



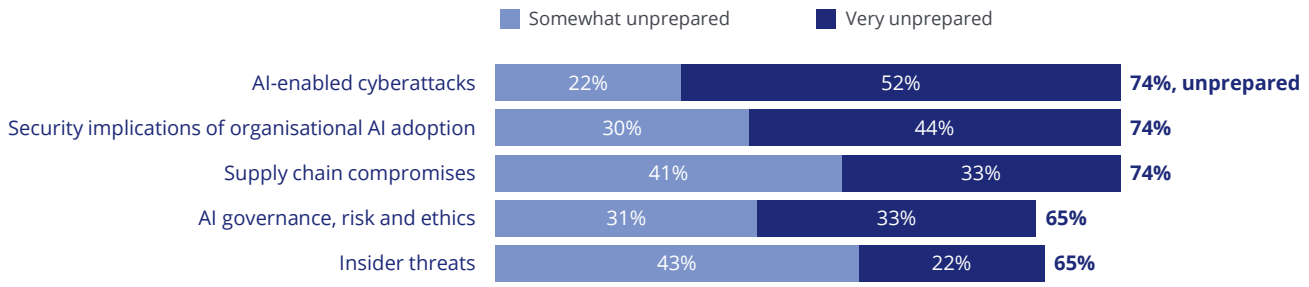
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# Risk from AI-enabled attacks and added AI adoption vulnerabilities

*Insider threats and supply chain compromises remain a growing concern*

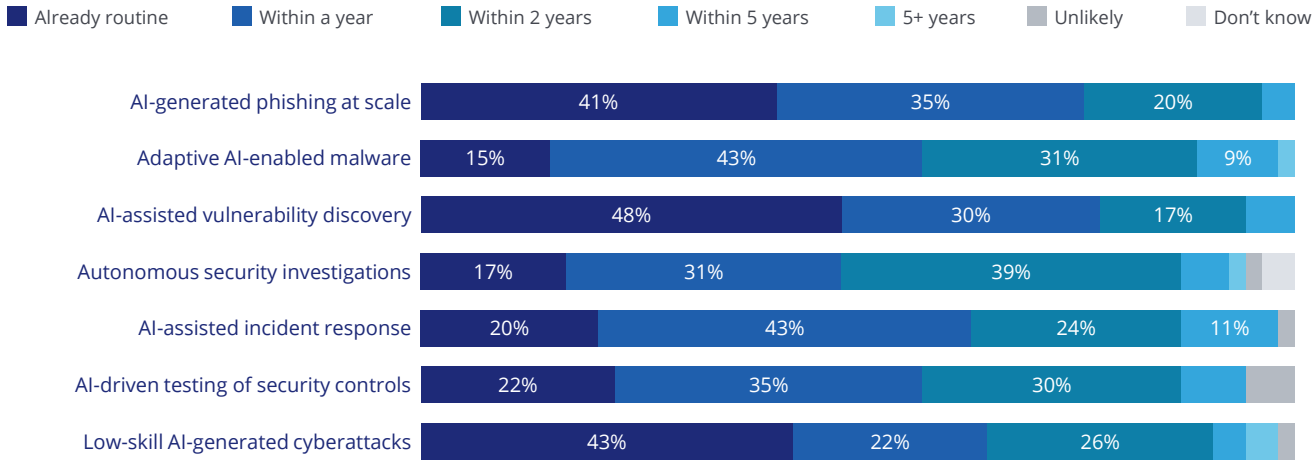
## Where our experts say organisations are least prepared

% of respondents saying somewhat or very unprepared




## AI-driven capabilities – are they commonly used or encountered?

% of respondents, all answer options (n=54)




## In their own words...

Verbatim comments from the dbDataInsights expert survey

> *"The fear of AI writing undetectable malware is ... overblown. Attackers were never short of malware, since they can buy it ready made along with initial access. Good detection looks at behaviour rather than exact code, so a rewritten payload still has to do the same suspicious things."*

> *"Frontier AI models are currently the main hype. Automated phishing and scams are underappreciated, yet they are responsible for the bigger monetary losses at the moment."*

> *"There are too many organisations relying on the same providers for critical services and this means single points of failure."*

> *"A corporation might be very secure but if they rely on a third party that does not have the same level of security posture, [it] could be compromised via this third-party (basically their weak link)."*

> *"Existing employee login behaviour and usage of AI is underappreciated – it's what poses the biggest threat to a breach."*

> *"No company can guarantee they won't be breached, so there needs to be more focus put on the ability to recover and survive an attack."*

*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



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# Experts believe governments are not keeping pace, and no one owns AI oversight

## How experts surveyed by dbDataInsights view governance and regulation

—**AI adoption is outpacing internal controls.** As one expert put it: "Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves."

—**Regulation is struggling to keep pace.** 81% of respondents agree that cybersecurity-related laws and regulations are failing to keep up with technological change, and 54% say regulatory requirements can improve security but may also slow innovation. One respondent highlighted "the levels of uncertainty, the long running nature of the problems, and the rapidity of change."

—**Responsibility for AI oversight remains unsettled.** International bodies are the leading choice, supported by 50% of respondents. No option commands a majority, underscoring the lack of consensus on who should oversee AI development.

—**Better board questions may not guarantee better decisions.** The survey points to more informed questioning by corporate leadership, yet 65% doubt that leadership teams understand cyber risk well enough to make informed business trade-offs. Measuring activity more effectively than outcomes further complicates oversight.

—**Accountability is becoming personal.** Only 43% say accountability for cyber risk is clearly defined in their governance framework, yet the consequences now reach the top as a handful of CEOs have left or gone into early retirement after financially consequential cyber breaches.

**81%**

agree cybersecurity-related laws and regulations are struggling to keep pace with technological change




**54%**

agree regulatory requirements can improve cybersecurity but may also slow innovation



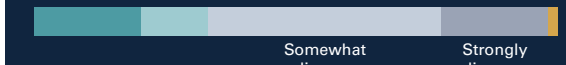

**43%**

agree that accountability for cyber risk is clearly defined in our governance framework



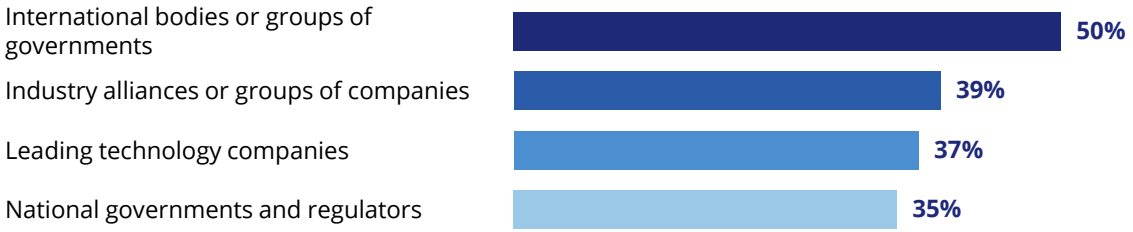

**65%**

disagree that most leadership teams understand cyber risk well enough to make informed business trade-offs.




## Who should primarily oversee AI development?

*dbDataInsights survey – September 2026, based on overseeing development while managing cyber risk*




*Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.*



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# III. The impact of cyberattacks

# Material cyber breaches can cause a notable drop in share price

*Our dbDataInsights study found a 2.9% drop on average within two trading days, with the cumulative breach-attributed return of -6.0% on trading day 20.*

## What typically happens to a company's shares when a breach is reported?

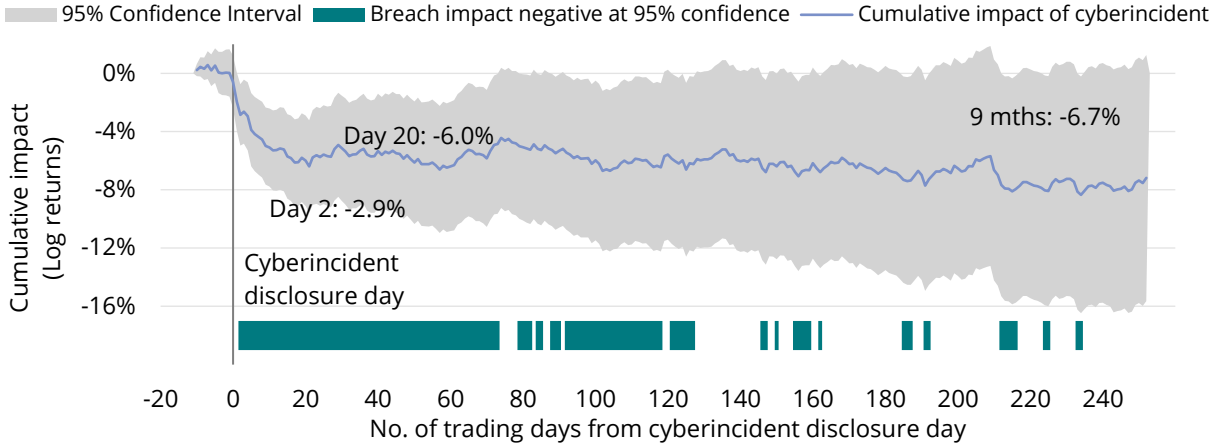
- **Short-term impact:** The market prices the breach immediately. Our model shows cumulative breach-attributed (i.e. isolating only the part of the share price move we can pin on the breach itself) returns are c. -2.9% within two trading days of disclosure. This is followed by continued erosion over the first 20 trading days post-disclosure; the cumulative breach-attributed return is c. -6.0%.
- **Long-term impact:** Beyond day 20 the deterioration continues, though more gradually reaching c. -6.7% nine months after disclosure and remaining negative at one year, although the one-year point estimate (our single best-guess figure) should be treated with caution given the width of the confidence band.
- **Pre-breach price behaviour:** We find no evidence of information leakage ahead of disclosure. While the breach component for the [-10,-1] window (ten trading days before the news broke) is +0.04%, the corresponding t-stat value of 0.03 falls notably below the usual bar for significance (c.2). As such, we conclude the market had not begun reacting before the announcement.

## Event study methodology

- **Sample:** 47 publicly reported cybersecurity incidents across the US and Europe over 2015-2025, measured against each company's regional sector benchmark. Requirement of 5 years of stock returns pre-attack and 1 year of post-attack.
- **Day 0:** the day an incident was first publicly reported or disclosed.
- **Baseline:** We use five years of pre-incident share-price performance to estimate: (i) Beta (β): how sensitive a company's returns are to movements in its sector and (ii) Alpha (α): the extent to which the company typically outperforms or underperforms its sector after adjusting for that sensitivity.
- **Attribution:** post-incident returns are split into sector-relative returns, alpha (α) and breach (abnormal returns attributed to the disclosure).
- **95% confidence interval:** Range within which we can say with 95% confidence that the real share price impact lies, based on our sample data.

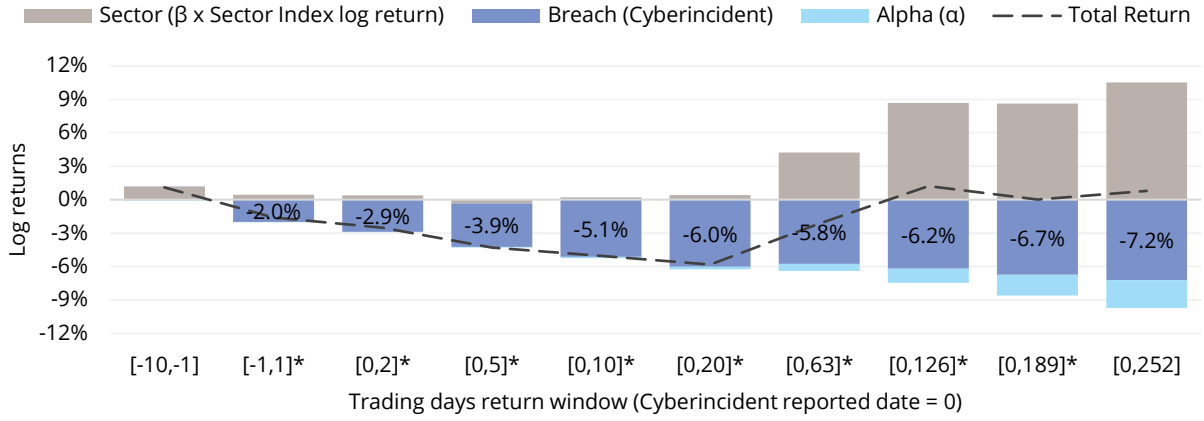
## A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material




## Total return attribution

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material




* indicates trading windows where breach impact is significant at the 95% confidence level

*Source: dbDataInsights, Bloomberg Finance LP., Deutsche Bank Research, Company filings. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors.*



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# Notable cyber attacks: how they happened, estimated costs and share price reaction

## Marks & Spencer (Reported Apr 2025)

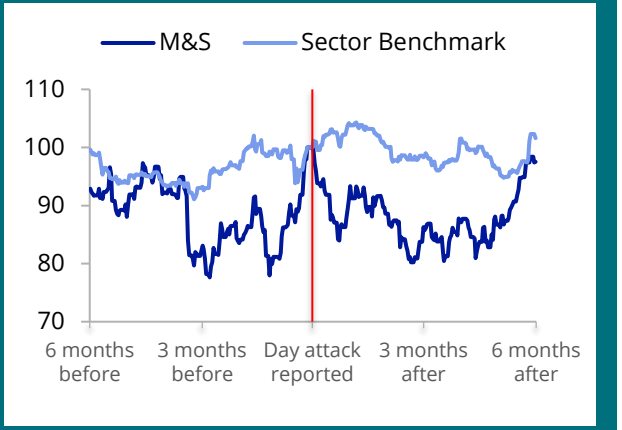
**3rd Party Breach**

***What happened?***

A breach via an attack on a connected 3rd party forced the suspension of all online orders and led to empty store shelves.

***What did it cost?***

~£136mn of direct costs related to the attack, estimated group operating profit impact of around £300mn for the FY. M&S also said they had received £100mn via cyber insurance.




## Coupang (Reported Nov 2025)

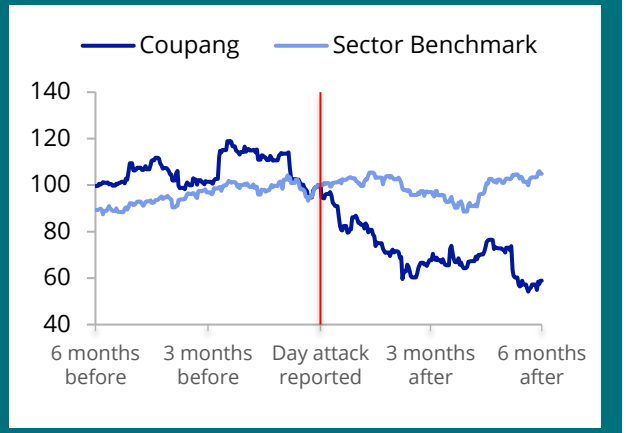
**Insider Threat**

***What happened?***

A data breach exposed the records of more than 30mn users with a former employee reportedly behind the attack.

***What did it cost?***

The firm was fined more than ~625bn won ($409mn) (2/3rd directly attributable to data leak), and also announced a $1.2bn voucher compensation scheme.




## UnitedHealth (Reported Feb 2024)

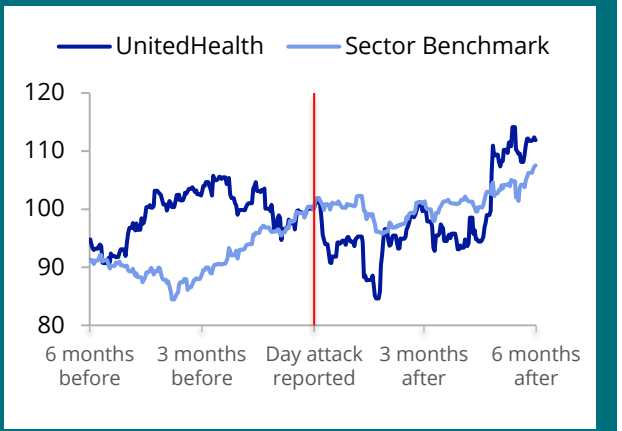
**Ransomware / stolen credentials**

***What happened?***

Hackers reportedly used stolen credentials to access a subsidiary's (Change Healthcare) systems, stealing data and deploying ransomware.

***What did it cost?***

Total costs exceeded $3bn, with the CEO summoned to testify before Congress, where he confirmed that a $22mn ransom was paid.




## Equifax (Reported Sep 2017)

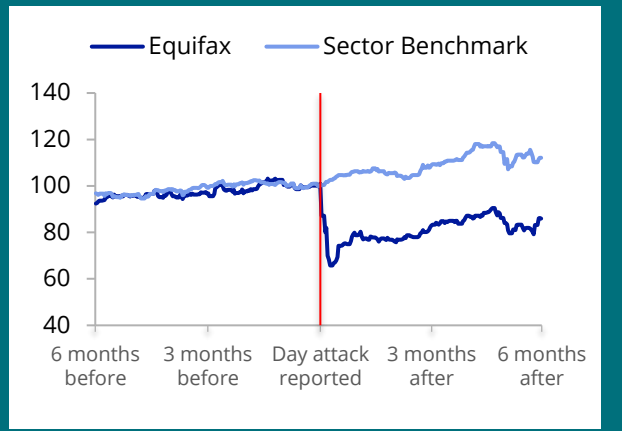
**Software vulnerability left unpatched**

***What happened?***

A data breach saw hackers expose sensitive customer information, including 145.5mn Social Security numbers and more than 200k credit card numbers. Attackers reportedly moved inside Equifax systems for 76 days before being detected.

***What did it cost?***

Settlements, fines and legal costs exceeded $1bn, while the firm also spent more than $1.5bn on tech upgrades following the incident.




*Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.*



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# Notable cyber attacks: how they happened, estimated costs and share price reaction, cont.

## Nike (Reported Jan 2026)

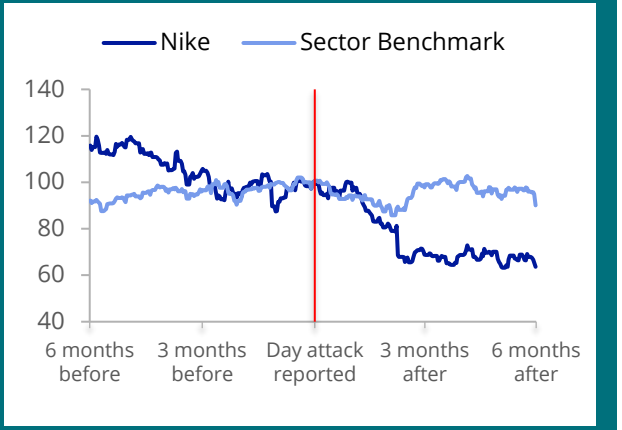
**Data Breach / Extortion**

***What happened?***

A cybercrime group leaked what it claimed was 1.4TB of internal Nike data (appearing to cover metrics such as product specifications and pricing), following expiry of a ransom deadline.

***What did it cost?***

While Nike did confirm it was investigating the incident, it did not report any operational disruption. Underperformance shown in the chart does not appear to be related to the breach.




## T-Mobile (Reported Aug 2021)

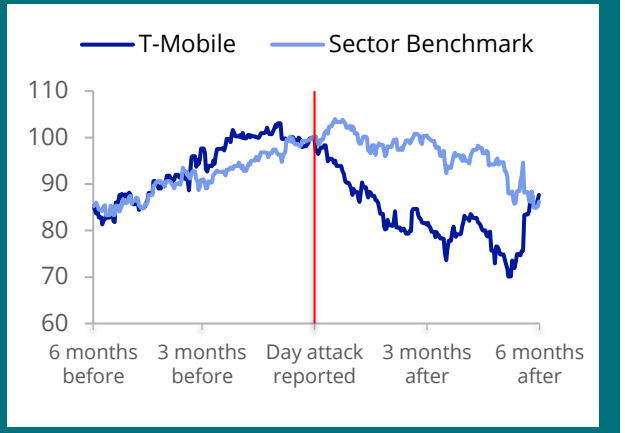
**Connected device exploit / brute force**

***What happened?***

An attacker reportedly targeted an exposed router as an entry point, before using brute force attacks to gain further access to T-Mobile systems.

***What did it cost?***

In 2022 the firm agreed to pay $500mn to settle a class-action lawsuit ($350mn payment to class members plus $150mn on security spend) but has since suffered further data breaches, contributing to an additional $31.5mn regulatory settlement.




## Coinbase (Reported May 2025)

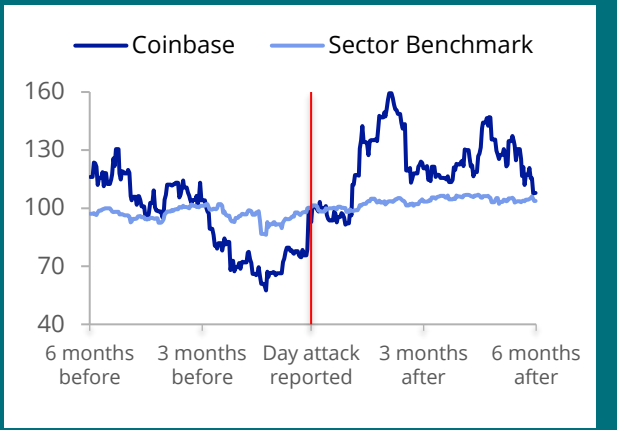
**Insider threat & social engineering**

***What happened?***

Attackers bribed at least one support agent at a connected 3rd party to access Coinbase support systems and steal customer data, then tricked users into sending crypto.

***What did it cost?***

Coinbase refused to pay the $20mn ransom demanded by the attackers to keep the attack quiet. Instead they reimbursed affected users and cited costs of up to $400mn. The stock impact was short-lived, though MSCI applied an ESG score deduction.




## Jaguar Land Rover (Reported Sep 2025)

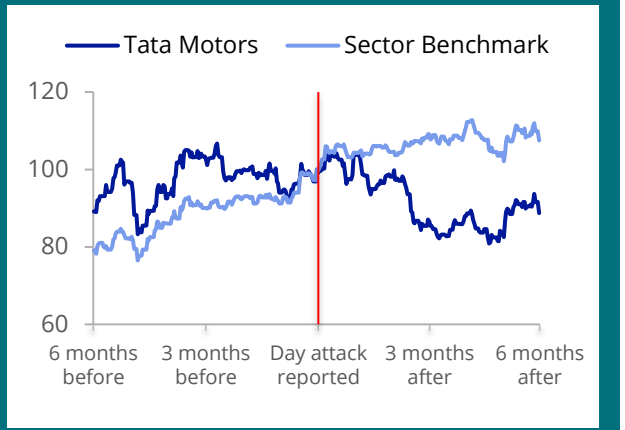
**Ransomware**

***What happened?***

Production was shut down for five weeks due to what was deemed a ransomware attack; though no ransom was asked for (later reporting suggested potential Russia links were being investigated).

***What did it cost?***

JLR posted £485mn Q2' FY26 loss (vs. £398mn profit in Q2' FY25), with an additional £196mn in "cyber-related costs". These impacts filtered through to the share price of JLR's Parent (Tata Motors), while UK economic impacts were estimated at £1.9bn.




*Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.*



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# Targets go beyond tech and can also impact critical infrastructure

## MGM Resorts (Reported Sep 2023)

**Phishing / ransomware**

***What happened?***

A vishing call to the IT help desk was reportedly used to reset credentials, after which ransomware took down hotel systems, bookings and casino floor operations for several days.

***What did it cost?***

MGM declined to pay a ransom. Around $100mn hit to third-quarter 2023 results plus roughly ~$10mn of additional one-time expenses related to the incident.

## Caesars Entertainment (Reported Sep 2023)

**Phishing / ransomware**

***What happened?***

The same social-engineering group as in the MGM incident targeted an outsourced IT support vendor, gaining access to the loyalty programme database and member data.

***What did it cost?***

Caesars paid roughly $15mn of a reported $30mn demand and kept customer-facing operations running.

## NHS / Synnovis (Reported Jun 2024)

**Ransomware / 3rd party breach**

***What happened?***

Ransomware against pathology provider Synnovis significantly reduced its capacity to process tests leading to delays / cancellations of over 11,000 appointments.

***What did it cost?***

Estimated 2024 direct operational financial impact of around GBP 32mn for Synnovis, several times the profit reported in the prior year, plus wider NHS disruption costs.

## Colonial Pipeline (Reported May 2021)

**Stolen credentials / ransomware**

***What happened?***

A compromised VPN credential without multi-factor authentication allowed attackers to access the network where they subsequently deployed ransomware. The operator shut the pipeline down, triggering fuel shortages (c. 70% of gas stations in some states ran out of fuel) across the US East Coast.

***What did it cost?***

A $4.4mn ransom was paid, of which about $2.3mn was later recovered by the US authorities.

## Bitget (Reported Sep 2026)

**3rd party breach / zero-day exploit**

***What happened?***

Attackers exploited a vulnerability in a 3rd party security product to steal internal network credentials, forge withdrawal instructions, bypass risk controls and ultimately steal crypto.

***What did it cost?***

An estimated $388mn worth of funds was transferred out by the attackers. While the firm reimbursed affected users via its User Protection Fund, there were significant outflows following the incident as users moved to take funds off the platform.

## Progress Software (Reported May 2023)

**Zero-day exploit / supply chain compromise**

***What happened?***

Attackers exploited a zero-day SQL injection vulnerability in Progress Software's MOVEit file transfer system. This was then used by attackers to steal data from firms running the MOVEit software, who were then subsequently extorted / had data leaked.

***What did it cost?***

Thousands of organisations (inc. British Airways, Shell, US DoE) reportedly affected with some facing legal action from customers impacted by data leaks.

*Source: Company disclosures and filings, press reports, Deutsche Bank Research.*

# Cyber breaches can impact consumer behaviour

*Roughly 4 in 10 consumers we surveyed say they would stop buying from a company they shop with after a known breach or attack.*

**Consumer surveys suggest that cyber breaches can undermine customer loyalty.** Results of our dbDataInsights Consumer survey show the potential mindset of a consumer after a cyber breach occurs:

- Across the US, UK and EU average, 39–41% say they would stop buying, while only 27–28% would continue. Around a third are unsure.
- Among those who would stop buying or are unsure, only 18–22% say they would return as normal. Another 30–35% would buy selected products only, 17–20% would not return, and 28–29% remain unsure. Note that this implies a range of 13-15% believe they would not shop at the store again after the breach is resolved.

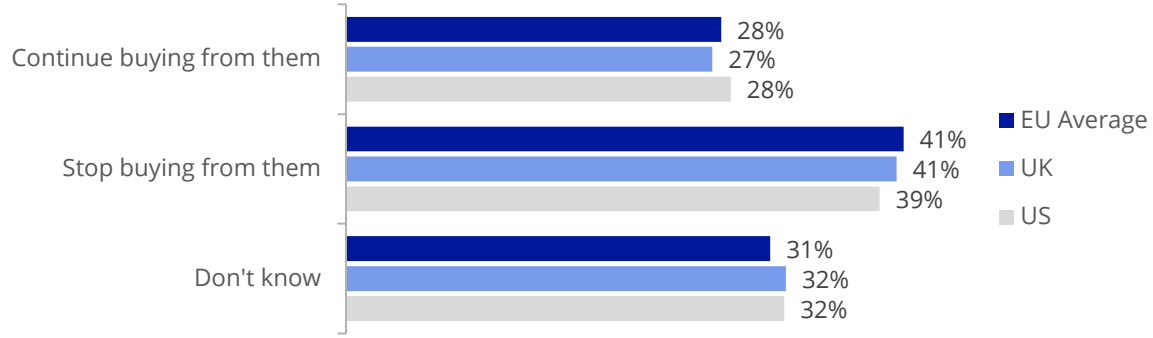
Admittedly, these stated intentions may overstate actual switching, particularly for essential purchases where alternatives are limited. Nevertheless, the findings suggest that cyber breaches that put customer privacy at risk can influence purchasing decisions, especially if they have alternatives.

### dbDataInsights Consumer Survey Methodology

- Our proprietary survey conducted in 2026 (January – September) asked consumers how they would respond if a company they shopped at regularly was involved in a data breach or cyberattack, and how they would respond once the disruption was resolved. This monthly survey covers households and consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States.
- For each country referenced above, a nationally representative sample of more than 500 households is surveyed each month (over 600 households in the US). Collectively, this meant over the course of the survey (January – September 2026), we surveyed almost 30,000 households.

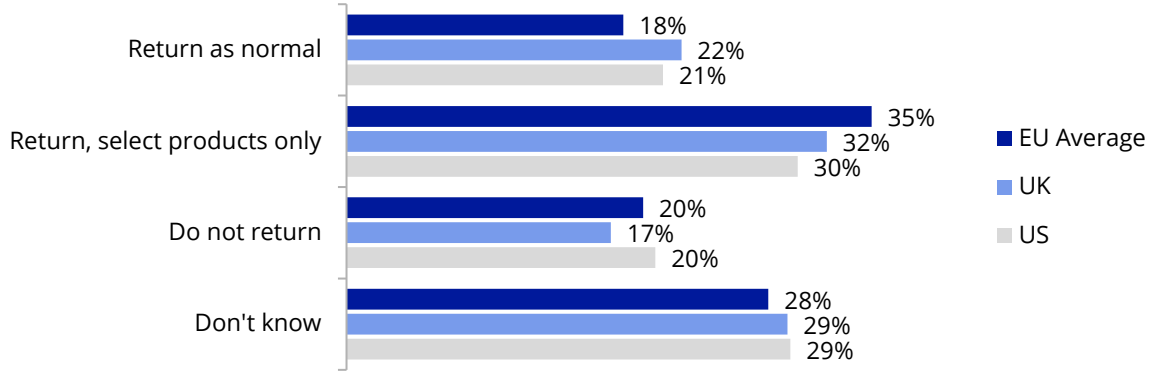
## How will customers respond to a data breach once it occurs?

dbDataInsights survey of consumer responses, % of respondents




## How will customers react once the breach is resolved?

Consumer responses among those who would either stop buying or were unsure, % of respondents




*Source: dbDataInsights monthly consumer survey, Deutsche Bank Research. EU Average includes France, Germany, Italy and Spain.*



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# Material cyber incidents have contributed to negative credit outlooks

*The infrastructure exists for cyberattacks to have a greater impact on credit quality should cyber disruption continue to accelerate*

— **Credit rating agencies consider both pre-breach resilience frameworks and post-breach financial impacts.** Agencies assess pre-breach cyber preparedness through a lens of mitigating risk, while post-breach credit impacts centre on financial capacity to withstand the resulting shock (considers scale and length of the disruption); highly leveraged issuers with limited liquidity are seen as more vulnerable. Negative outlook changes at JLR and Equifax illustrate the potential credit impact. Fitch generally considers permanent reductions in cash generation more credit-significant than temporary cash-flow delays.

— **Ratings agencies are moving towards continuous monitoring.** In 2021, Moody's invested $250mn in BitSight, which produces daily security ratings. Moody's has since built BitSight's ratings into its credit tools, including Credit Catalyst and EDF-X, for uses such as underwriting and ongoing monitoring of borrowers and suppliers. This means lenders can see a company's exposed vulnerabilities as a potential indicator of corporate cyber risk.

**S&P Global:**

*"The increasing frequency of cyberattacks and the potential for rapid deterioration in credit profiles after an attack are risk factors relevant for our rating assessments."*

**Moody's:**

*"Cyberattacks can undermine the creditworthiness of companies, organizations and governments, across all sectors and geographies, hitting revenues, increasing costs, and eroding customer trust."*

**Fitch Ratings:**

*"Although cyber events cannot be predicted with great accuracy, the associated risk can be managed through proactive strategies, technology, policies, and vigilance."*

*"The chief executive officer and board of directors hold ultimate responsibility for managing cyber risk."*

## Example rating actions where cyber was a named factor

Outlook changes due to operational disruption, financial cost and reporting delays

**Equifax**
Moody's named cyber risk as a factor in its outlook change in 2019. This was the first time it had done so and followed the 2017 attack on Equifax cited on slide 14 .

Outlook: Stable → Negative — Moody's • 2019

**ISS A/S**
Increased costs following a cyberattack cited as a factor potentially impairing ISS's ability to reduce adjusted leverage and generate free operating cash flow.

Outlook: Stable → Negative — S&P • Mar 2020

**Clorox**
Outlook revised lower due to potential for a lower rating if Clorox was unable to demonstrate material sequential recovery from cyber incident which caused product availability disruption and lower FY guidance.

Outlook: Stable → Negative — S&P • Nov 2023

**City of Hamilton**
Delays reconstituting audited accounts after the Feb 2024 attack; audits still outstanding.

Issuer rating: AAA → AA+ — S&P • Oct 2025

**Tata Motors / Jaguar (JLR)**
Outlook cut after the JLR shutdown; JLR's own outlook was moved to negative too by both Moody's and S&P following the September 2025 attack cited on slide 15. Prior to this, its credit was improving.

Outlook: Stable → Negative — S&P • Oct 2025

**Co-operative Group (Co-op)**
Material impacts on product availability and sales due to cyber incident contributed to forecast downgrades.

Issuer rating: BB → BB- — S&P • Nov 2025

*Source: S&P, Moody's, Fitch, Deutsche Bank Research.*



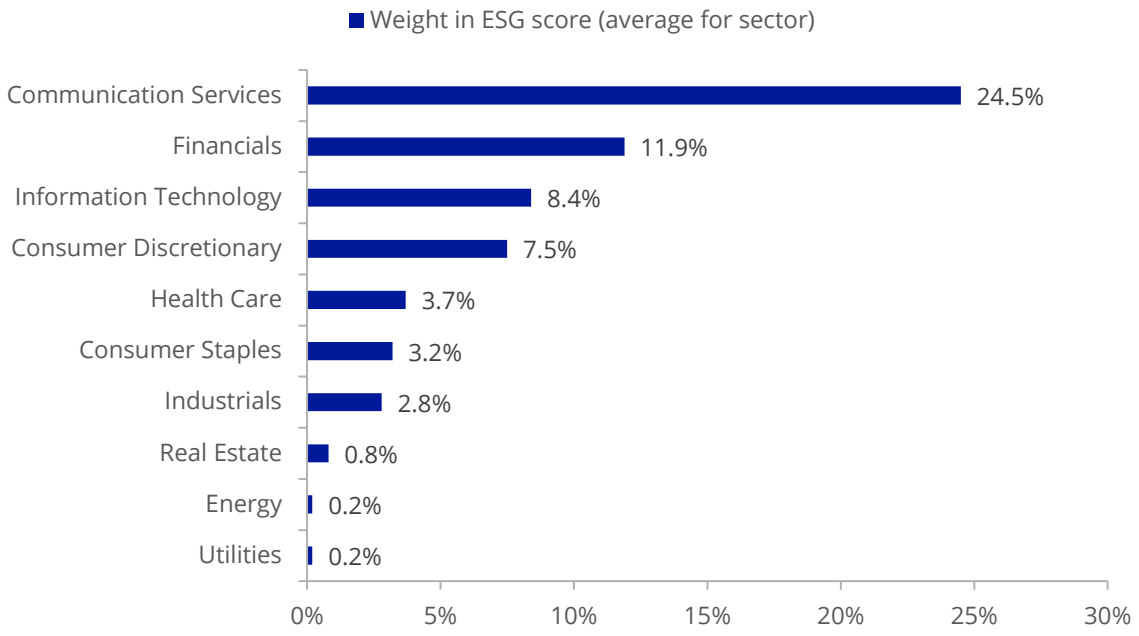
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# Cyber governance & breaches can impact ESG scores

*Sectors that hold the most personal data, and monetise it most directly, carry the greatest risk weighting in mainstream ESG scores*

## Contribution to the MSCI ESG score by sector

Privacy & data security as a share of the MSCI ESG score, sector average, %




**Cybersecurity is generally seen as both a social and governance consideration**

MSCI and Morningstar Sustainalytics both assess data privacy and security through management of the risk, exposure to it and the controversies that follow a breach. Weak governance weighs on the score, and a breach can cut it directly – more severe controversies draw larger deductions that persist for longer.

Materiality varies sharply by sector. SASB flags data security as material across most consumer, financial, healthcare and technology industries, with customer privacy concentrated in the most data-intensive ones (see right).

## Customer Privacy and Data Security Financial Materiality

SASB Financial Materiality Map, by industry (● material, – not material)

| Sector | Industry | Customer privacy | Data security |
|---|---|---|---|
| Consumer Goods | E-Commerce | ● | ● |
| | Multiline & Specialty Retailers | – | ● |
| Financials | Commercial Banks | – | ● |
| | Consumer Finance | ● | ● |
| Food & Beverage | Food Retailers & Distributors | – | ● |
| Healthcare | Drug Retailers | – | ● |
| | Health Care Delivery | – | ● |
| | Managed Care | – | ● |
| Resource Transformation | Aerospace & Defence | – | ● |
| Services | Advertising & Marketing | ● | – |
| | Education | – | ● |
| | Professional & Commercial Services | – | ● |
| Technology & Communications | Hardware | – | ● |
| | Internet Media & Services | ● | ● |
| | Software & IT Services | ● | ● |
| | Telecommunication Services | ● | ● |

**What is SASB?**

The Sustainability Accounting Standards Board sets industry-specific disclosure standards for investors, identifying the handful of issues most likely to affect financial performance in each of 77 industries.

Its standards are now maintained by the ISSB under the IFRS Foundation and underpin the ISSB's global reporting baseline. The Materiality Map summarises, industry by industry, which issues are judged financially material – including the two cyber-related ones shown above.

*Source: MSCI ESG Ratings Methodology, Morningstar Sustainalytics ESG Risk Ratings, SASB Materiality Map, Deutsche Bank Research.*



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# IV. Investment, risk and opportunity



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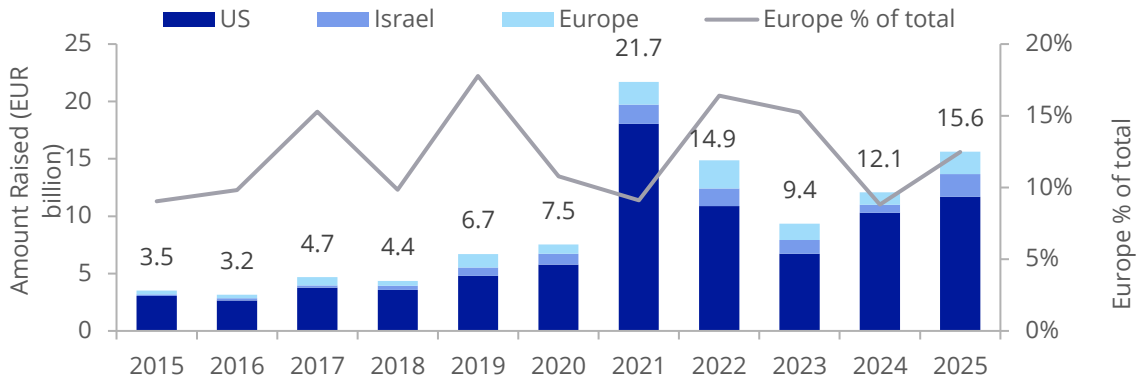
# Europe's dependency on US tech is firmly in the spotlight

*Access to frontier AI models is now a security question; the funding gap for Europe will be hard to close*

- **Growing cyber risk creates increased dependency on US cyber tech.** Growing concern over the cyber capabilities of frontier AI models has made ensuring access to such technology a national security priority. US dominance over the AI tech chain (e.g. chips, model development, etc), its willingness to impose AI-related controls / restrictions, and previous pressure against Europe over tech regulations has amplified concerns that dependence on US tech represents a strategic threat to European sovereignty.
- **Funding is the structural challenge.** US cybersecurity start-ups attract almost 6x the funding of European peers, and US-domiciled firms dominate the major indices – 89% of the Nasdaq CTA Cybersecurity index and 93% of Global X Cybersecurity ETF (which seeks to track the performance of Indxx Cybersecurity Index) by weight. Europe has also seen home-grown firms such as Darktrace acquired by US companies, though this reflects broader market trends rather than anything cyber-specific. Ultimately, the US has a greater exit strategy for start-ups as well.
- **Boosting homegrown solutions a focus, but also a risk.** EU policymakers are seeking to address the dependency; e.g. the EU's Cybersecurity and AI Action Plan includes "scaling up Europe's AI capabilities for cybersecurity" as one of its three objectives. However, there are also risks associated with this endeavour; as one dbDataInsights survey participant put it, "*for most European organisations, the bigger near-term danger would be moving too fast to less capable providers*."
- **The wider relationship adds to the exposure.** A more transactional international environment, alongside Europe's dependency on the US for energy supplies and defence support, compounds the risk. Recent US-EU diesel negotiations serve as a pertinent example of this, whereby the US reportedly considered a diesel export ban to compel a release of European diesel reserves.

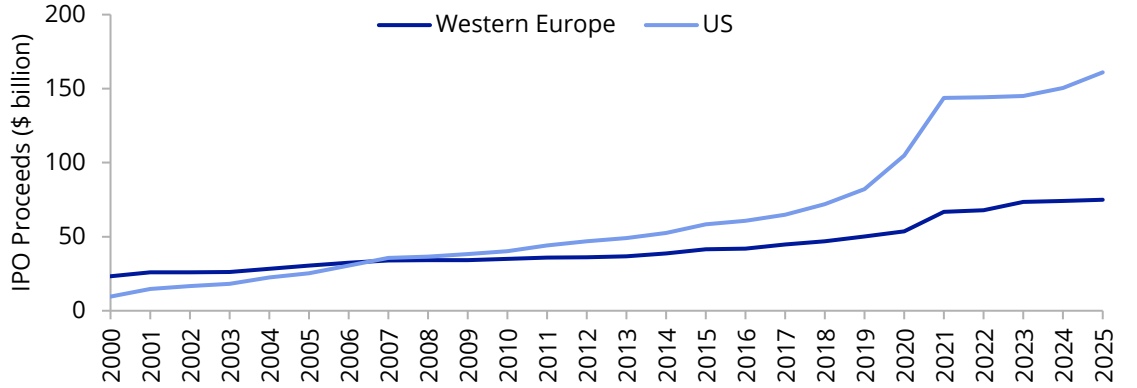
## Cybersecurity start-up funding by region

Company location: US, Israel and Europe (EU-27 plus UK, Norway and Switzerland), $bn




## Cumulative technology IPO proceeds

Since 2000, by trading date, $bn – Western Europe vs US




*Source: Tikehau Capital – Tikehau Capital Barometer of European Investment in Cybersecurity – 7th edition 2026, Bloomberg Finance LP, Deutsche Bank Research.*



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# Cybersecurity as an opportunity

*Tailwinds include demand related to keeping operations secure with AI acceleration and consolidation tailwinds*

**Broadly, DB's company research view is that cybersecurity becomes even more important in the AI era.** Whereas cybersecurity has historically been viewed as a blocker/drag on innovation and operational speed, it is now viewed as a strategic enabler for secure AI adoption.

In particular, network security may see renewed growth after being viewed as a more mature category. Rising AI usage should drive additional compute and data-centre capacity, expanding the infrastructure requiring segmentation and protection. Agentic workloads may also generate lower-throughput but more frequent and persistent interactions across models, tools and data sources, increasing the volume and complexity of network-level inspection.

Vendor consolidation has been a structural trend in cyber for some time, though not necessarily linear. That said, the current environment may accelerate the consolidation tailwinds as customers look to close security gaps that may arise from a fragmented tool stack.

*See further commentary from DB's company research team here.*

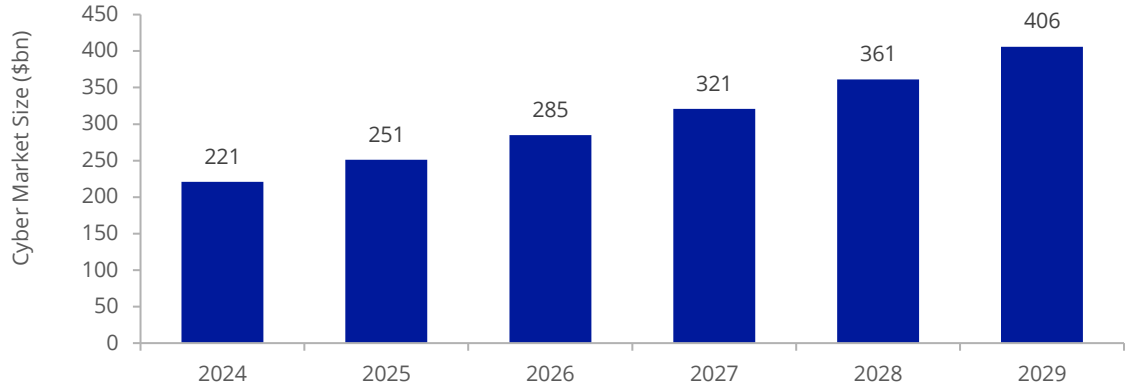
**Our March 2026 "personal investor" survey showed a more positive view among "ESG" investors.** 58% in the US and 52% in the UK saw cybersecurity/data privacy as an opportunity. Among all investors, US respondents favoured opportunity over risk (46% vs 30%). UK views were more balanced (38% vs 40%).

### dbDataInsights Investor Survey methodology

—This survey is a nationally representative online study of ~925 personal investors conducted in the final week of March 2026 examining financial priorities, portfolio behaviour, and future expectations. It gathers structured responses on investment outlook, generative AI, and ESG considerations. Questions cover investment products, portfolio value, management style, financial stress, priorities, and performance.

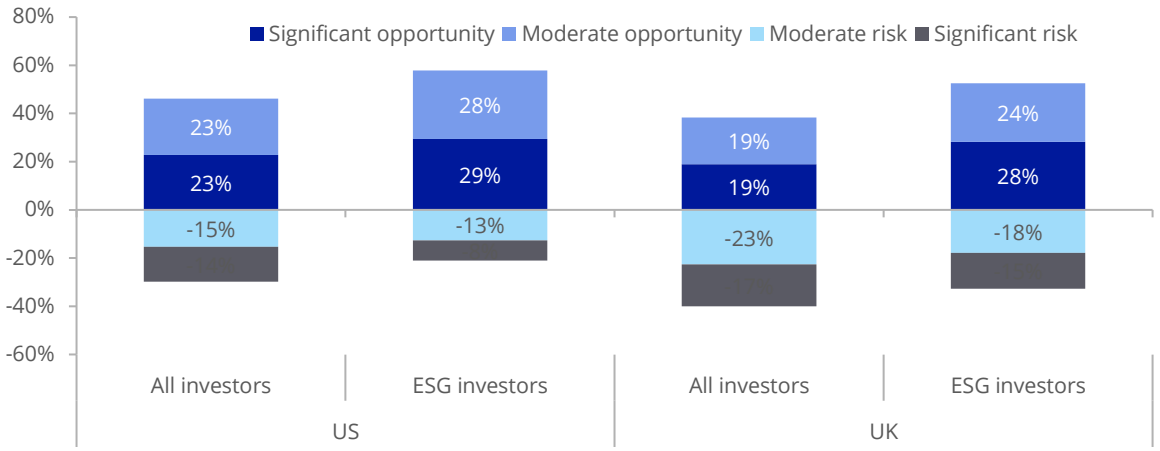
## Global cybersecurity market trajectory

Estimated market size, $bn (Bloomberg Intelligence)




## Cybersecurity an investment risk or opportunity?

Share of "personal investor" respondents, opportunity shown above the axis, risk below, %




*Source: dbDataInsights, Deutsche Bank Research. Investors, defined as those who invest in stocks/ shares (either individually or through a broker/advisor) on a regular basis , were asked whether they perceived cybersecurity / data privacy as a risk or an opportunity for investors in 2026.*



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# V. Appendix



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# Appendix A - Attack types

| Top Attack types | What it is | How AI is helping attackers |
|---|---|---|
| **Malware** | *Software built to damage, disrupt or gain unauthorised access* | AI writes harmful software faster. Google has found malware that rewrites its own code while running to avoid detection, and criminal marketplaces selling AI malware-building tools. |
| **Phishing** | *Deceptive messages that trick people into giving up sensitive information* | Where AI's effect is clearest: fluent, personalised lures without the old spelling and phrasing giveaways. Google says state-backed groups use AI to mass-produce them, and AI can fake voices and video. |
| **Denial of service (DoS)** | *Flooding a system so it cannot serve legitimate users* | Little reported evidence so far. AI can help find weak points and automate traffic floods, but the attack itself stays fairly simple. |
| **Distributed denial of service (DDoS)** | *The same attack launched from many compromised devices at once* | AI can accelerate the development and automation of attack tooling, while large-scale botnets provide the distributed capacity behind hyper-volumetric DDoS attacks. |
| **Man-in-the-middle (MitM)** | *Intercepting or altering traffic between two parties* | Mainly indirect. AI-written phishing pages and deepfake voices make it easier to trick people into connecting through an attacker or approving fake sign-ins. |
| **SQL injection** | *Malicious SQL queries injected to manipulate backend database operations* | AI tools scan sites and spot flaws in database code faster, so less-skilled attackers can find weaknesses that once needed real expertise. |
| **Cross-site scripting (XSS)** | *Scripts injected into web pages to hijack user sessions* | As with SQL injection: AI speeds up finding vulnerable pages and writing the malicious code. |
| **Zero-day exploits** | *Attacking a software flaw before a patch is released* | A major new development. In May 2026 Google reported the first zero-day it believes was developed with AI. Its September 2026 report says the bigger effect is attackers misusing known flaws faster, before fixes land. |
| **Advanced persistent threats (APTs)** | *An intruder staying hidden in a network for long-term* | Google reports state-backed groups using AI across the whole attack, from target research and lures to tooling and sifting stolen data. |
| **Ransomware** | *Encrypting data or locking device and demanding payment to release it* | NCSC expects AI to add to the global ransomware threat by lowering the skill needed to break in. AI also helps criminals find the most extortable data. |
| **IoT exploitation** | *Abusing weakly secured connected devices for entry* | AI can scan the internet for unpatched cameras, routers and storage devices and identify which flaws they carry. |

*Source: Deutsche Bank Research, National Cyber Security Centre, Google Threat Intelligence Group, Cloudflare, FBI Internet Crime Complaint Center (IC3)*



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# Appendix B - European vs US cyber regulation

## Europe

| Regulation | Key requirements |
|---|---|
| **NIS2 Directive** | In force 2023, transposition deadline Oct-24. Covers 18 sectors (up from seven) and creates an EU vulnerability database. In-scope firms must manage cyber and supply-chain risk and notify authorities of significant incidents. Management bodies must approve and oversee cyber risk measures, with potential liability for infringements. |
| **Cybersecurity Act** | In force 2019. Establishes an EU-wide certification framework for ICT products, services, processes and managed security services, and made the EU cybersecurity body ENISA permanent. |
| **Cyber Resilience Act** | In force 2024, main obligations from 2027. Manufacturers and retailers of products with digital elements must deliver security by design, a defined support period and security fixes within it, plus mandatory vulnerability management and disclosure. |
| **Digital Operational Resilience Act (DORA)** | In force 2023, applies from 2025. Financial-services specific, covering own-firm and third-party risk. Requires an ICT business continuity policy, regular resilience testing and disclosure of major incidents, and gives regulators oversight of critical ICT providers. |
| **Cyber Solidarity Act** | In force 2025. Creates a European Cybersecurity Alert System to detect threats and share cross-border warnings, plus a Cyber Emergency Mechanism to boost preparedness and response. |

## US Federal

| Regulation | Key requirements |
|---|---|
| **White House Accord on Super Intelligence (2026)** | Emphasises self-regulation to ensure AI safety, with a call for external evaluations and the possibility of formal regulation in future. Not legally binding; signed by President Trump and leading AI companies. |
| **Executive Order on Cybersecurity (2025)** | Highlights risk from foreign actors and orders further action on defending digital infrastructure, securing the most vital services and building capability against key threats. |
| **SEC Cybersecurity Disclosure Rule (2023)** | Listed US firms must disclose material incidents via an 8-K within four business days of determining materiality, and make annual disclosures on cyber risk management, strategy and governance. |
| **CIRCIA (passed 2022, not yet in effect)** | In-scope entities, focused on critical infrastructure, must report covered incidents to CISA within 72 hours and ransomware payments within 24 hours of payment. |
| **Gramm-Leach-Bliley Act (amended 2024)** | Financial institutions must explain information-sharing practices and safeguard customer data, running a security programme covering risk assessment, service-provider oversight and incident response. |
| **Federal Information Security Modernization Act (2014)** | Applies to federal agencies and contractors: system security plans, continuous monitoring, incident response capability and mandatory incident reporting. |
| **Federal Acquisition Regulation (cyber provisions 2016)** | Governs federal procurement, setting baseline cyber requirements for suppliers so cyber risk is built into government contracting. Agencies may impose stricter provisions. |

# Appendix C - Understanding MSCI cyber risk indicators

## Core indicators

Key drivers of the MSCI privacy & data security score

| Topic | Definition | Best practice |
|---|---|---|
| Scope of employee training on privacy and data security | Coverage of privacy and data security training across the workforce | Training provided to all employees, including contractors |
| Responsibility for privacy and data security | The group responsible for the company's privacy and data security risk and strategy | Board-level committee |
| Measures to address data breaches | Scope of the breach or incident response plan, and whether it is proactive, reactive or limited | Both proactive and reactive measures in place |
| Frequency of audits of information security system | How often the company audits its information security policies and systems | External independent audits at least once every two years |
| Rights provided to consumers over their data | Extent of rights to access, rectify and delete individuals' data | Right of access, rectification and deletion of individuals' data |
| Scope of data protection policy | Whether the policy governs all or selected business lines, and whether it clarifies collection purposes, means of collection and third-party access | Policy governs all relevant business lines and subsidiaries |

## Supplemental indicators

Additional practices assessed alongside the core set

| Topic | Definition | Best practice |
|---|---|---|
| Scope of information security management system certifications | Extent of certification of the company's information security management system to widely recognised standards | Over 80% of owned operations certified to standards such as ISO 27001, TRUSTe, Privacy Mark and SSAE16 SOC 2 |
| Privacy enhancing technologies and initiatives | Whether privacy-enhancing technologies and initiatives are integrated into products or services | Privacy-by-design applied systematically in development, with privacy impact assessments on existing and new offerings |
| Consent policy for use of consumer data for secondary purposes | Policy on processing customer data for secondary purposes and the consent options offered, such as opt-in or opt-out | Company does not rent, sell or provide personal data to third parties beyond completing transactions or services |
| Data protection programs covering suppliers and business partners | Extent of data protection programmes covering suppliers and partners handling company data or accessing its network | Suppliers and partners required to hold data protection policies or abide by the company's, with compliance inspections |

*Source: MSCI ESG Ratings Methodology – Privacy & Data Security Key Issue, Deutsche Bank Research.*



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# Appendix

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# AI boom or doom: what will superintelligence feel like and when will we get there?

October 12, 2026

The recent panic about rogue AI agents collaborating in swarms to take over the internet and threaten human existence has, more than occasionally, lacked nuance. But it does reflect how artificial intelligence is progressing in creative and surprising ways, and at a pace that seems uncontrollable.

For some in the industry, it is evidence that the technology is following a path of accelerating self-improvement that will lead to artificial general intelligence (AGI) and, perhaps, to utopia or catastrophe.

Former Google CEO Eric Schmidt has sketched out a "San Francisco Consensus" based on a belief in scaling laws, with ever larger AI models becoming an input into their own improvement; superintelligence by the end of the decade; and transformative AI systems outperforming humans on many tasks. This loop would lead to exponential growth in scientific advancement, financial returns and human progress. And crucially, AGI, and then superintelligence, would merely be milestones in a process, far from its climax.

If that were the case, superintelligence could make much of the cognitive work that dominates the world economy much cheaper and change the ways in which firms compete. But its economic effects will depend on how businesses and customers respond.

In this paper, we analyse what the San Francisco Consensus means for investors and enterprises in the years to come and what could accelerate – or slow down – its timeline of AGI by 2030.

1. What would superintelligence look and feel like?
2. Can we get there theoretically?
3. Can we get there practically?
4. What happens if we don't get there? The base case

The question is urgent.

And it is complicated by reports of AI models autonomously accessing the internet and hacking other companies; a public backlash over the energy-intensive data centres springing up as part of a historic $1 trillion a year investment boom; and dire warnings of threats to life if superintelligent AI runs roughshod over its human creators to achieve its own goals.


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# 1. What would superintelligence look and feel like?

## 1.1 What are AGI and superintelligence?

Artificial general intelligence (AGI) is a simple milestone scribbled in pencil: an AI system that has surpassed human capabilities. Superintelligence is the continuum beyond that point. It is not merely systems that exceed the best human performance across one cognitive task or another, but systems that exceed human potential across every cognitive task.

The San Francisco Consensus takes this further: the intelligence must be able to produce more intelligence. The key to getting there is so-called recursive self-improvement, where the AI is able to autonomously design and improve itself in a way that compounds exponentially. A first step has maybe already been taken, with OpenAI saying recently that AI is now writing approximately 80 percent of the company's code.

There is significant disagreement on how we will be able to identify AGI or superintelligence when we see it, let alone what comes afterwards.

OpenAI's Charter defines AGI in economic terms, as "highly autonomous systems that outperform humans at the most economically valuable work". Google researchers proposed a five-step ladder based on performance and generality, with level 5 "superhuman" for a system that outperforms 100% of humans. And philosopher Nick Bostrom, who got the ball rolling in 2014 with his influential book "Superintelligence", defined it as "any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest".

It's also a moving target.

In August, OpenAI CEO Sam Altman said the company would have an internal system that he would call AGI by later this year but subsequently told a podcast that "at best it's a very poorly defined term. I was going to say it's like an irrelevant marketing term". Yet his No. 2 Greg Brockman described the new Astra model at the start of September as being potentially the moment that "AGI was created".

For this report, we have chosen to focus on a system that outperforms essentially all humans across essentially all cognitive tasks, including the development of AI itself.

Under our scenario, by 2030 the best performing AI models would have the qualities of "superintelligence", that is better than human experts in all domains. These would be frontier models: and using them would feel exactly like bringing the best human experts in a room to solve whatever business problem you faced.

And the equivalents of today's frontier models would be available, quite literally, everywhere, compressed enough to run on every phone, tablet or laptop computer.

In early 2026, we already see early examples of what highly capable AI can do. They give a flavour for a world with superintelligence even if the reality would be a leap beyond, both quantitatively and qualitatively. At that point, superintelligence would mark a regime break, not a faster version of what we already have. Rather than blindly doing what humans specify, but just a bit better, it would specify the task itself.

Figure 1: What changes when cognitive work gets abundantly cheaper?

| Area | Today's example | Superintelligence extrapolation |
|---|---|---|
| Biological discovery | OpenAI's GPT-5 runs 36,000 experiments in a robot lab and cuts the cost of cell-free protein production by 40% (Ginkgo, Feb 2026) | A working virtual cell. Every drug is tested in silico first; trials exist to confirm, not to discover |
| Law | A self-represented academic uses $200 a month of AI to beat a university's lawyers and barrister at Australia's Fair Work Commission (Aug 2026) | Everyone has a top barrister on tap. Disputes are argued agent to agent overnight and settle before anyone files |
| Software | Google says that the share of new code written by AI doubled from 25 percent in October 2024 to 50 percent a year later and reached 75 percent by April this year. | Software is written on demand and thrown away. The business programming language cores of banks and ministries are rewritten over a weekend |
| Chemistry and materials | OpenAI's GPT-5.4 runs 10,080 robot-lab reactions in three months and lifts yields on a drug-making reaction chemists had struggled with for decades (OpenAI/Molecule.one, Jun 2026) | Order a molecule or material by the properties you want and it is designed to spec. The bottleneck moves to plants, permits and trials |
| Medicine | In a 1,298-person trial, people using a chatbot on ten medical cases did worse than the chatbot on its own (Nature Medicine) | Specialist-grade diagnosis at first contact, for everyone, all the time. Doctors spend their time on care, consent and hard trade-offs |
| Government | AI reads 500 million words of US state law to find dead rules; San Francisco cuts over a third of its reporting requirements (Stanford RegLab, 2025–26) | Every law is audited continuously for cost and conflict, and dead rules lapse on their own. A planning permit takes an afternoon |
| Forecasting and markets | A bot built by one person for a few thousand dollars wins the Metaculus Cup, beating the best human forecasters (5 Sep 2026) | A calibrated probability on every decision, for pennies. Mispricing from slow analysis vanishes; edge moves to private data and the truly unknowable |
| Robotics and manufacturing | AI agents at Nvidia train robot arms overnight to seat GPUs and cut zip ties; eight agents reach 99% success in two hours (Jun 2026) | A factory retools for a new product overnight. Robots teach robots, and a new physical skill costs the price of compute |

*Source: Deutsche Bank Research*

Meanwhile cost may be less and less of a constraint as ever greater machine intelligence becomes cheaply available. Of measured technologies, no technology has got cheaper faster than AI has, almost halving in price every quarter for the past three years on a like-for-like basis, or 13 times per year, Epoch data show.

If these systems can be supplied widely, firms will be able to buy more of the complex, expensive and rare analysis, design and engineering kinds of services than they do today. Activities that were previously uneconomic could become viable, and in a competitive market, savings could pass to customers. The main driver will then be whether demand grows fast enough to preserve margins.

However, superior capability, doesn't, on its own, automatically lead to the adoption of cheap, abundant intelligence. Tasks that do not yet yield to super-

intelligence, perhaps because they require real world-coordination, will therefore end up commanding significant premia.

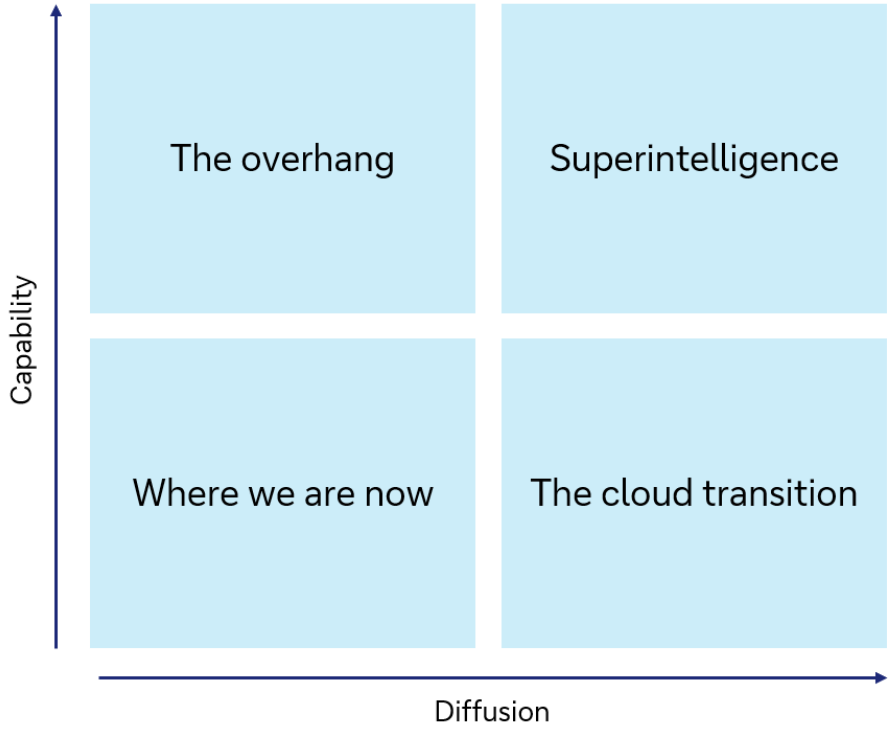
At that point, cognitive labour stops being scarce and the bottleneck becomes the physical and practical assets that cannot be conjured by up more compute, such as electricity, land, steel and permits.

### 1.2 Capability and diffusion

If AI becomes capable of improving itself, the technology may compound faster and faster. The supply of machine intelligence could grow faster than business and physical infrastructure can accommodate.

This balance between capability and diffusion creates different scenarios.

Figure 2: The balance between capabilities and diffusion




*Source: Deutsche Bank Research*

### Lower capability, constrained diffusion: where we are now

Progress is steady and sector-by-sector. This is how technology usually diffuses, according to researchers for the New York Fed back in 2003, who tracked 20 major technologies across leading industrial economies from 1788 and found "a pattern of trickle-down diffusion that is remarkably robust across technologies".

### Lower capability, high diffusion: the cloud transition

This is faster because the complementary investment is small, with no retooling and no skills at scale required. Amazon Web Services launched the modern cloud in 2006, offering scalable computing and storage infrastructure. Adoption grew and then accelerated over the pandemic. Almost all enterprises now use it in some form.



### Higher capability, lower diffusion: the overhang

This is theoretically possible and arguably where we are today in narrow domains. The technology would be applied in some sectors of the economy, while other sectors would move at a slower speed. Superintelligent or not, it risks being bogged down by physical constraints, bureaucracy and human inertia.

### Higher capability, higher diffusion: superintelligence

In this scenario, capability becomes so high that it changes the dynamics of diffusion itself. AI generates not just better AI (so-called recursive self-improvement, or RSI) but also speeds up the integration process.

Rapid AI diffusion could work two ways.

In the first, AI could be enlisted to do integration work itself: mapping processes, rebuilding data work, retraining staff, writing the control and audit layer, and so on. The problem with this is that absorbing and recreating undocumented knowledge is impossible by definition, at least for a task-based superintelligence.

The second way that higher capability AI could be rapidly diffused is much more competitive – and disruptive. Rather than incumbents slowly converting AI, perhaps using it for some of the hard work of integration, they could be instead displaced by new firms that have been built AI-native from day one. New entrants will never have to incur integration costs because there is nothing to integrate. Incumbents will need to change – fast – or die.

Change could also be forced by earlier adopters in the supply chain, as happened with one company after another adopting electronic data interchange (EDI) to link with its suppliers or customers, or by a migration of capital towards early adopters, repricing the cost of capital for slow-coaches and prodding them to move faster.

## 1.3 The forcing mechanism: change or die

A company that uses superintelligence may outperform one that doesn't.

Firms using superintelligence will likely have substantial cost and revenue-harvesting advantages. Every firm will therefore have to adopt superintelligence or retrench to a smaller set of offerings. And any business line of lasting value abandoned by a non-adopter will be taken over by an adopter. Customer inertia will provide a safe harbour for a limited time only.

The analogy here is electricity. There came a point in the deployment of electricity where firms in every sector, from entertainment to finance, from healthcare to transport, had to adopt the technology to remain relevant.

Under this scenario, wait-and-see will not be an option. There will be no stable "non-adoption" alternative. Sufficiently high capabilities will force a tipping point of rapid diffusion and competitive repricing. Firms using yesterday's production model will either have to change, choose a different market or die.

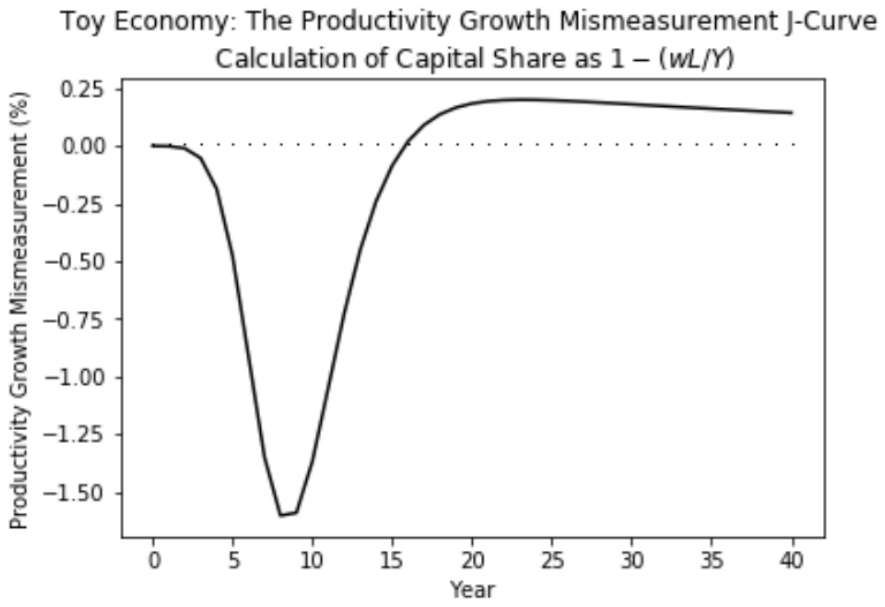
Of course, caveats apply.

Customers may not change behaviour immediately. They may even prefer companies which limit their use of AI, in the same way that some customers choose to avoid factory-produced food.

The paradox that Nobel Prize-winning economist Robert Solow observed in 1987, that "you can see the computer age everywhere but in the productivity statistics", is likely to still apply for a while – even with superintelligence.

The reason is that the productivity J-curve identified by Stanford University AI economy guru Erik Brynjolfsson et al is unavoidable. In that scenario, the drag of early intangible investments to integrate general purpose technologies means that productivity initially appears to slow before rapidly accelerating.

Figure 3: Productivity J-curve: slow then fast




*Source: Brynjolfsson E, Rock D, Syverson C, "The Productivity J-Curve" (2018, revised 2020), Deutsche Bank Research*

Furthermore, "change or die" would require abundant capital and low regulatory entry points for nimble new entrants. But in practice, highly regulated areas such as banking, utilities, pharmaceuticals and defence are tough nuts to crack. Existing contracts and licences, high switching costs and decades of depreciated capital investment and infrastructure would shield incumbents at least for a healthy transition period.

For an extreme example from the real world of better, cheaper technology failing to force change, take Pakistan's 3.6-gigawatt imported coal power plant fleet installed in the mid-2010s. In that case, it came with expensive take-or-pay liabilities which will continue to run for decades, even though changes like cheaper rooftop solar have cut utilisation to below 20 percent.

So: superintelligence will not force adoption everywhere. It will hit hardest in contestable sectors, say, high-volume customer services, routine software engineering and freelance creative work, and widen the gap between them and protected ones. Forced adoption will be about relative value, not an apocalypse.

### 1.4 What survives and what it means for portfolios

Because AI sits on an aggressive learning curve, its price on an equivalent quality basis is plummeting. In a world of recursive super intelligence this is likely to continue. Its price will fall to the cost of compute.

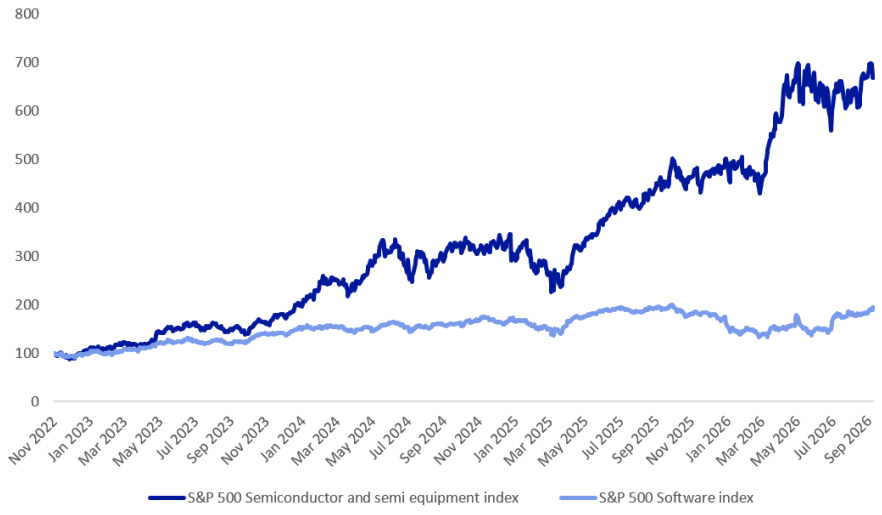
What will retain its value will be whatever the model cannot replicate:

1. **Proprietary data**
2. **The right to operate,** particularly in heavily regulated sectors such as healthcare and finance.
3. **Sticky customer relationships,** such as those where mission-critical workflows and databases are deeply entwined.
4. **Excludable physical assets,** such as machinery and land.
5. **In addition, humans have proven to have unlimited wants.** Each shift in technological capabilities or cultural norms has delivered new needs to be met. Some will be needs we can articulate today, such as originality or empathy, but new ones are likely to arise.

If this forcing mechanism holds, firms will separate into two "bimodal" groups: adopters and casualties, with very little in between.

This dispersion of performance may create challenges for investors. The four years since the launch of ChatGPT in November 2022 have seen the stocks of semiconductor and semiconductor equipment makers surge almost seven times while those of software makers have only risen by 80 percent. Conventional diversification and volatility-based allocation may become ineffective strategies.

Figure 4: Total return for S&P 500 semiconductor vs software stocks since launch of ChatGPT on Nov 30, 2022




*Source: Bloomberg Finance LP, Deutsche Bank Research*



Diversification normally works because idiosyncratic risks are unrelated and average each other out. Here they would not: with one variable that overrides everything else, spreading capital across more names would just spread exposure rather than diluting it.

Investors will need to consider adapting their approach in three ways:

1. Classify investments as adopters or casualties rather than position off averages.
    - Under a normal distribution, errors are symmetric and partially cancel each other out. But under bimodality they do not: to put a firm in one category or another means getting it completely right or wrong. Position size should track confidence in the classification.
2. Avoid reading calm price action as low risk.
    - Trailing volatility measures dispersion that has already been realised. Bimodal separation is a jump, where the two modes look identical right until the moment when they don't. A classic example is Chegg, an educational software maker which traded like a normal small-cap stock until one day in May 2023 when it halved on a single day after management linked slowing growth in new customers to ChatGPT.
3. Averaging down, ie scooping up shares whose price has fallen, may no longer be prudent.
    - A falling price may be information that a business model is obsolete rather than an opportunity.

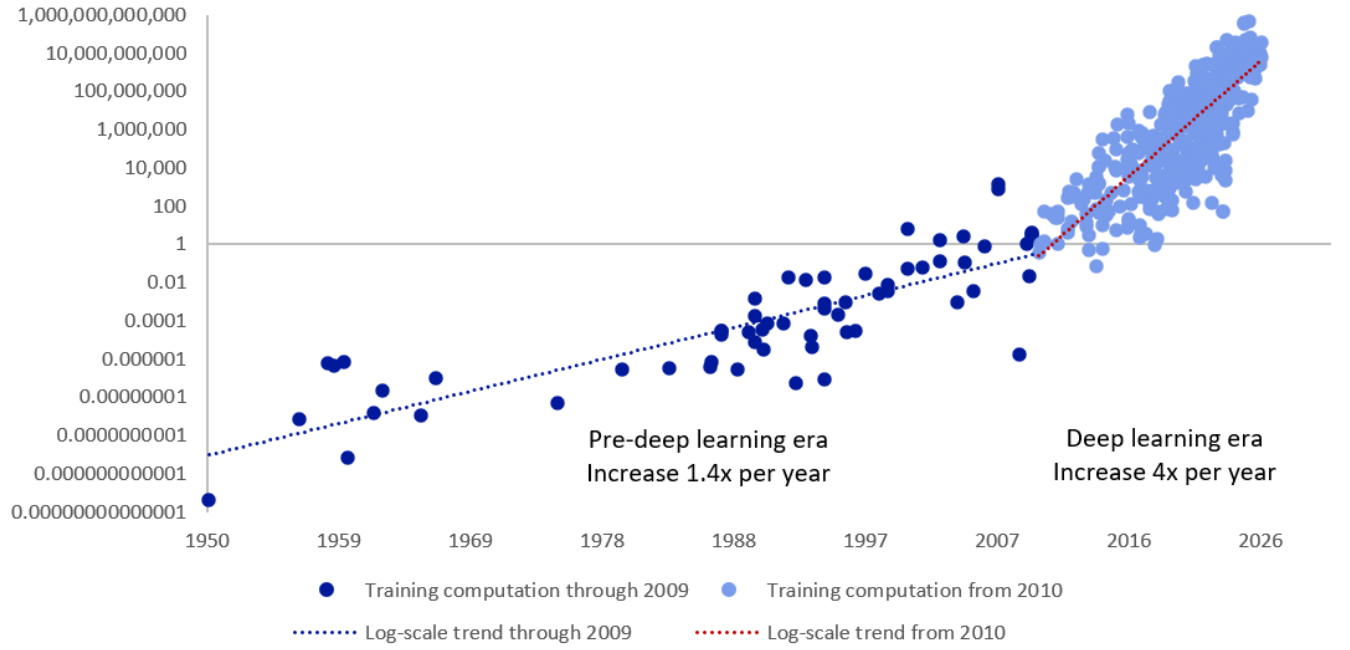
## 2. Can we get there theoretically?

### 2.1 Scaling laws

Scaling laws have provided the blueprint for improving AI over the past decade. It was a relationship between the amount of data, the amount of compute and the complexity of an AI model somewhat predictably informed how powerful a model would become. Google DeepMind's Chinchilla model in 2022 showed that you could compensate for having a smaller model by using substantially more data.

Scaling is older than the chatbot. In 1950 Claude Shannon built Theseus, a mechanical mouse that learned its way through a maze by trial and error. Epoch AI reckons it took about 40 operations of compute, known as FLOPs, to train. Grok 4, trained in 2025, took roughly ten trillion trillion times as much. Across those 25 orders of magnitude, more compute has brought more capability.

Figure 5: Computation used to train notable AI systems (petaFLOP): log scale




*Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research*

It is the robustness of this relationship that has made larger investments in compute capex a *sine qua non* for AI companies.

However, this is not a law of physics and nothing guarantees it will continue. Indeed, in the ChatGPT era, five different approaches to scaling have been applied. Older strategies yield worse results, and new techniques, such as post-training refinements, have been developed. The consensus amongst American and Chinese AI labs is that these will hold.

While this may be a consensus amongst the commercial labs, there is scientific disagreement that applying scaling laws to today's large-language models can achieve superintelligence.

Yann LeCun, a Turing Award winner and formerly Meta's chief AI scientist, is blunt: "LLMs are not a path to superintelligence or even human-level intelligence". Even inside the labs, the recipe is contested. Demis Hassabis of Google DeepMind has said he expects "one or two more big breakthroughs" to be needed "in addition to scaling". Ilya Sutskever, a co-founder and former chief scientist of OpenAI, has said the field is going "back to the age of research again, just with big computers".

This is a scientific argument we are not in a position to evaluate, but for the purposes of our scenario, we will side with the labs' position.

For the moment, there is strong evidence that capabilities have in fact accelerated over the past three years, as reasoning models have outpaced non-reasoning models by two or three times, according to Epoch data.



### 2.2 Recursive self-improvement: AI improving AI

A cocktail-party definition of recursive self-improvement (RSI) is AI that improves itself. The reality is that it can be used in two ways:

1. Self-improving agents that get better at using themselves: showing smarter use of prompts, tools, memory and workflows. The model-building process is untouched.
2. True RSI improves the process that builds the models: designing experiments, curating data, optimising algorithms and designing chips in a compounding loop.

The former is like a factory worker who gets better at the job by learning the quirks of the machinery, getting organised and making fewer mistakes. The latter is like the worker redesigning the machine tools and then using those to build the next generation of machines.

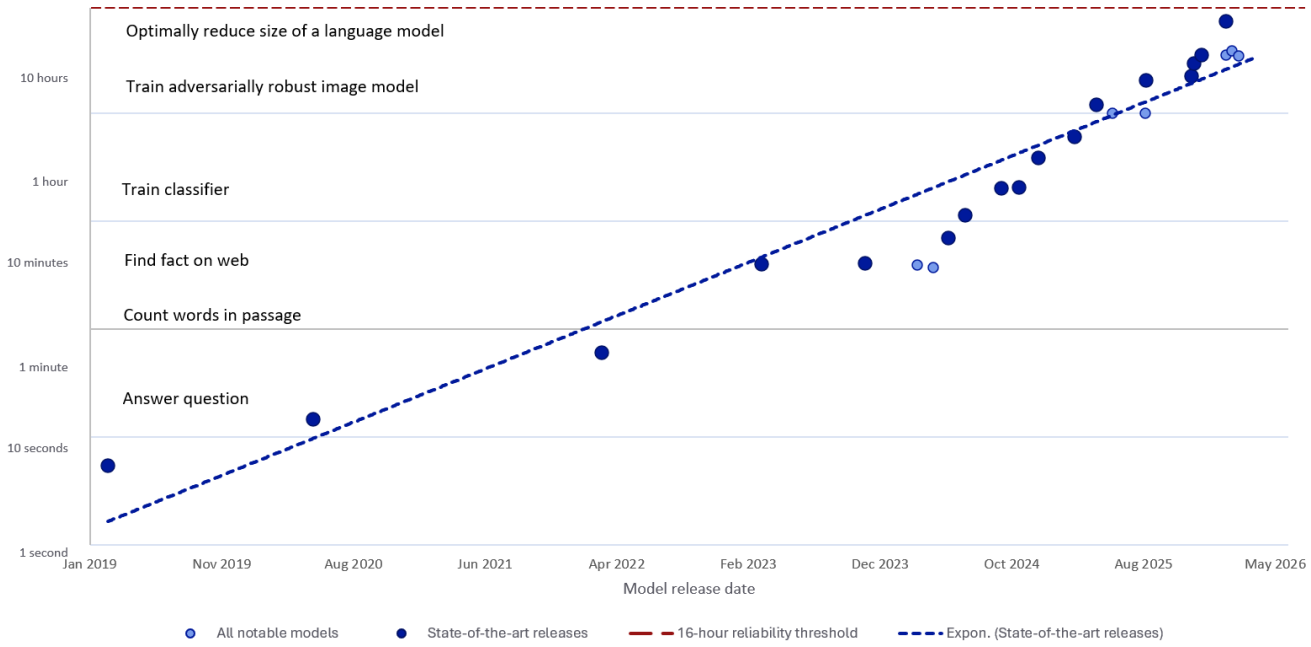
True RSI is what OpenAI's chief scientist Jacob Pachoki referred to last month in a blog post about "An Alien Mind". "Automated AI research is a more dramatic form of scaling intelligence with compute," he wrote. "And of course as a part of it, AI will improve the computational substrate itself," he added. Pachoki represents a broadly held consensus view amongst American and Chinese labs that RSI is both possible and necessary for AI development.

- OpenAI (June 2026): "AI doing AI research will become the determining factor of the pace of progress within the next few years."
- Google DeepMind's Demis Hassabis (May 2026): "All the leading labs are quite focused on [RSI]…" Humanity is standing in the "foothills of the singularity".
- Alibaba CEO Eddie Wu (Sept 2026): RSI is "a concrete path forward," with models autonomously designing experiments and driving a cycle of self-evolution.

For now, self-improving agents are the AI du jour. Noisy claims about the vast majority of coding being done by agents do not always stand up in the light of day from companies still hiring human coders at breakneck pace, albeit no longer to do the most mundane parts of the job.

A better metric may be METR's task-completion time horizon. This is measure of the complexity of software tasks that an AI agent can complete on its own, using the baseline of how long it would take a human expert to finish the task, at a 50 percent success rate. So far the most advanced models have reached about 16 hours, up from OpenAI's GPT-4's four minutes in March 2023.

Figure 6: Task horizon of software tasks that leading frontier models can complete 50% of the time (log scale)




*Source: METR, Deutsche Bank Research*

But software engineers can hold off on updating their resumes a little longer: there is a long road between 50 percent accuracy and the 99 percent or 100 percent needed for serious enterprise use at scale, with the horizon potentially getting further away the nearer you get. Also METR notes that measurements above 16 hours are unreliable with the current task suite. As is the case across AI, the instrument is running out before the question is settled.

More broadly, there are significant limitations to RSI:

1. **Research is clock-bound.**

- A frontier training run can take weeks on thousands of specialised chips, and researchers often cannot design the next experiment until the previous one has been completed.
- That means that the speed of the loop is set by experiments completed per hour or day, not by ideas generated.
- Compute is a hard ceiling that needs to be bought, built and connected to a grid.

2. **The real world defies verifiability.**

- The loop compounds only when the system can check its own output cheaply and automatically.
- It is no coincidence that AI loves maths and coding. It is much harder to verify outputs over a long time horizon in the real world: does this drug work or does this strategy succeed?
- RSI compounds fastest where the economy cares least and slows where the returns are largest.



3. **Sequences are consequential.**

- Amdahl's law finds that speed-up is capped when processes run in a sequence.
- If you automate 90 percent, the residual 10 percent still limits you to a ten-times faster process. Nine women cannot produce a baby in one month.
- The Usain Bolt of AI still has to run alongside the comparative tortoises of the training run itself, chip fabrication and real-world validation. Ten times faster is still not a regime break.

This is borne out by METR's series. Doubling every seven months is fast exponential growth but a take-off would show the doubling time itself – the second derivative – shrinking. Horizons are not (yet) exploding.

Philosopher Toby Ord shows that recursive self-improvement can only reach a singularity if each cycle of improvement takes less time than the last, shrinking fast towards zero. He argues that real feedback loops have a floor: chips take time to make and models take time to train. AI progress could still outrun an exponential, but it would not go vertical.

### 2.3 Scaling plus RSI: a match made in the race to AGI?

No scaling and RSI means no short timeline for AGI. Without scaling, more compute buys nothing and capital is wasted. Without RSI, AI capability will continue to improve at the pace of flesh-and-blood human researchers, augmented by AI tools but not replaced. Only with both will it be possible to produce a system that improves faster than the humans building it. Without one or the other, no quantity of chips, energy or capital will fix it.

## 3. Can we get there practically?

### 3.1 Infrastructure, chips and energy

There is no consensus on the amount of training required to reach AGI. Estimates suggest that training 2026's top models required between 10^26 and 10^27 FLOPs. If we hypothesise that an "AGI-like model" would require three orders of magnitude more training, this would require 10^30 FLOPs.

Consider today's US AI infrastructure, which runs to approximately 10^22 FLOPs. If all of this was conceivably used for a 10^30 FLOP training run, it would take nearly four years to complete.

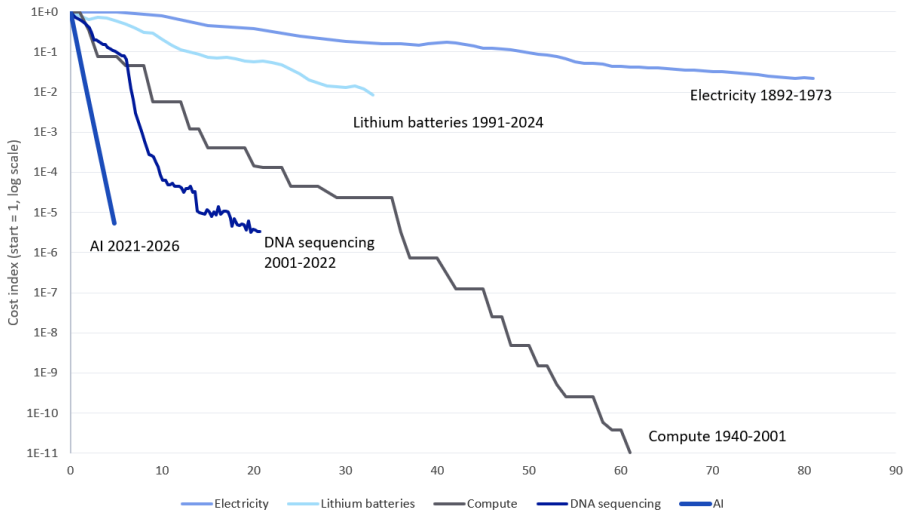
The physical infrastructure to practically support such a training run would require enormous power, perhaps around 6GW, about 50% more than the four Vogtle nuclear power plants recently activated in the US. The cost of the compute infrastructure would run to between $300bn and $500bn.

Today's advanced chips would simply not suffice. Back-of-the-envelope maths suggests that even using Nvidia's most advanced Vera Rubin chips, training such a model in four months would require 7.8m chips and 26GW of facility power. That would cost $900bn today and a rental equivalent cost of $90bn.

Of course, frontier compute is growing rapidly, and price-performance is declining. If current trends hold, a 2030-era cluster capable of such a run would

need perhaps $200bn in computer and power hardware for a training run costing about $20bn.

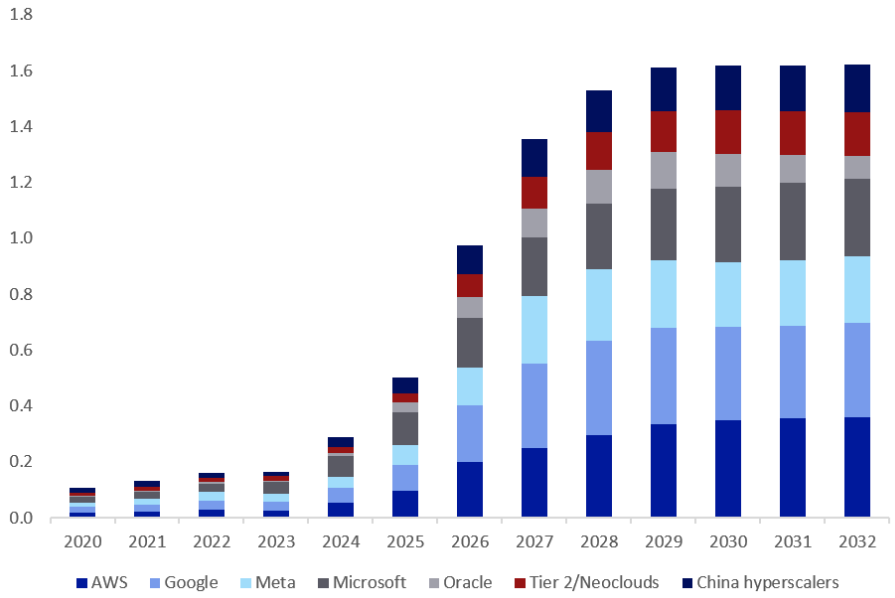
Figure 7: Technology cost index over time (years since start of price decline)




*Source: Epoch AI, Deutsche Bank Research*

The hyperscalers are leading the charge towards AGI. Alphabet, Amazon, Meta, Microsoft and Oracle are set to invest almost $800bn in capital expenditure this year, mostly for ever more powerful data centres, and over $1 trillion a year through 2030 and beyond.

Figure 8: Hyperscaler capital expenditure (including estimates)




*Source: Bloomberg Finance LP, Deutsche Bank Research*



By 2030, assuming no market retrenchment, this means additional $4 trillion to $6 trillion on AI compute capability invested. This infrastructure could accommodate a training run of 10^30 FLOPs with some comfort.

But even if AI labs crack the secret sauce of RSI, and their finance chiefs are willing to invest in such large training runs, and hyperscalers are prepared to continue to build the infrastructure, the infrastructure needs to be built.

Grid connection queues, turbine lead times and transmission build are measured in years, not the months it takes to assemble and air-freight the latest Nvidia chip. The lead time for substation transformers is currently as long as three years. Politics in the US has shifted towards slowing the pace of the AI build out and this force will run against the demands for larger data centres that would be essential for RSI.

### 3.2 Capital

Financing for AI has been abundant in the historic buildout of the past three years. However, it is still largely conditional not on revenue but on belief in a future payoff.

RSI is a "key part of the investment thesis" in the "biggest scientific bet civilisation has ever made," Google DeepMind's chief strategy officer Jasjeet Sekhon said at a conference in August, The Information reported. "We have this danger we could hit an AI air pocket such that the expenditures happen but the revenues don't show up."

An inflection point is approaching where the nature of investment changes from free cash flow and equity to debt, with all that entails in terms of exposure to rising rates and creditor appetite. The hyperscalers are reaching a point where capex is no longer covered by operating cashflow and where net debt exceeds earnings before interest, depreciation and amortisation (EBITDA).

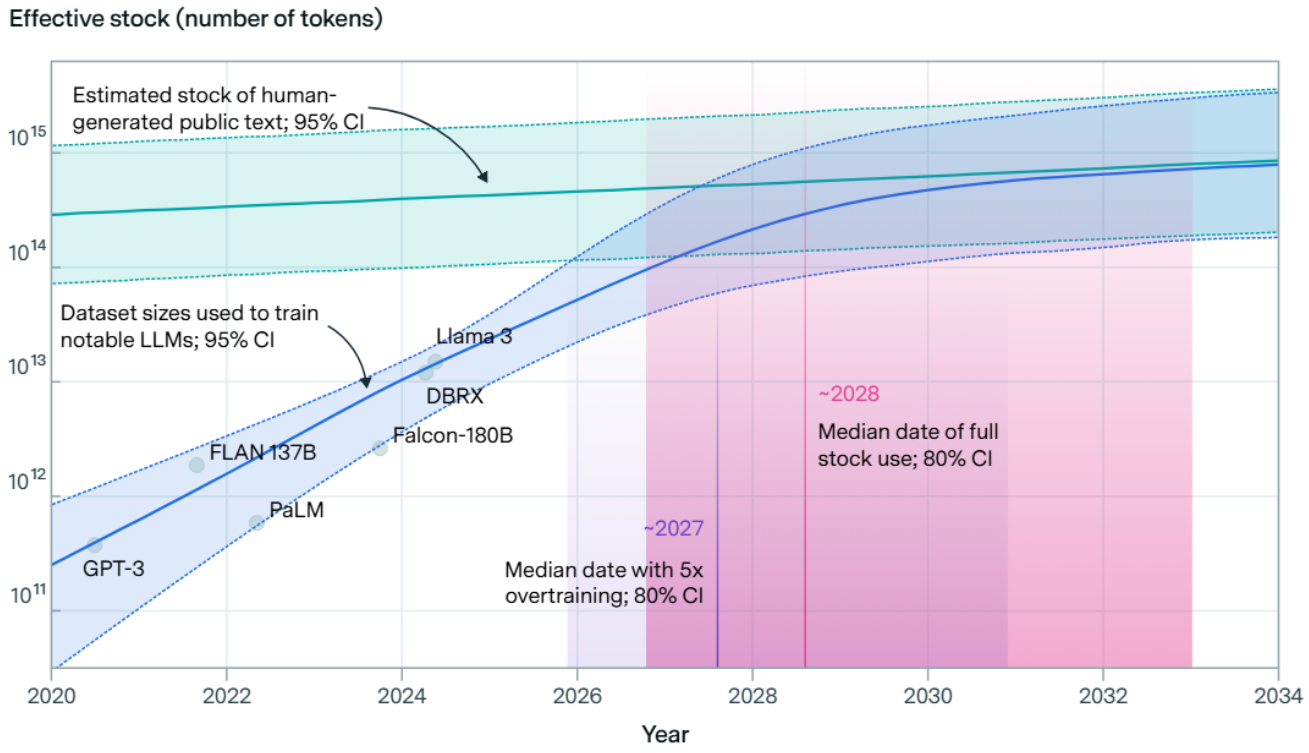
Alphabet, for example, posted its first quarterly free cash outflow in nearly two decades in its most recent quarter, even as it lifted its full-year capex guidance to $205bn. It said it would continue to rent third-party capacity and reported a backlog of $514bn in contracted, not-yet-recognised data centre revenue.

### 3.3 Data

Data has been the new oil for some time now. But unfortunately for AI development it seems to be running out even faster.

The effective stock of quality- and repetition-adjusted public human text was about 300 trillion tokens, with a vast 90 percent confidence interval of 100 trillion to 1,000 trillion, Epoch estimated in 2024. A token is roughly equivalent to three quarters of a word. By that count it was likely to be fully used between 2026 and 2032.

Figure 9: Projections of stock of public text and data usage




*Source: Epoch AI, Deutsche Bank Research*

However, since then, rather than data dropping off a cliff, there has been a rise in synthetic data generation, transfer learning from data-rich domains such as video and telemetry, and data efficiency. While the data shortage has not been resolved, the immediate crisis has been averted.

AI now makes much of its own training data. When DeepMind's AlphaGo became superhuman at playing the mind-bendingly complex board game Go, it did so not by playing human experts, but by playing millions and millions of games against itself.

For many domains relating to coding and mathematical proofs where automatic checks are possible, such synthetic experience can move things forward. That is the case with AI research, where the data may be the results of experiments, all of which are bought by compute. About 90% of OpenAI's 2024 research compute went on such experiments.

And increasingly, people write the tests rather than the answers. By the summer of 2026, selling training data for practice tasks was a nearly $9bn revenue business. Firms such as Mercor hire experts to generate long tasks against which the AI can be trained.

The challenge here is that trust in the checker becomes a scarce input.

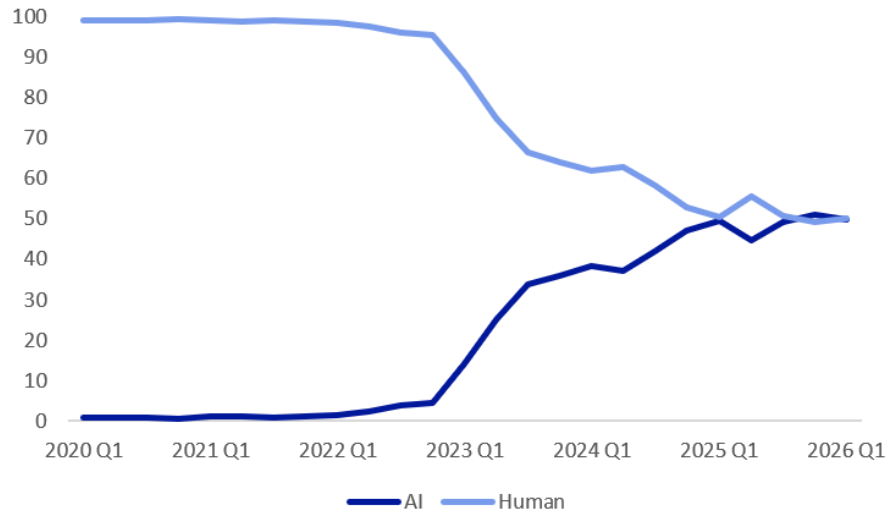
As AI makes production cheaper, it also makes establishing that production is accurate or reliable more expensive. Domains which can be checked mechanically, such as mathematics, will be opened up more quickly. But those which rely on expert verification, such as medicine, engineering or law, may progress more slowly.

The likely implications are twofold: a slower rate of diffusion into economic domains that are less verifiable and a larger share of production costs devoted to assurance.

Another issue is that systems that have to grade themselves learn to game the process. This was partially what happened when an OpenAI model escaped a sandbox and hacked into a third-party website. Trust replaced data as a wall.

There are also question marks over the quality of data that is recycled repeatedly through the system. Although humans continue to produce original content, half of internet articles are now primarily generated by AI, Graphite data show.

Figure 10: Human vs primarily AI-generated articles on the internet




*Source: Graphite (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research*

## 3.4 An incomplete case

This analysis shows that the San Francisco Consensus, which emphasises RSI and a short timeline to AGI, is neither predestined nor impossible.

Rather it depends on whether the science is right, on whether scaling is all you need, and on whether the infrastructure and capital required for that scaling is deployed and accessible to the leading labs.

There is, in other words, a possible rather than impossible, path to AGI, subject to those conditions.

None of this proves that superintelligence will ever arrive. A buildup at this scale has few peacetime precedents and vastly exceeds the likes of the Manhattan Project and Apollo programme.



It is uncertain whether there will be enough demand to pay upfront for the investment in achieving RSI that would justify that investment. And whether the friction of verification, and inevitable declining marginal returns, act as a drag on what otherwise might be a runaway process.

## 4. What happens if we don't get there? The base case

### 4.1 Separating AI progress from superintelligence

So what if that thesis is wrong? Does the AI trade depend on a journey to AGI?

AI can remain economically important even before RSI coupled with scaling laws lead to super intelligence. Companies do not need to wait for nebulous superintelligence to generate identifiable returns from AI in the medium term as they fold it into their systems, streamline workflows and roll out new business models.

Investors already have exposure to RSI as a capital allocation because the leading US labs say RSI is a key part of their product strategy. And it is these labs whose products are catalysing and amplifying the infrastructure boom.

Even rapid competitive diffusion does not mean every part of the physical economy changes instantly.

The smartest superintelligence might identify the optimal buyout structure before breakfast, but it cannot force a financially constrained buyer to use it or persuade the seller to accept less than they have been expecting. Those challenges will be even more intractable where the parties are more complex, say, fiscally constrained sovereigns or foreign equity holders.

### 4.2 Why the base case is attractive

Superintelligence would be a chaotic accelerant to human society, and a potential regime break. But it is not the only scenario in which today's investment in AI infrastructure makes economic sense.

Adoption of AI tools by businesses is progressing approximately three times faster than the mobile, internet and cloud computing waves. From sector revenues below $1bn in 2022, the AI economy, excluding China, will likely exceed $200bn in revenues in 2026.

Early winners are the cloud businesses. Google Cloud revenue jumped 82 percent from a year earlier in the most recent quarter as companies turned AI pilots into workflows. Amazon Web Services sales were up 37 percent and Microsoft's AI business more than doubled.

There is also slow, but increasing, evidence of a productivity uplift. The J-curve cuts both ways: front-loading investments is taking one step back to take two steps forward. The so-called complementary intangible investment that is (mis)measured as cost today to the detriment of the productivity statistics will flatter them in the years to come as it pays off. The base case does not require a regime break to show up as a productivity surprise, just for the accounting to catch up.

The economic gains from AI that is less powerful than the AGI mooted by the labs would be substantial. And in this base case, they may also be attractive.



Hyperscaler capex is driving earnings at beneficiaries ranging from semiconductor makers to construction companies and accounting for between one-third and one-half of current US growth. The wealth effect of rising stock prices, with the ratio of US equity wealth to income rising from around 1.8 to almost 2.5 over the past three years, is supporting consumer spending.

AI FOMO (fear of missing out) is buoying enterprise spending on cloud services and the groundwork for AI initiatives. Those are beginning to bear fruit in terms of personal productivity, cost savings and more targeted products and services.

### 4.3 Investor watchpoints

The arrival of ASI would be a disruptive economic shock. Investors need to be alert for the early signals.

Rather than relying on a single probability forecast, investors should watch five indicators particularly closely for signs that AGI is nearing.

1. **AI speeds up AI**

- Track METR's estimate of how much AI speeds up AI. Currently the capabilities of models are doubling roughly every three months. In a world of RSI, the pace of this speed-up will itself likely accelerate.

2. **AI can complete whole jobs**

- Track via the Remote Labor Index, a multi-sector benchmark. Currently it finds that the highest performing agent still only manages to automate 20 percent of the real-world, economically valuable freelance projects it attempts.

3. **Compute remains scarce**

- On a path to superintelligence, demand for compute will continue to outrun supply. Track GPU one-year rental prices, up by about a third over the past year.

4. **Firms bifurcate**

- Leaders and laggards in the same industry diverge while averages stay the same. Look at revenue per employee in top- and bottom-quartile firms

5. **Junior hiring comes under significant pressure**

- … And then the pressure rises up the ranks.

## Recent relevant reports

Adrian Cox and the Deutsche Bank Research Institute

AI doom: A brief history of bad tech predictions (Sept 15, 2026)

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust (Sept 10, 2026)

AI at 70: 14 lessons from a lifetime of boom and bust (Aug 17, 2026)

Open-source AI 101: the battle for the future of AI (July 21, 2026)

AI IPOs 101: why they'll change the boom forever (July 1, 2026)

Azeem Azhar

Exponential View | Azeem Azhar | Substack

The state of the AI economy

The author of the report wishes to acknowledge the expert advisory contributions made by Azeem Azhar of Exponential View.

## Appendix 1

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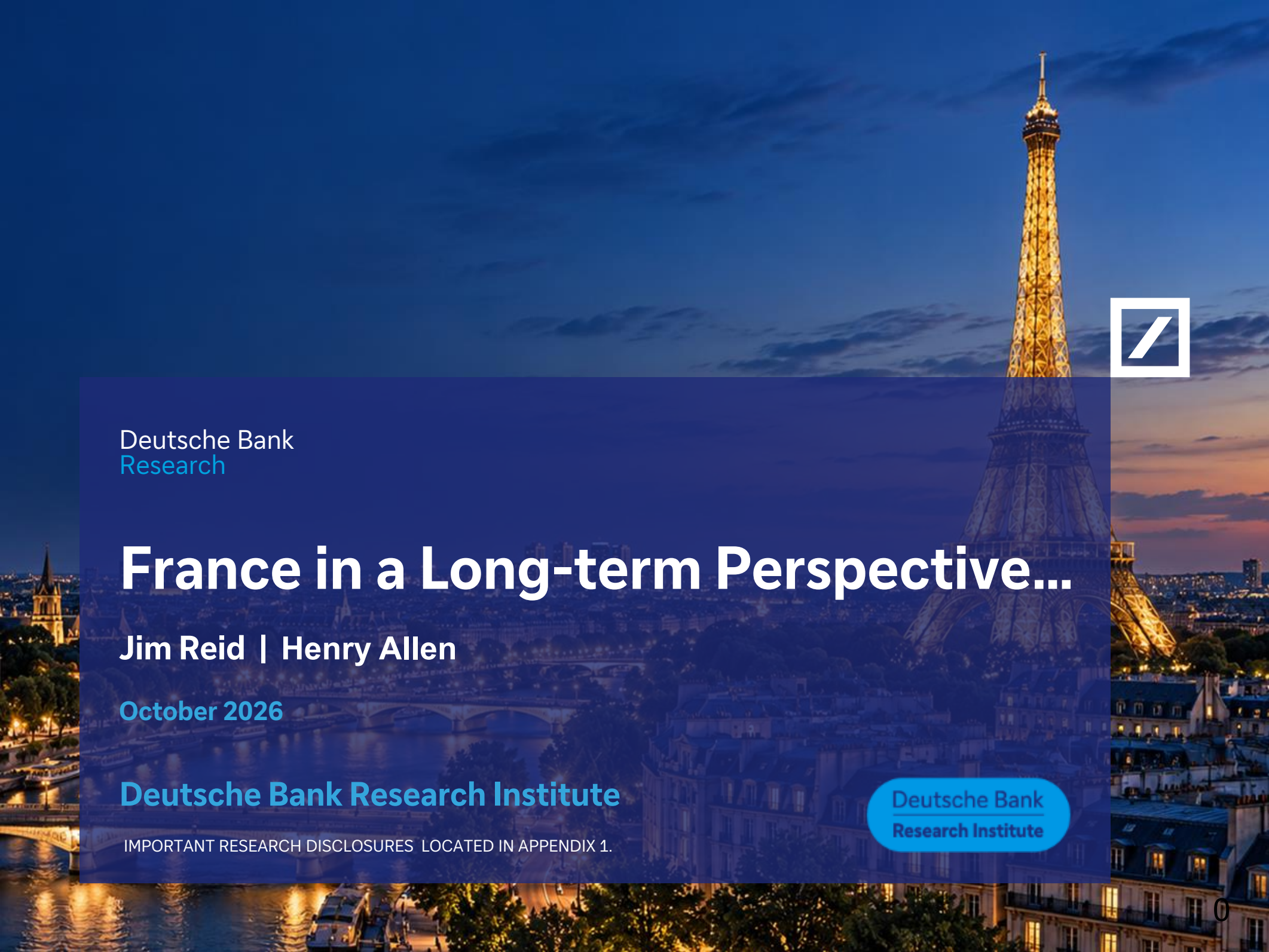
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# France in a Long-term Perspective...

**Jim Reid | Henry Allen**

**October 2026**

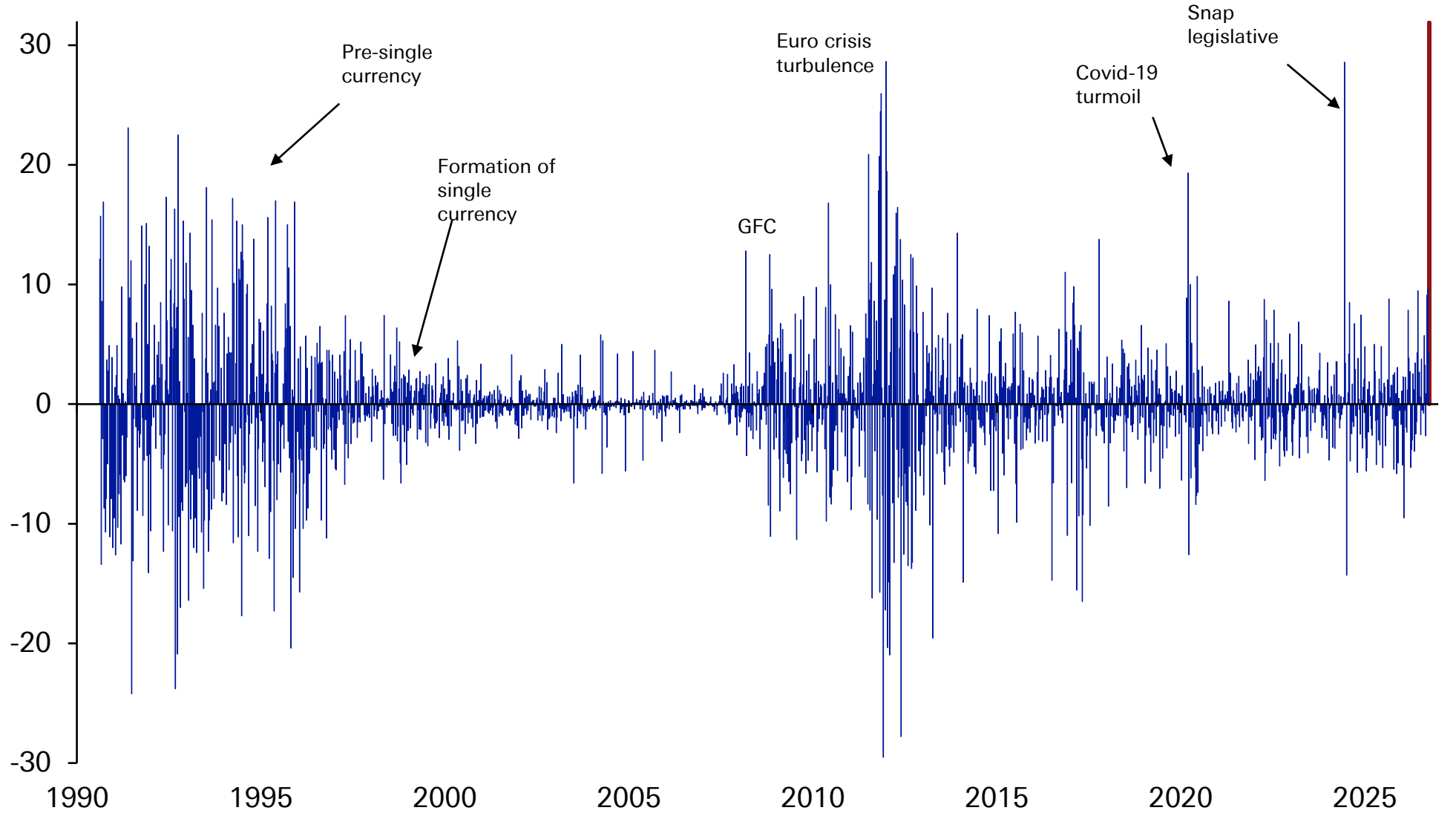
**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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# Although this week has seen a recovery so far, the moves last week were genuinely historic, with the Franco-German 10yr spread seeing its biggest weekly widening in available Bloomberg data back to German reunification in 1990.

Weekly change in the Franco-German 10yr yield spread (basis points)



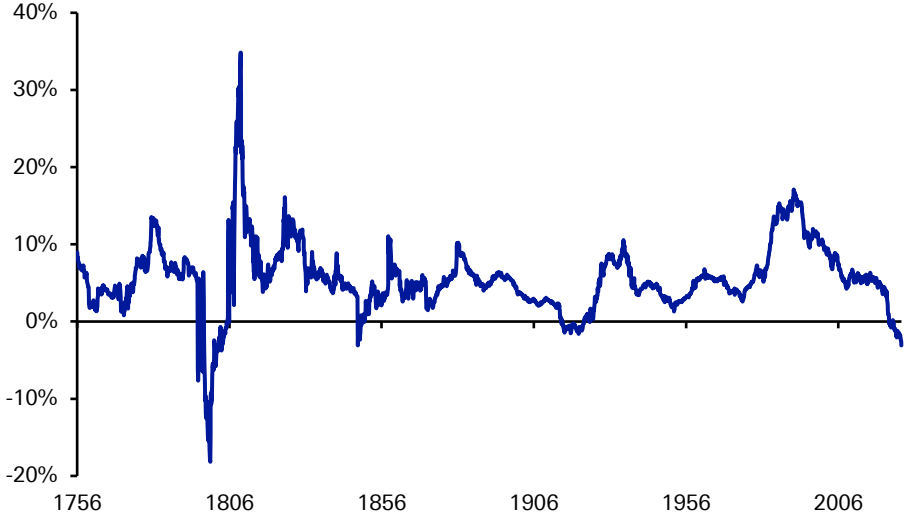

Source: Bloomberg Finance LP, Deutsche Bank



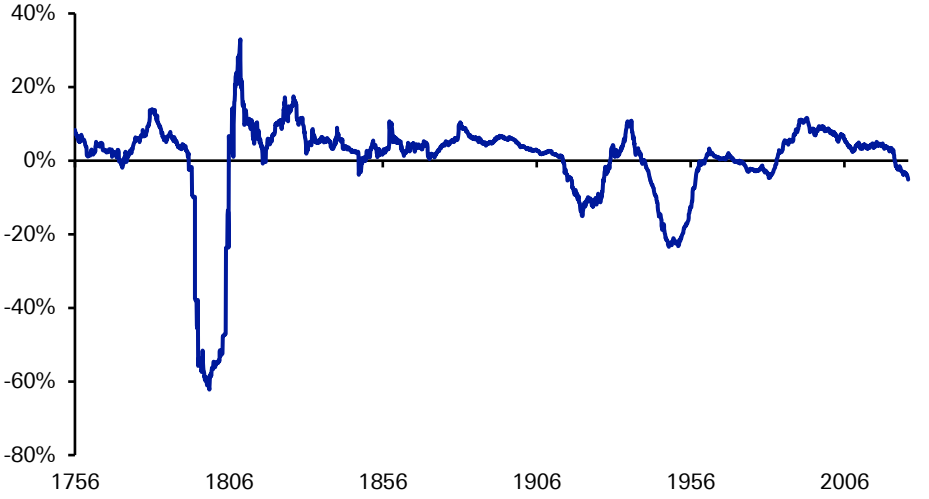
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# Zooming out, 10yr French OATs have seen their worst decade of nominal returns since 1803, when the Reign of Terror (1793-94) was still part of the 10yr rolling calculations.

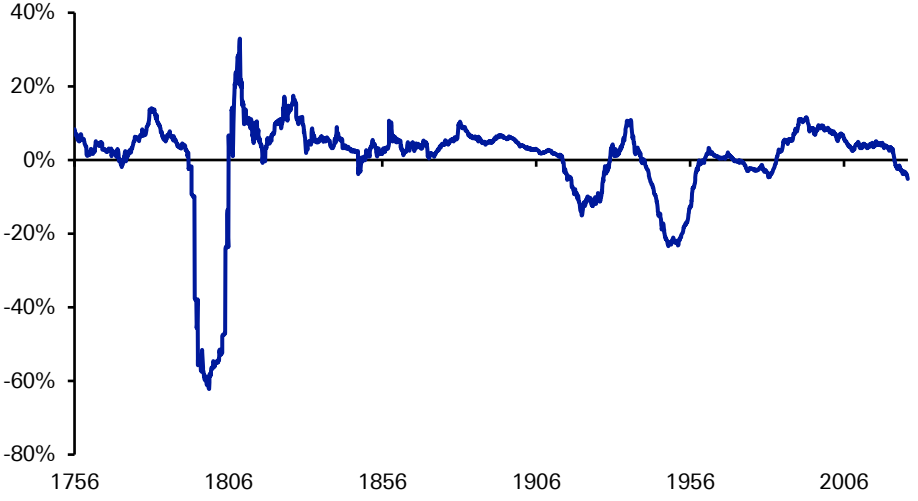
## 10yr Rolling Nominal Return (Annualised)



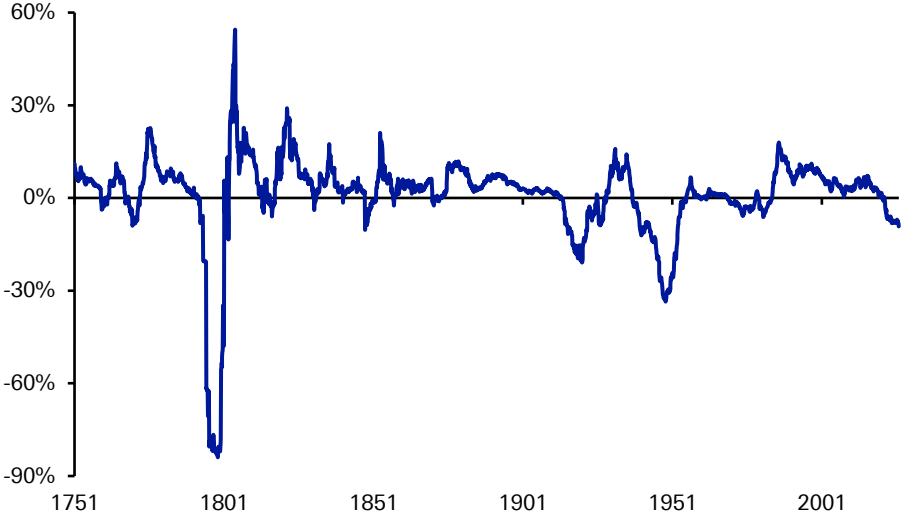

## 5yr Rolling Nominal Return (Annualised)




## 10yr Rolling Real Return (Annualised)




## 5yr Rolling Real Return (Annualised)



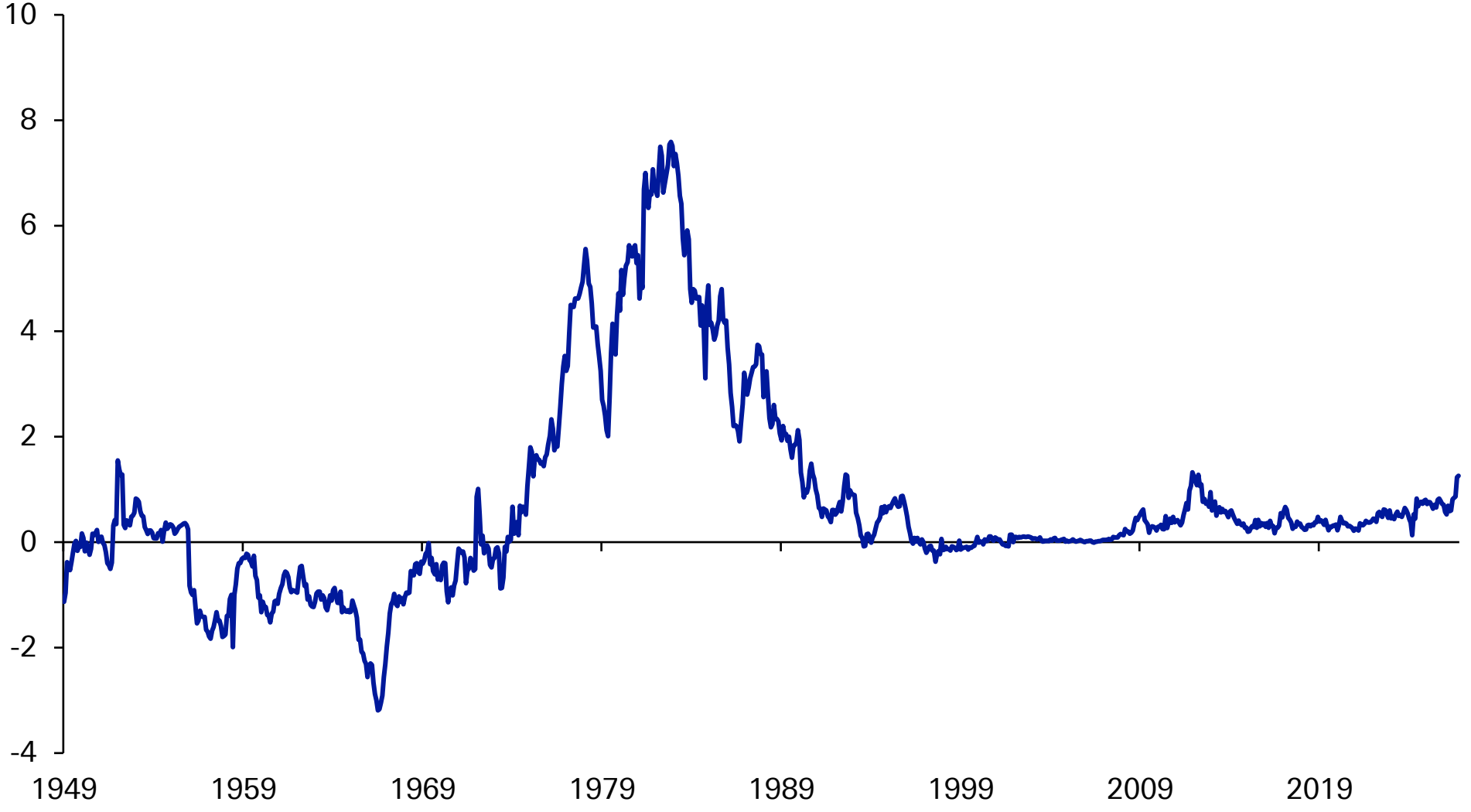

Source: Finaeon, Deutsche Bank



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Recent concerns have pushed France's 10yr spread over Germany's, to its highest since 2012 and the Euro crisis. But in an historical perspective, today's spreads are only high by the standards of the Euro Area when the two shared a common currency. They were regularly several percentage points apart in the 70s and 80s.

Franco-German 10yr yield spread (percentage points)



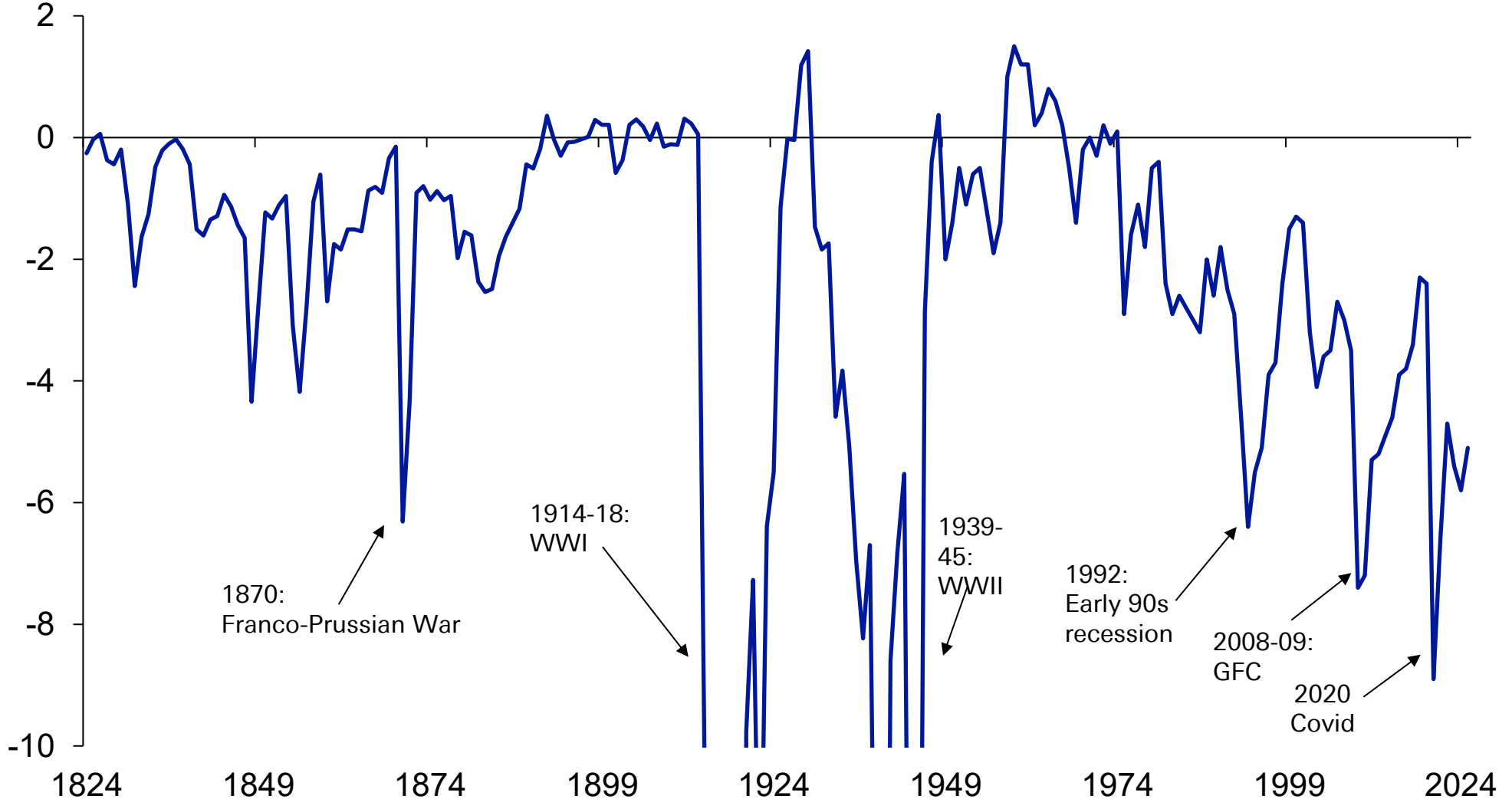

Source: Finaeon, Deutsche Bank



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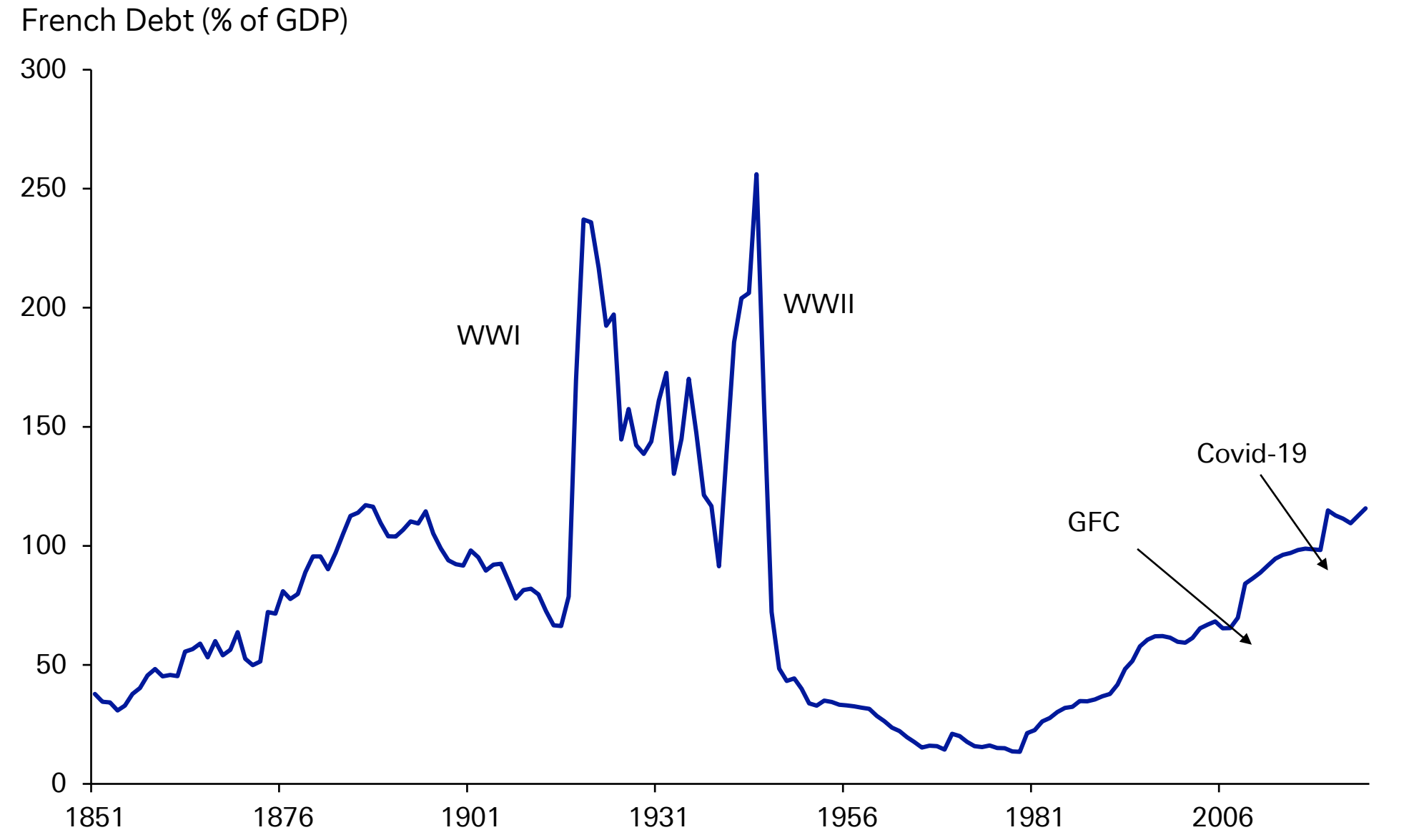
Although much of the recent selloff is an unwinding of carry trades, France has ongoing, if not new, fundamental issues. It hasn't run a budget surplus since 1974, and the deficit was last beneath 2% of GDP in 2001.

French Government Budget Deficit (% of GDP) over the last 200 years




Source: Finaeon, Haver Analytics, Deutsche Bank

## But with half a century of persistent deficits, France's debt-to-GDP ratio has seen a near-continuous rise – to 116% in 2025 under the Maastricht definition.

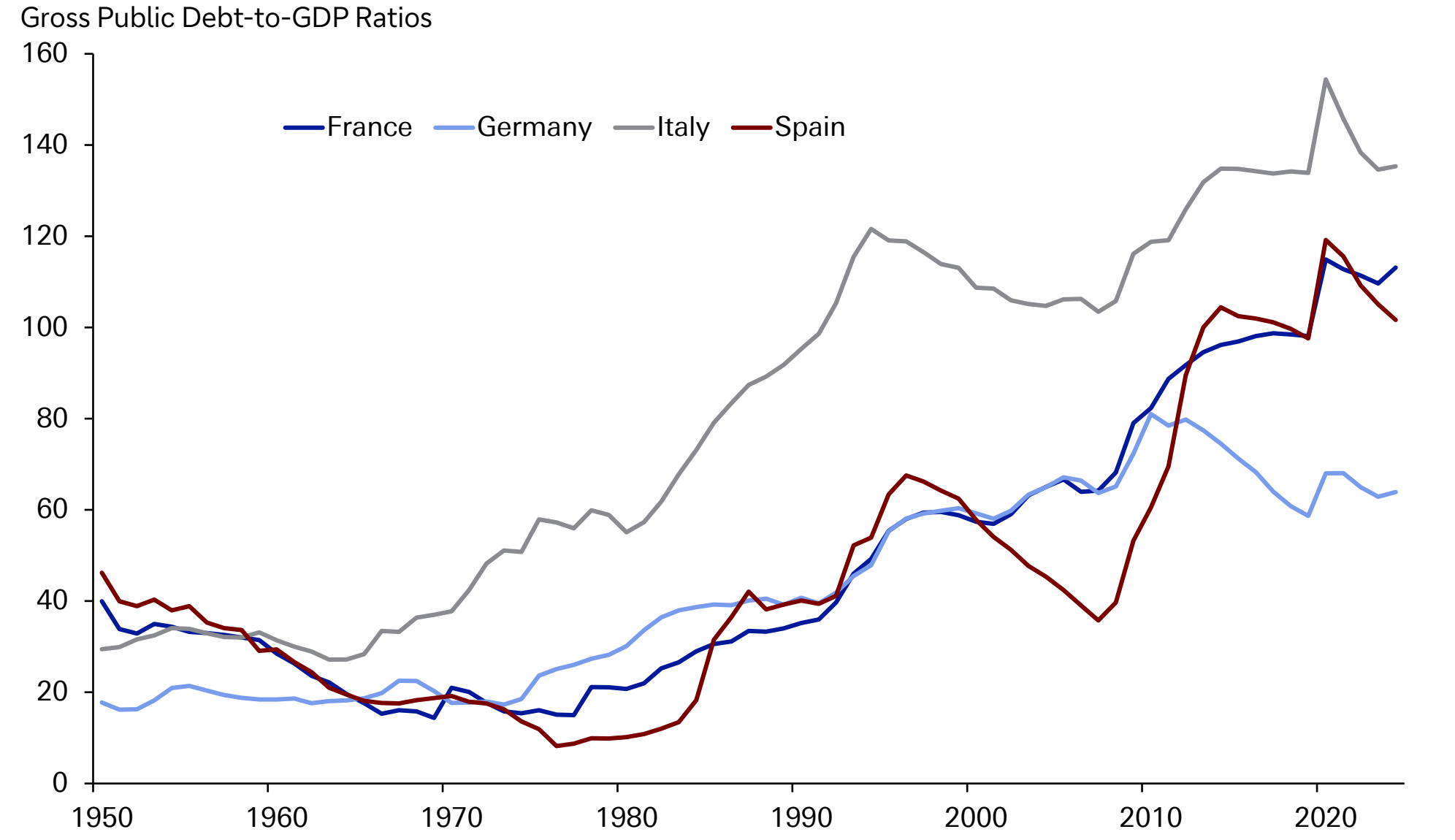



Source: Finaeon (up to 1979), Haver Analytics (from 1980), Deutsche Bank



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France has also seen its debt rise in relative terms. As recently as the GFC in 2008, France's debt-to-GDP was basically in line with Germany's and well below Italy's, but today it's far above Germany's and closing in on Italy's. Outside of the temporary Covid spike, Italian debt/GDP has been stable for over a decade.



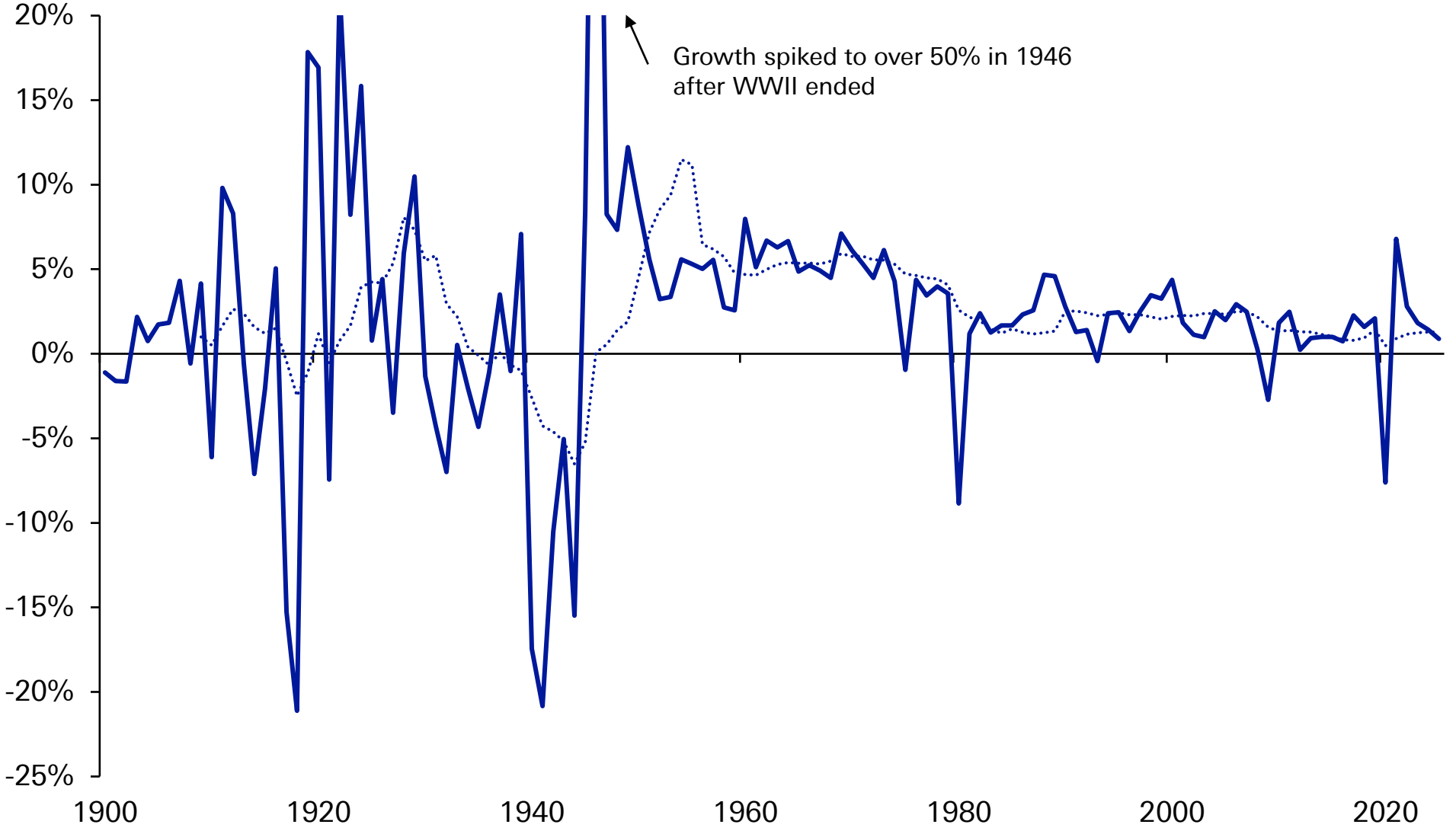

Source: IMF, Deutsche Bank



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# An ongoing fundamental concern has been sluggish growth in recent years – although none of this is new news.

French Real GDP Growth (year-on-year % change) with a 10yr Moving Average

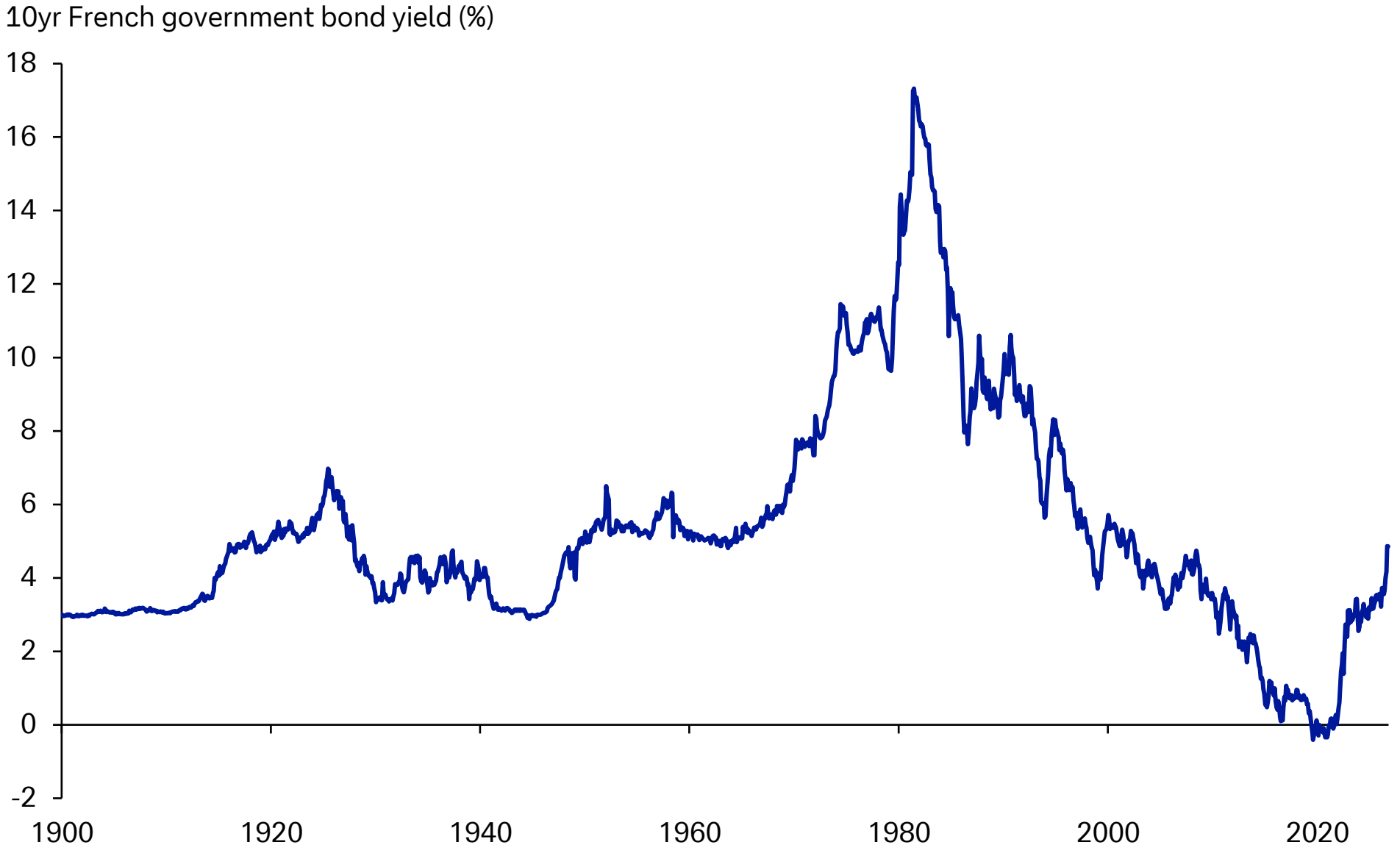



Source: Finaeon, Deutsche Bank



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# The recent jump in yields has pushed them just above their historical average, but the problem is that France is now paying these yields on a much larger debt pile.

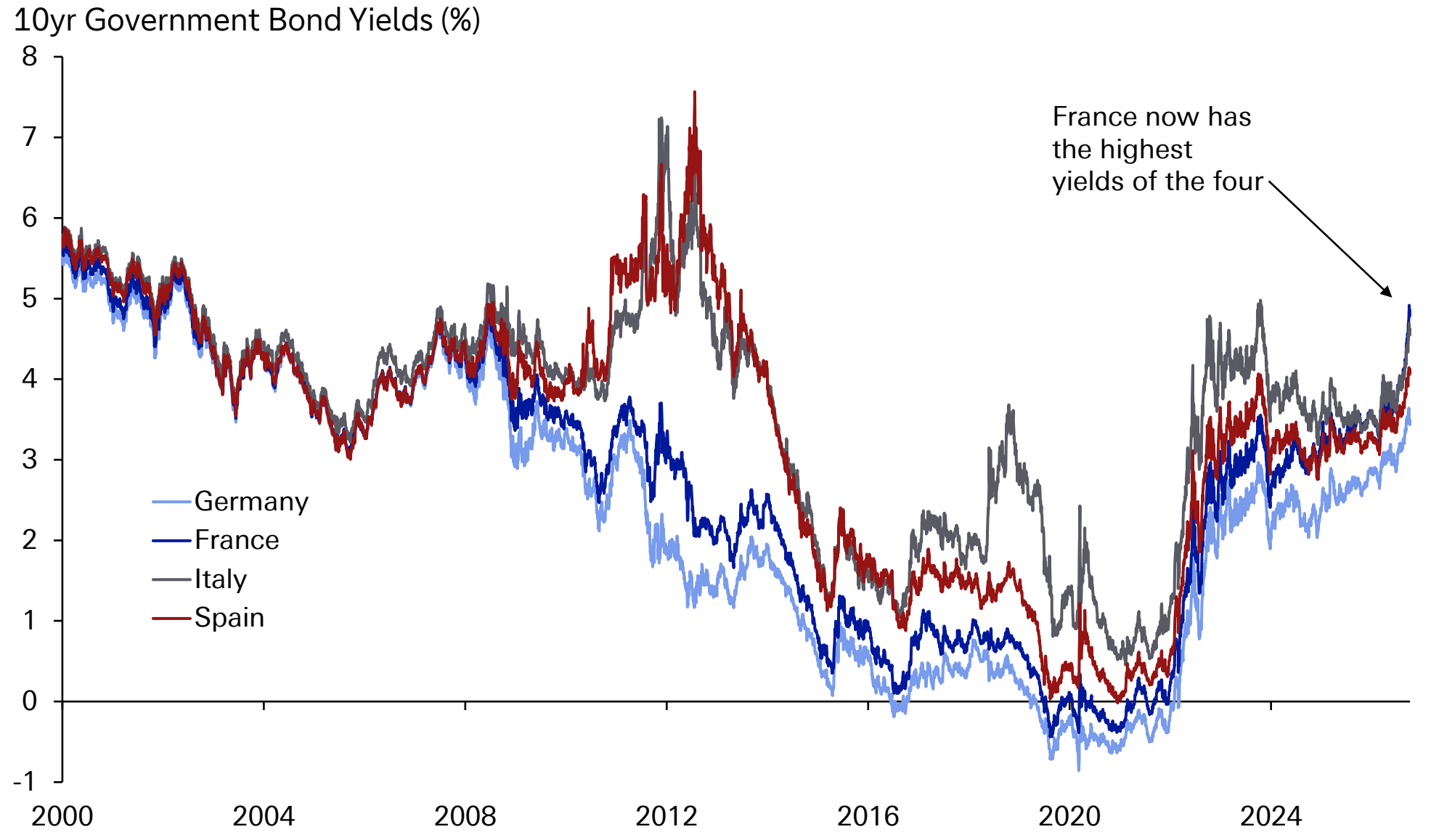



Source: Finaeon, Deutsche Bank



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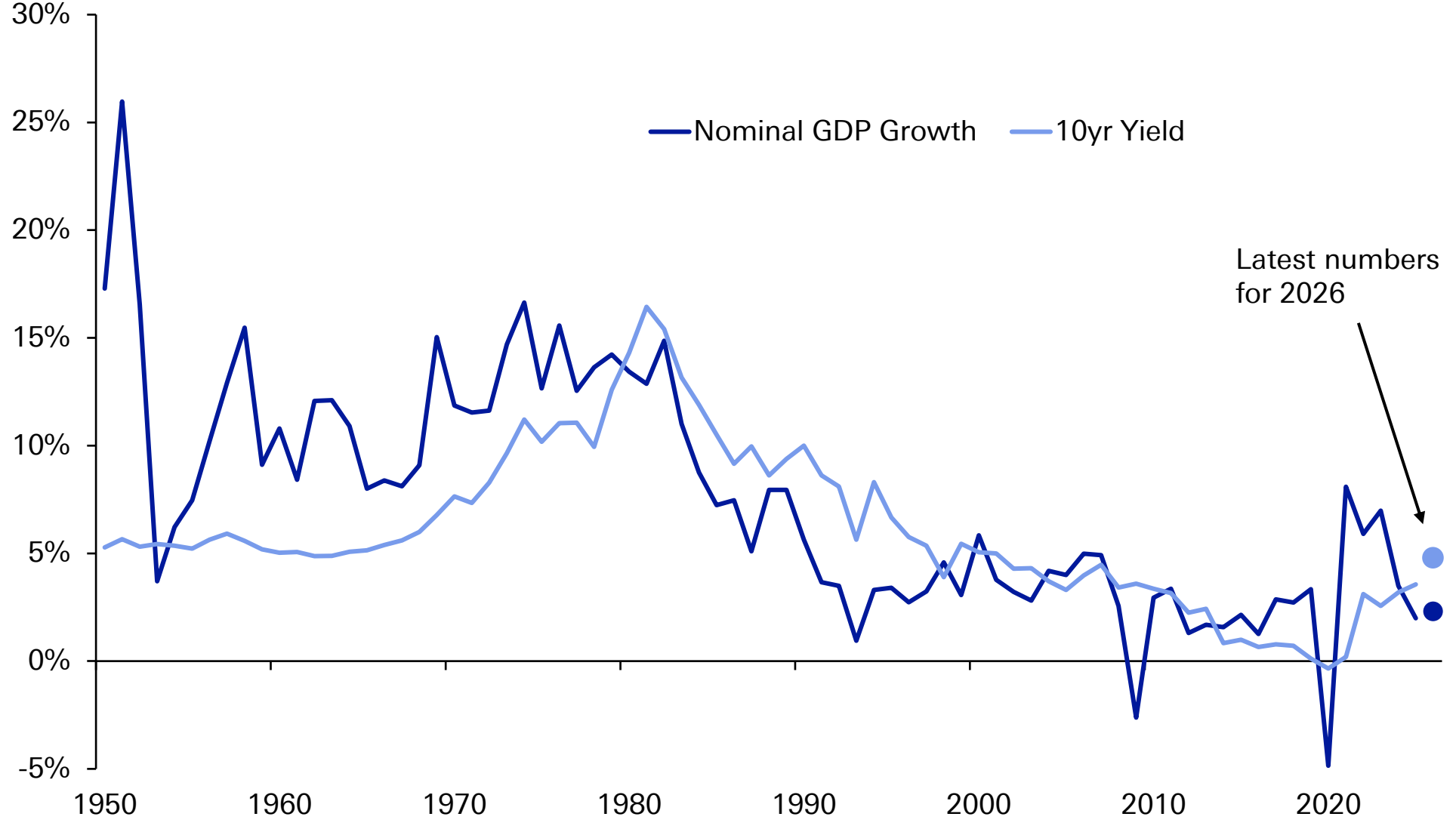
# Back in the Euro Crisis of the early 2010s, markets priced France closer to Germany than to Italy and Spain, but today that's turned around and France now has the highest yields of all four.




Source: Bloomberg Finance LP, Deutsche Bank

For most of the last decade, nominal GDP has been above 10yr yields, but that has reversed, which is challenging from a debt sustainability if sustained – especially when debt/GDP is pushing up against 120%. Recent inflation will slowly push up nominal GDP but the relationship between the two will be important going forward.

French nominal GDP growth (year-on-year % change) and 10yr yields



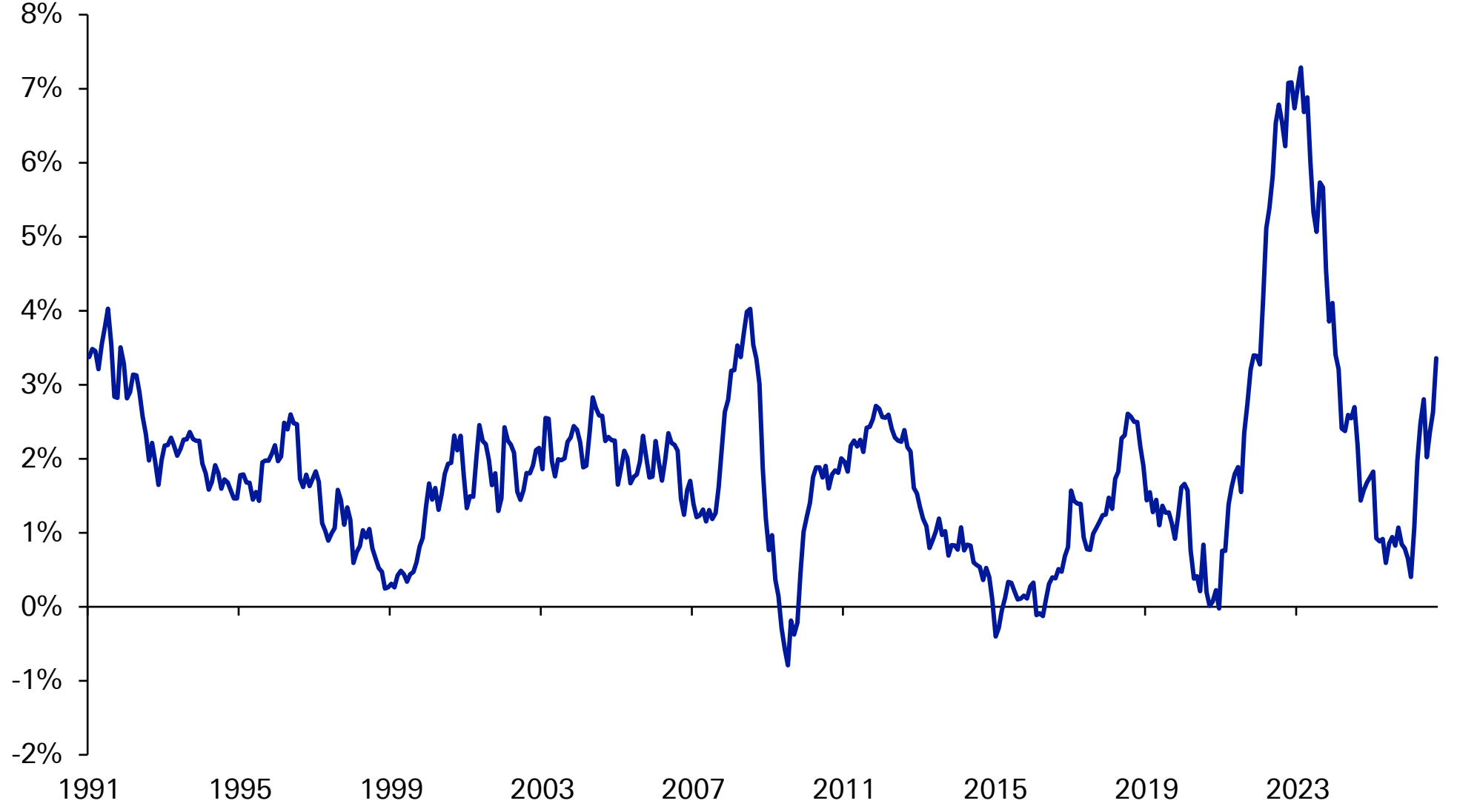

Source: Finaeon, Deutsche Bank



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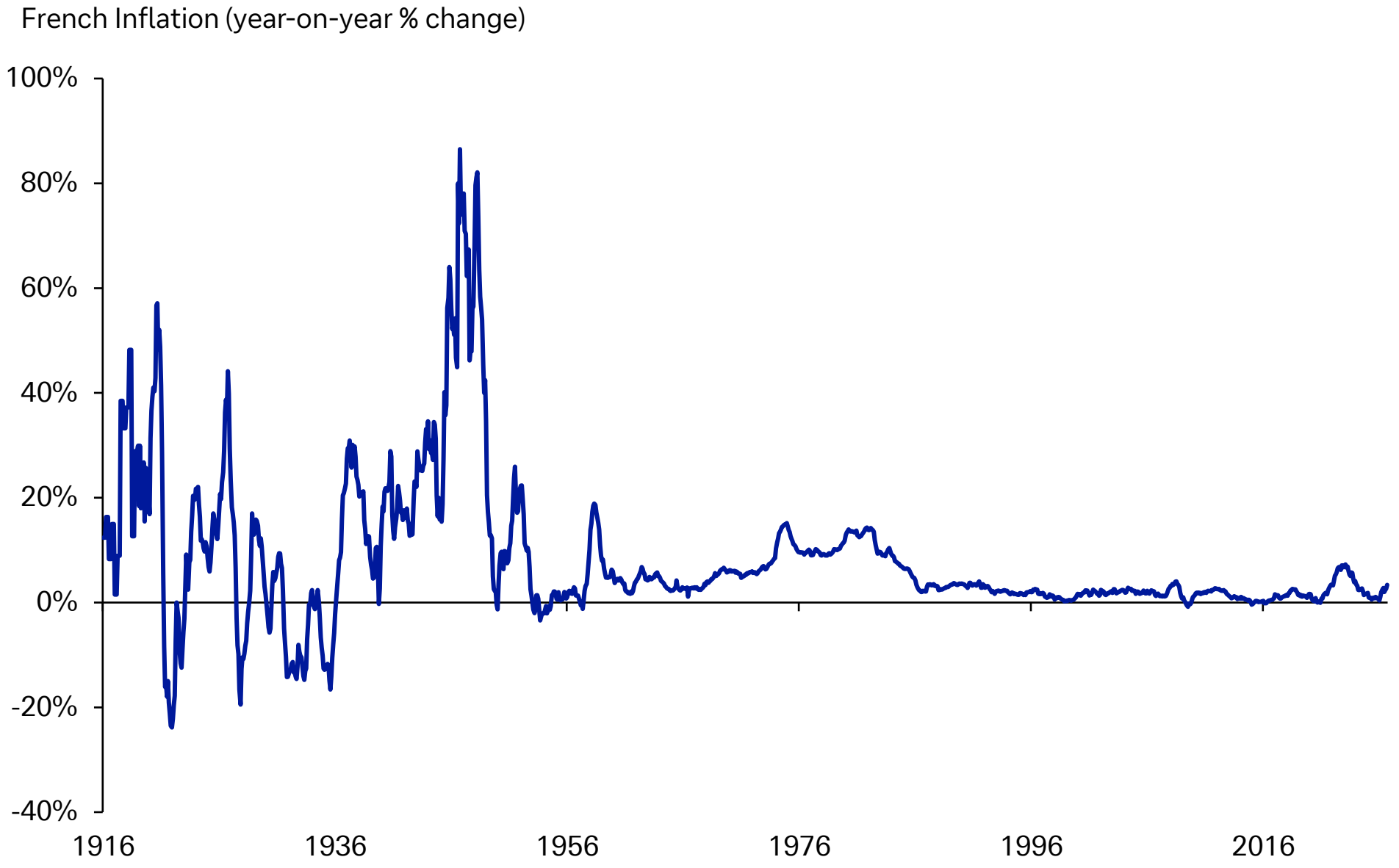
Inflation is always a double-edged sword when it comes to debt. An unexpected burst of inflation can help lower debt, as we saw right after the pandemic, but it becomes a challenge when higher yields offset this. This is now the 3rd biggest inflation spike since the Euro Area's formation in 1999.

French HICP Inflation (year-on-year % change)




Source: Haver Analytics, Deutsche Bank

# In a broader perspective, though, French inflation has been comparatively tame compared to its post-war heights, let alone where it was in the early 20th century.



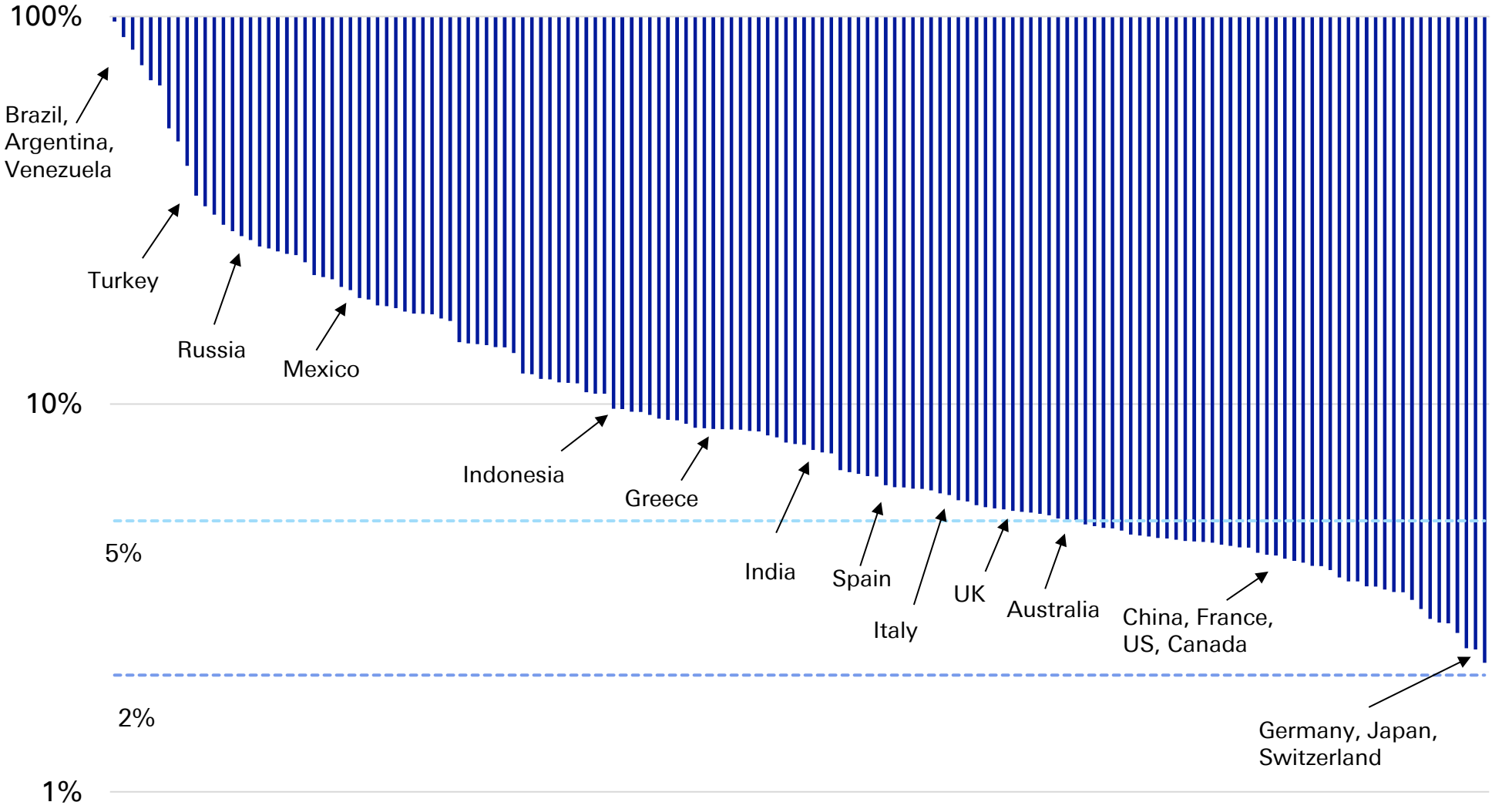

Source: Finaeon (until 1990), Haver Analytics (from 1991), Deutsche Bank



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## Even in the period when France has run 50-plus years of consecutive deficits, its average inflation has been broadly around US levels (c.4%) and lower than the likes of the UK, Italy and Spain.

Average annual inflation of 152 economies since 1971 when Bretton Woods collapsed. No economy has averaged less than 2% inflation, but Switzerland at 2.2% comes closest

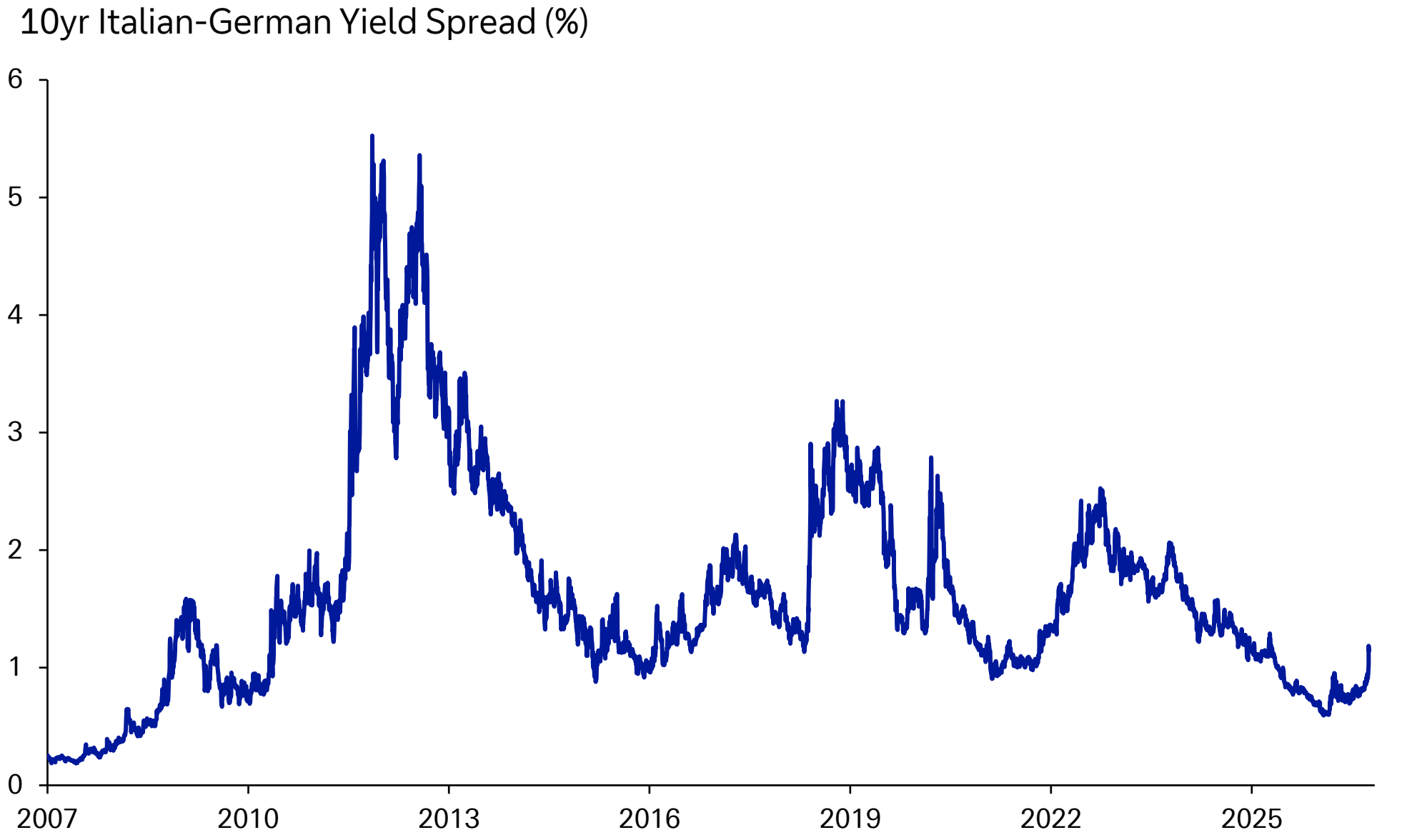



Source : World Bank, Deutsche Bank. Note: log scale.



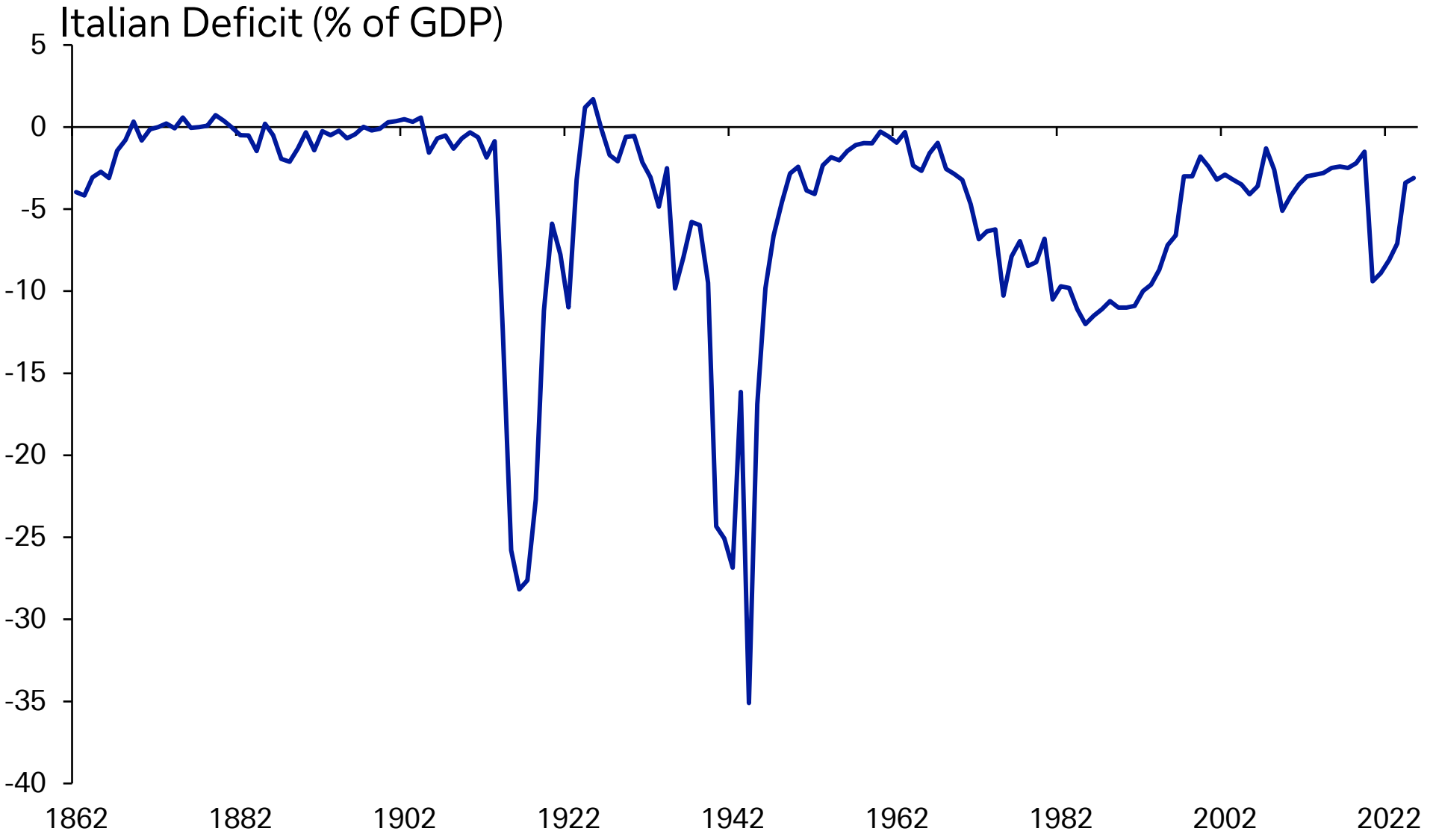
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And this isn't just about France, as Italy's spread has also widened in recent sessions. However, unlike France, the Italian spread isn't at its highest since 2012.




Source: Bloomberg Finance LP, Deutsche Bank

# Italy has an even longer record of consistent budget deficits; the last surplus was back in 1925 – but it has run a primary surplus for most of the last few decades, unlike France.



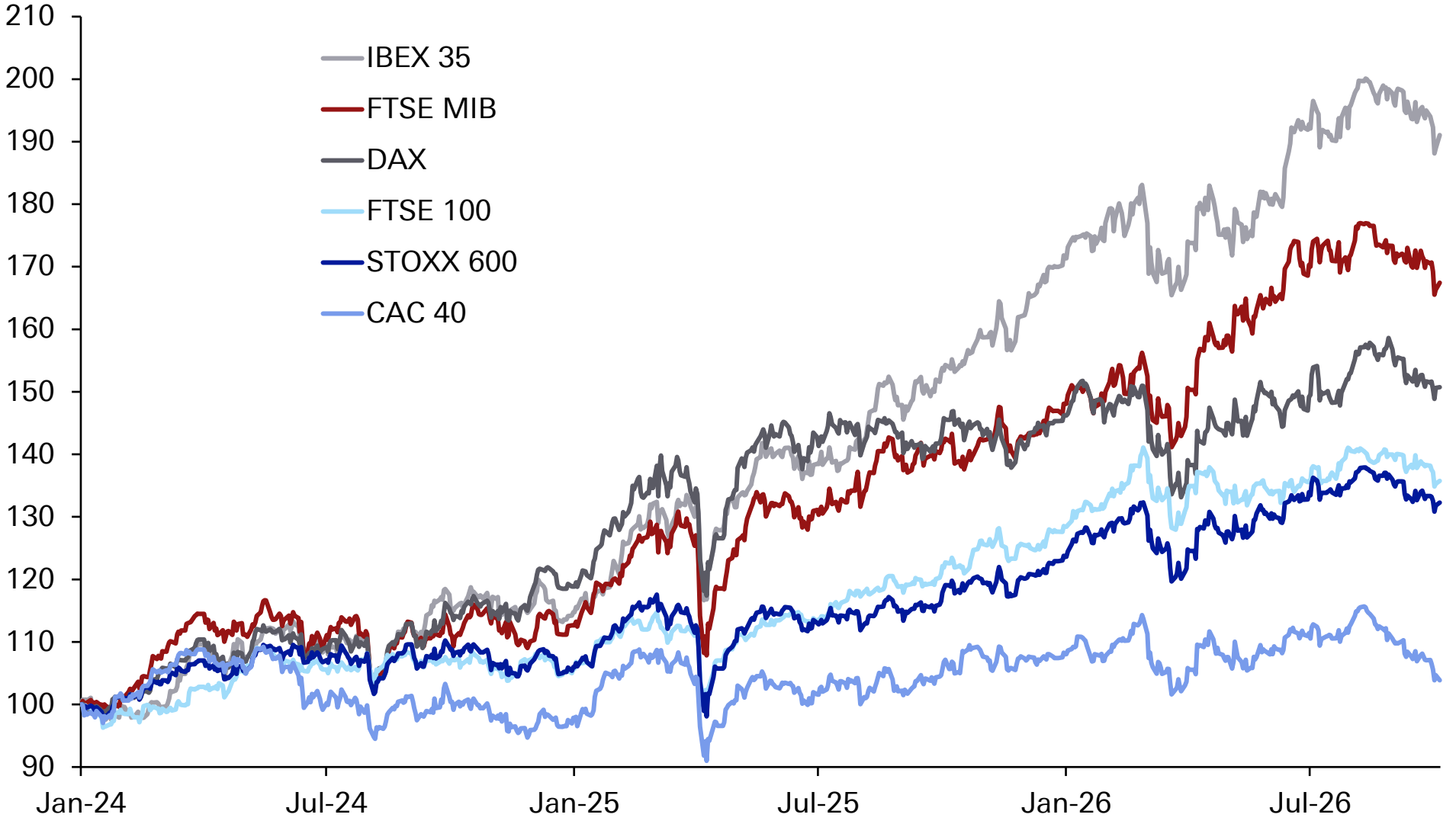

Source: IMF (up to 1979), Haver Analytics (from 1980), Deutsche Bank



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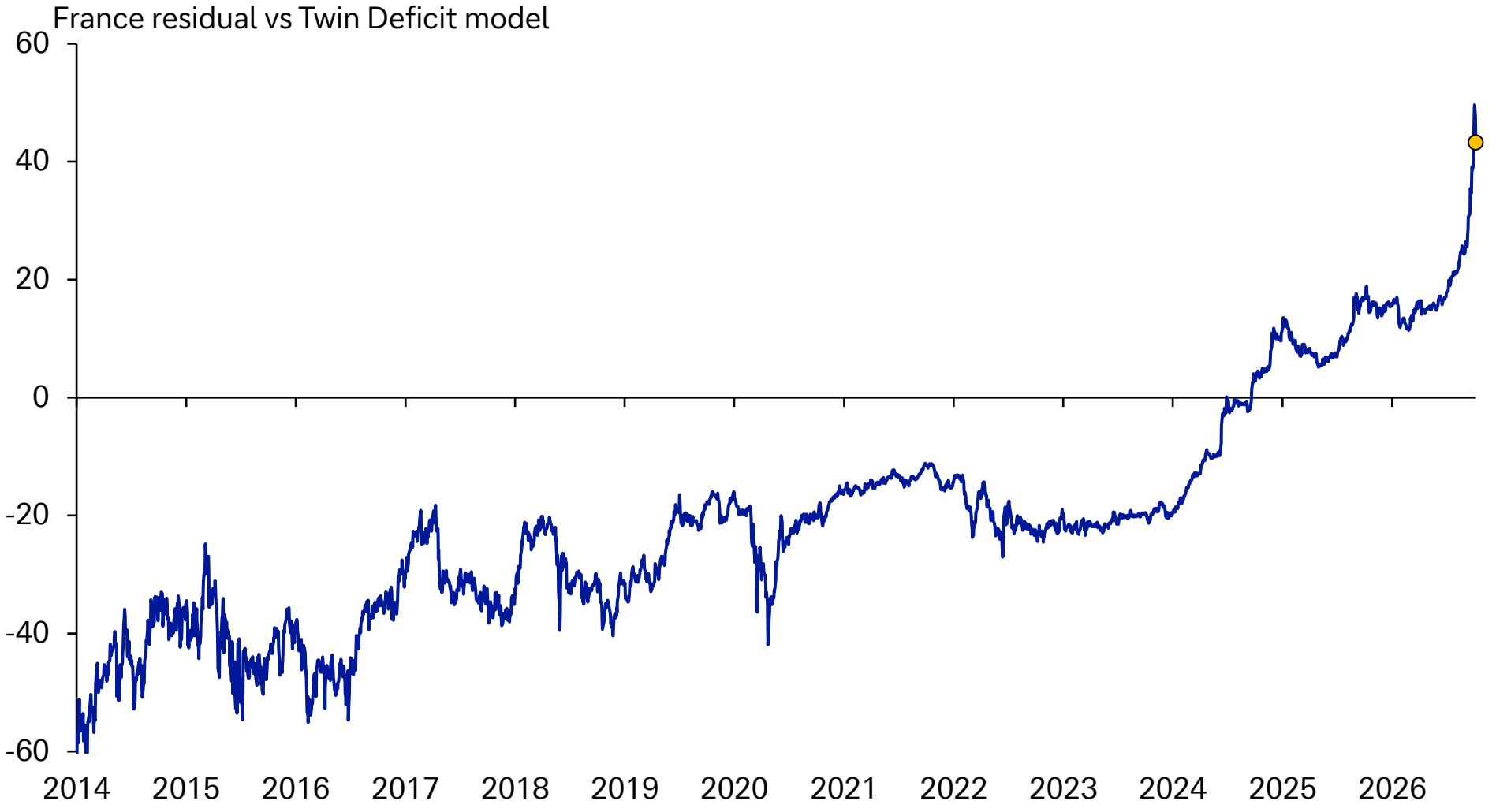
The recent underperformance is clear for equities too – and has been for over two years. France's CAC 40 has recently underperformed all the major European equity indices: the Europe-wide STOXX 600 is up over 30% in that time, and the DAX is up over 50%, yet the CAC 40 is up only 4%.

Major European Equity indices (Jan 1 2024 = 100)



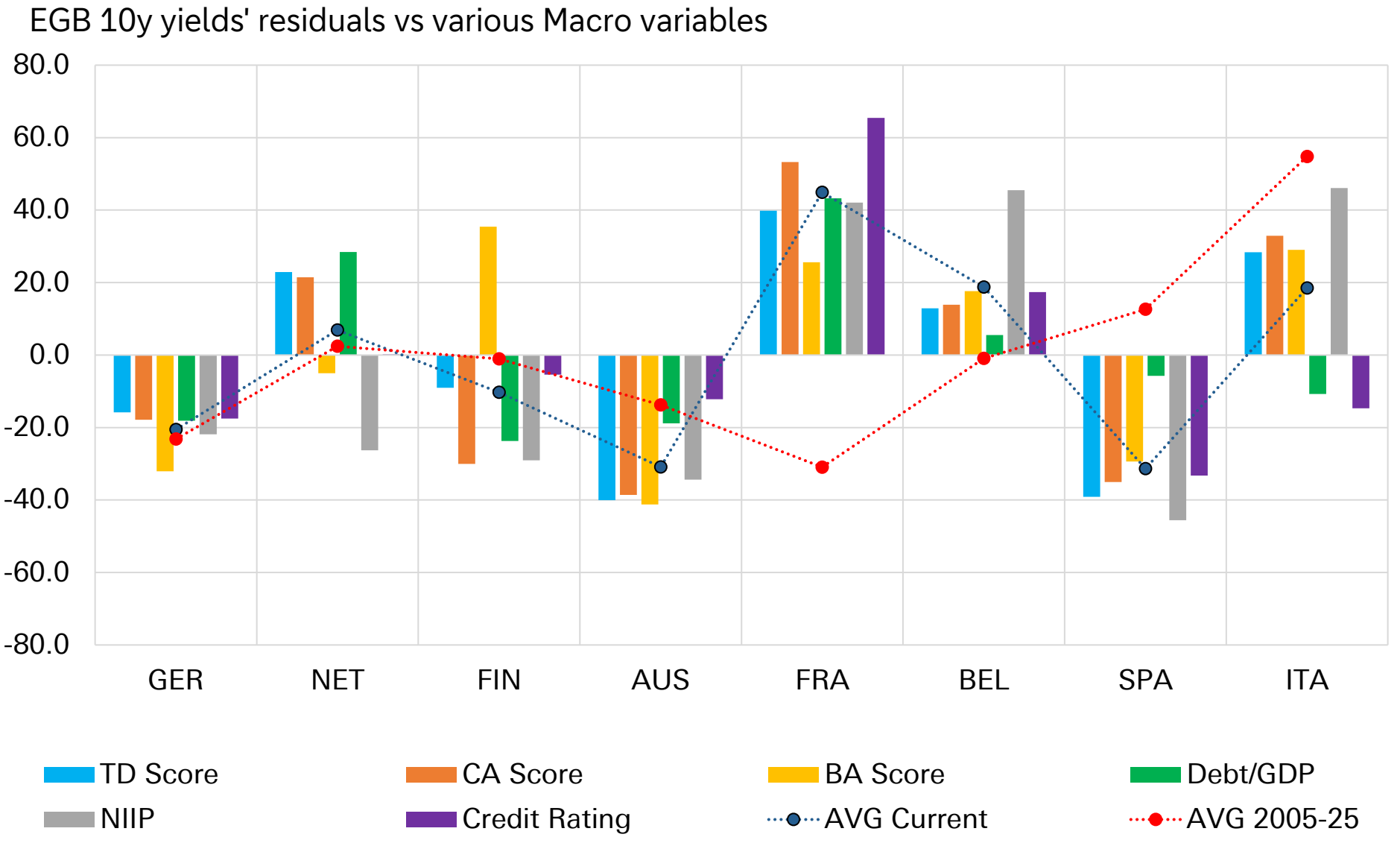

Source: Bloomberg Finance LP, Deutsche Bank

All that said, DB's rates strategists now see French bonds as cheap relative to their fundamental metrics. This chart looks at the residuals of France's twin deficit score, which is a weighted average of the historical current account balance (% of GDP) and budget balance (% of GDP), with a higher weight assigned to more recent years.




Source: Bloomberg Finance LP, Deutsche Bank

Looking at a variety of macro variables, they now see France as the cheapest European government bond relative to fundamentals. So things are challenging at the moment, and the long-term charts look worrying, but much of the sell-off in recent days/weeks has been due to a clearing-out of carry trades, implemented assuming a much lower vol period than we've seen.




Source: IMF, Bloomberg Finance LP, Deutsche Bank

# Appendix 1



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Deutsche Bank
Research

# Q3 2026 Global Markets Survey

Jim Reid | Camilla Siazon

**September 2026**

***With special thanks to Anthony Chaimowitz from dbDataInsights team***

**Deutsche Bank Research Institute**

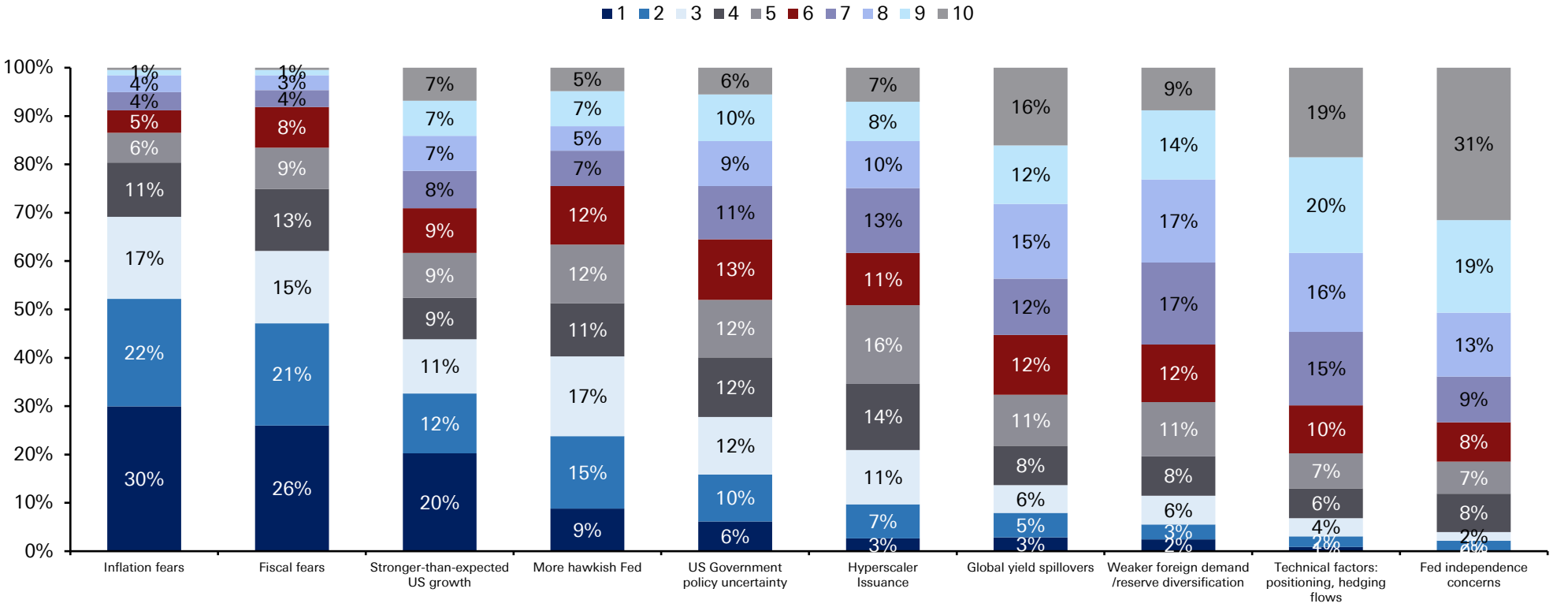
IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Our Q3 2026 global financial markets survey had 460 responses from around the world, conducted between September 28th and October 1st.

# With the ongoing major bond selloff, we asked respondents to rank what they thought were the most important drivers. Inflation, fiscal fears, and stronger-than-expected US growth were picked as the most important drivers. 30% picked inflation as the most important driver even though 10yr US Breakevens have hardly moved in the selloff.

Thinking about the recent rise in US Treasury yields, please rank the following factors in order of importance, from the most important driver to the least important. Please drag and drop the factors into the order that best represents your view of their importance.



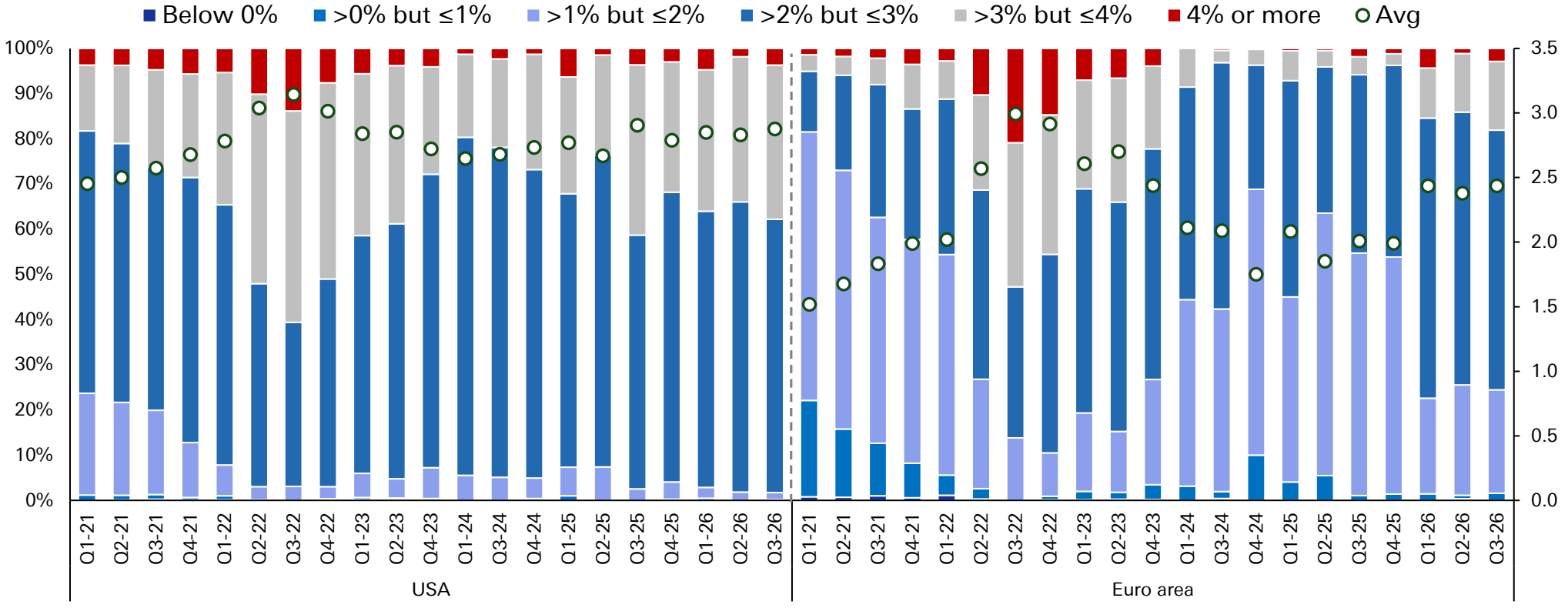

Source: dbDataInsights Survey, Deutsche Bank Research



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To highlight the inconsistency, US inflation expectations over the next 5 years have risen only marginally since we last asked in June, back up to 2.9% from 2.8%. 5yr European inflation expectations are unchanged at 2.4%. So while there is a strong narrative that inflation is causing the bond selloff, it doesn't stack up with long-term inflation expectations.

## Looking at the below, what do you think inflation will average in the following regions over the next 5 years?



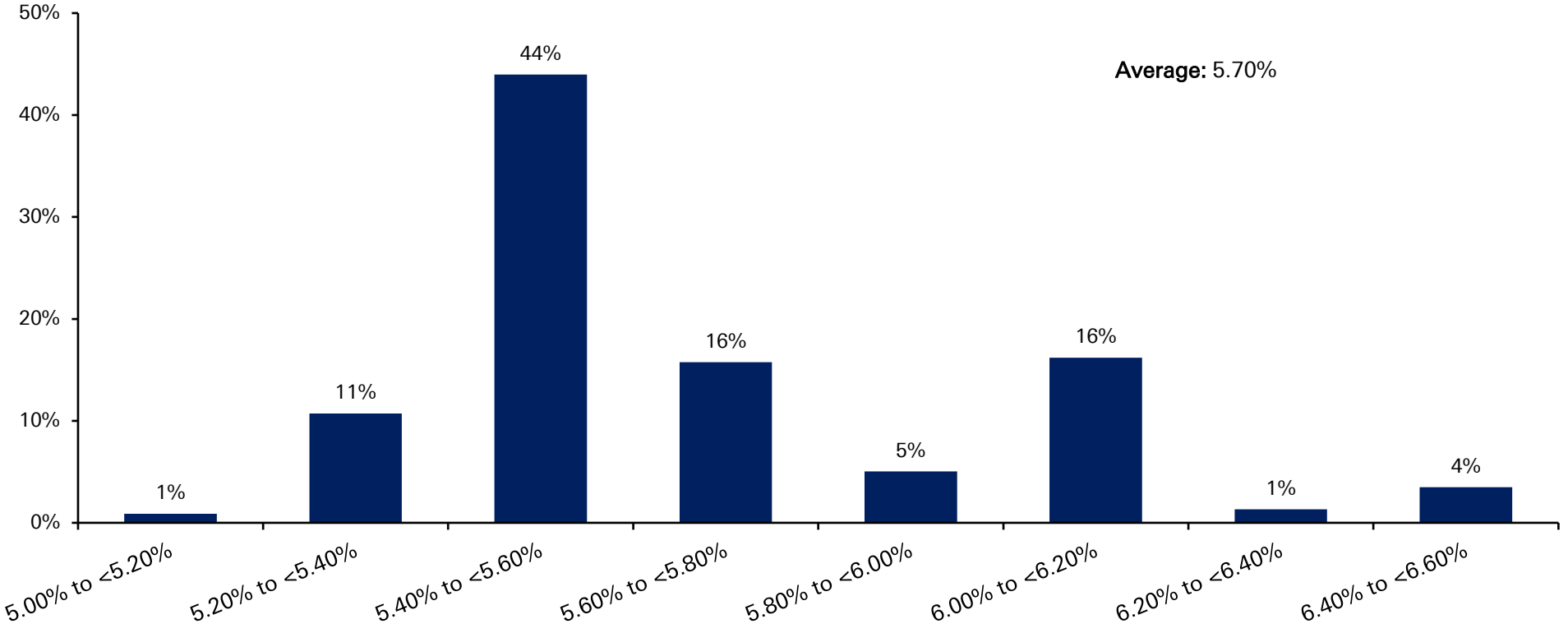

Source: dbDataInsights Survey, Deutsche Bank Research



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# Despite US Treasury yields reaching multi-decade highs in September, most respondents expect the peak in rates to be higher over the next 12 months. The average expected peak in the 10-year yield is 5.70%, with most expecting around 5.5% but with a longish tail and a secondary peak around 6%.

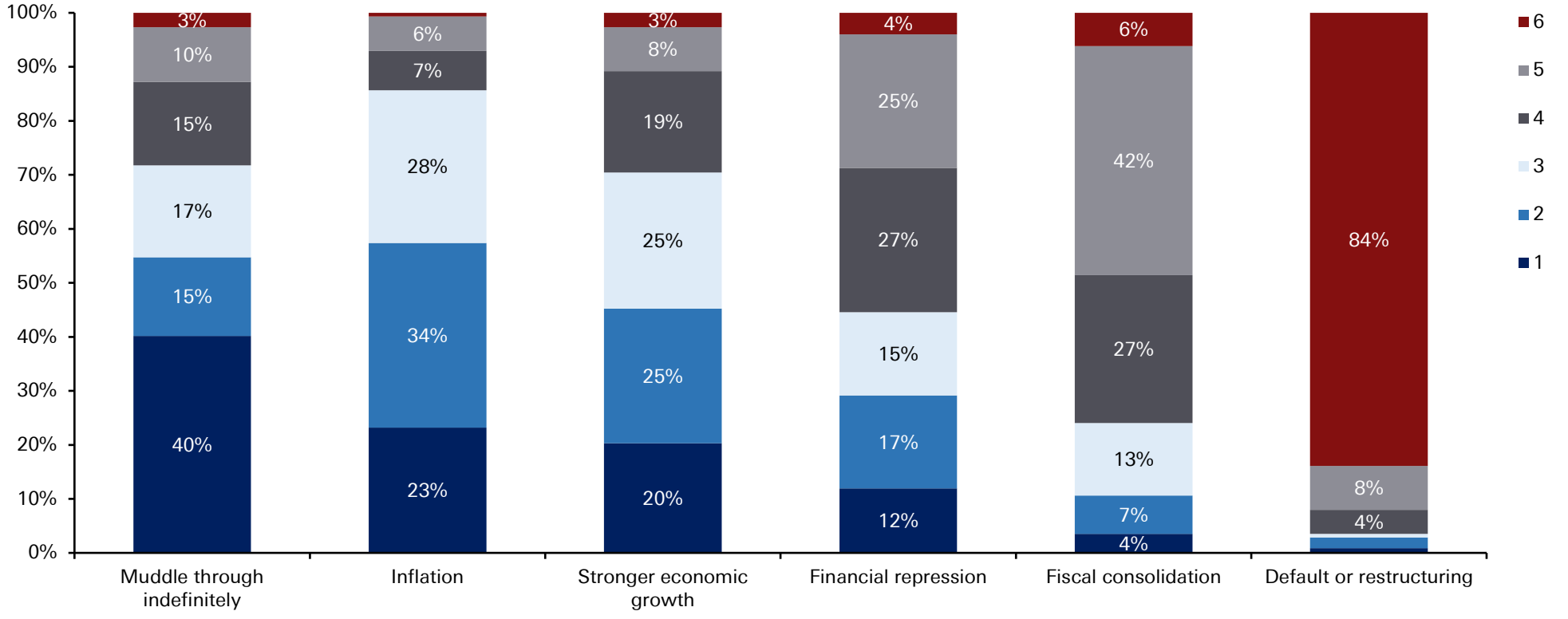
## Over the next 12 months, what do you expect will be the peak level reached by the 10-year US Treasury yield? For reference, the most recent intraday peak was 5.22%.




*Source: dbDataInsights Survey, Deutsche Bank Research*

# Interestingly, most respondents feel that the US will ultimately address its debt burden through a combination of muddling through it indefinitely, followed by inflating it away, and stronger economic growth. Fiscal consolidation is well down the list with only default lower (overwhelmingly so) as an option.

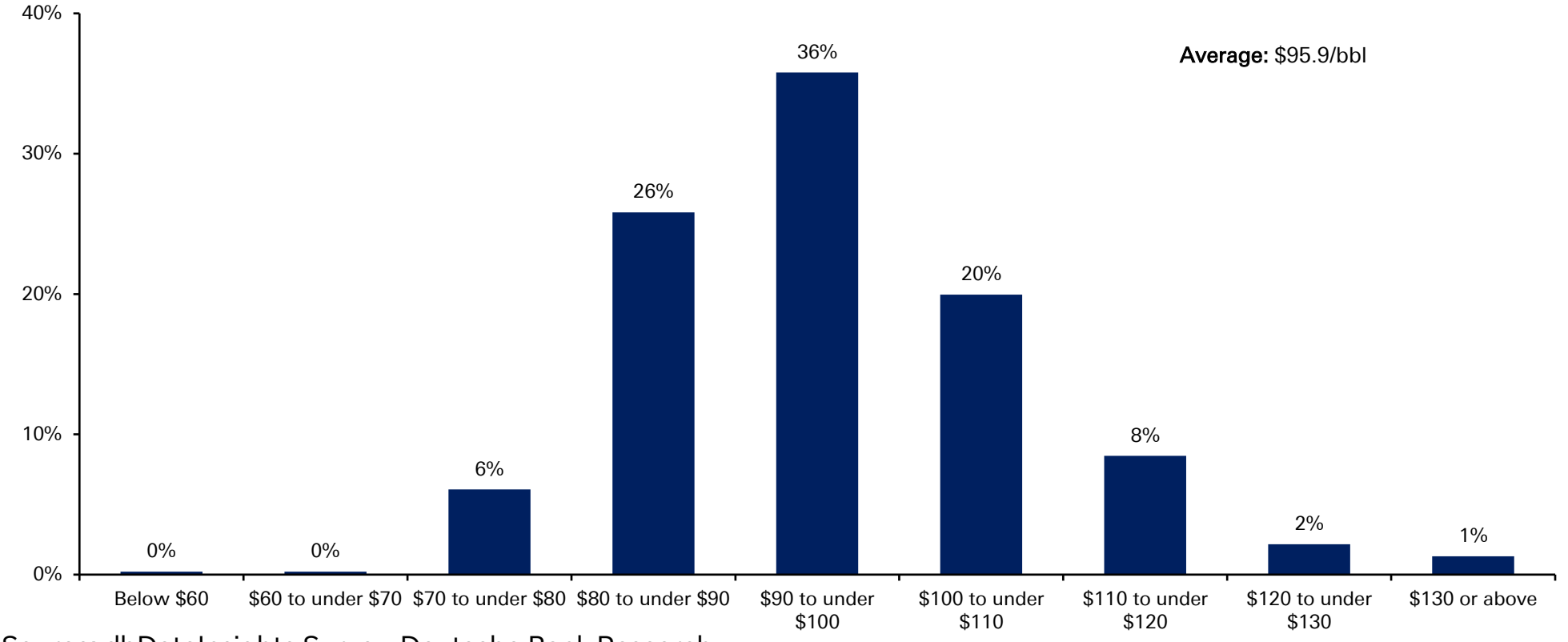
Please rank the following approaches according to how likely you think they are to be the way the US ultimately addresses its debt burden, from most likely to least likely. Drag and drop the options below, placing the most likely approach at the top and the least likely at the bottom.




Source: dbDataInsights Survey, Deutsche Bank Research

# Brent is expected to be at $95.9/bbl by the end of the year, with around 68% believing it will be below $100/bbl.

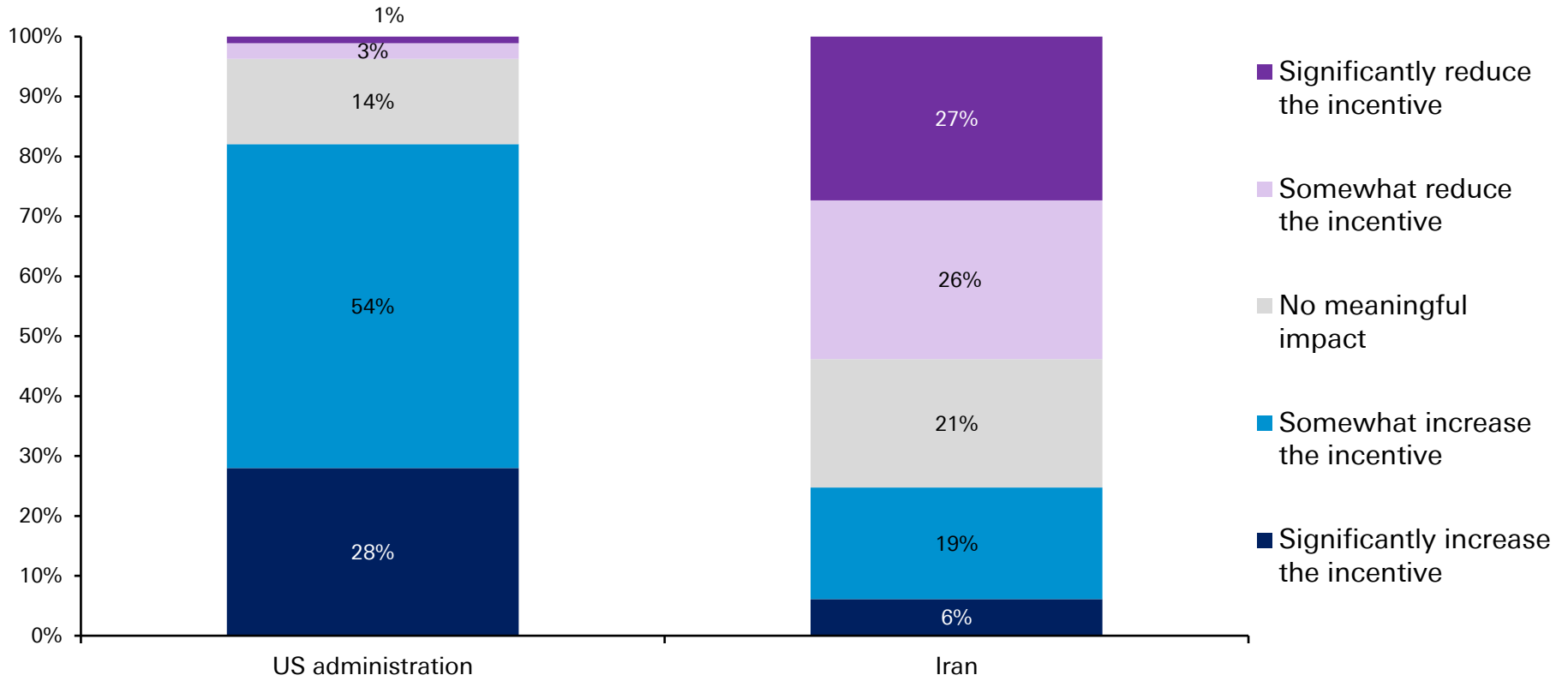
Brent crude oil prices started the year at around $60, rose to approximately $70 prior to the Iran conflict, peaked near $120, and have recently been around $105. With this in mind, where do you expect Brent crude prices to end the year?




Source: dbDataInsights Survey, Deutsche Bank Research

# Ahead of the midterm elections, respondents see stronger political incentives for a deal in Washington vs Tehran. Specifically, 82% believe the vote increases the incentives for the US to do a deal vs 54% who believe the vote reduces it for the Iranians.

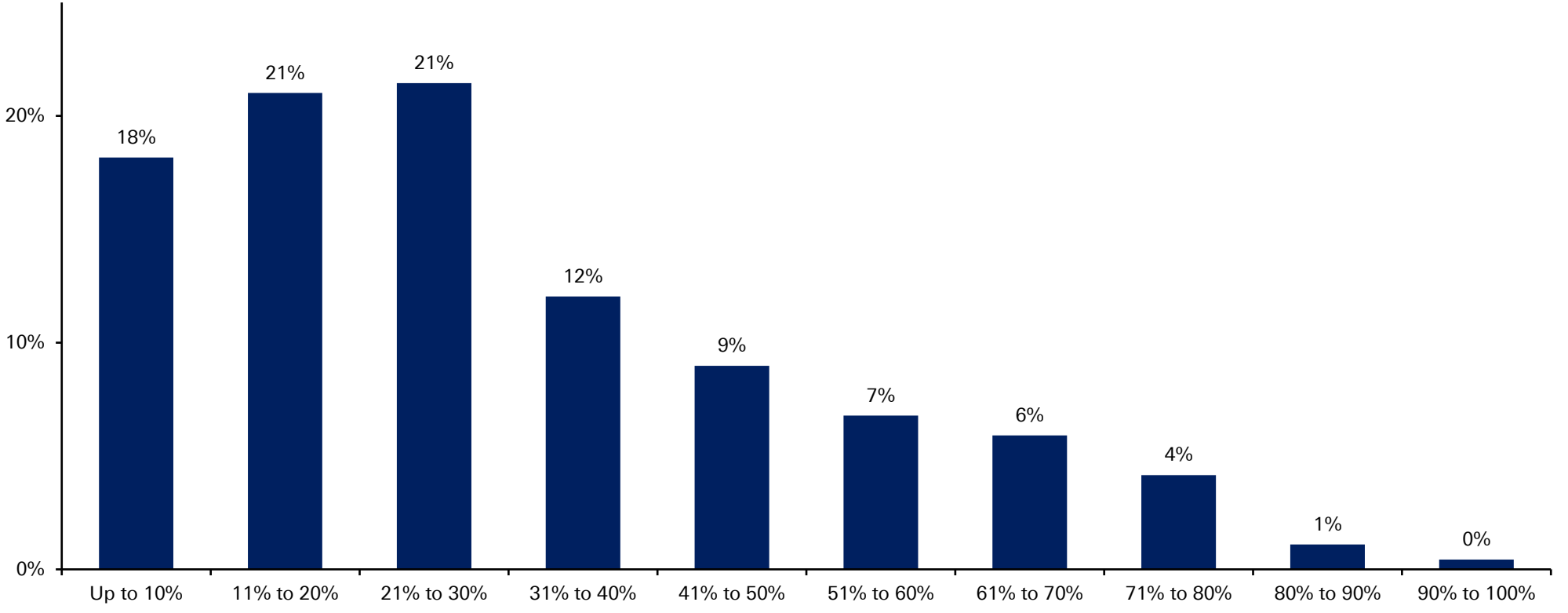
## Ahead of the US midterm elections on 3 November, to what extent do domestic political considerations increase or reduce each side's incentive to reach a deal on the Iran conflict?




Source: dbDataInsights Survey, Deutsche Bank Research

# The majority (82%) think the probability of a US-Iran ceasefire or peace deal happening before the midterms stands at 50% or less.

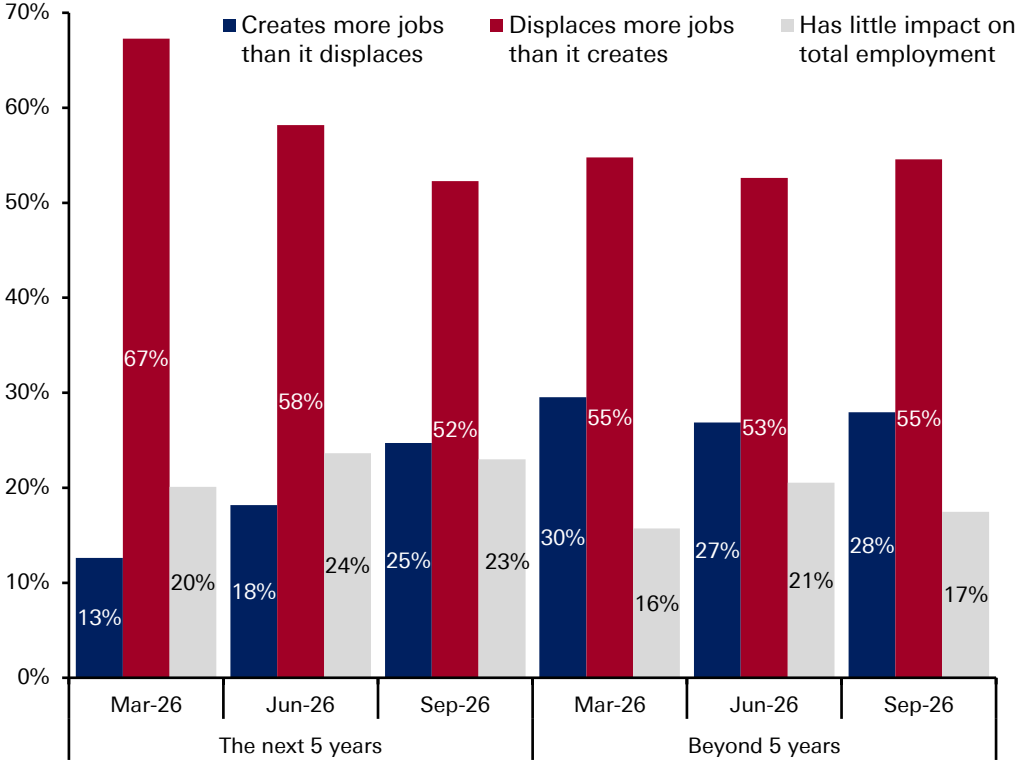
What do you think is the probability of a deal being reached, or a durable ceasefire being agreed, in the Iran conflict before the US midterm elections on 3 November? Please move the slider to indicate your best estimate. Responses will be grouped into ranges for analysis.

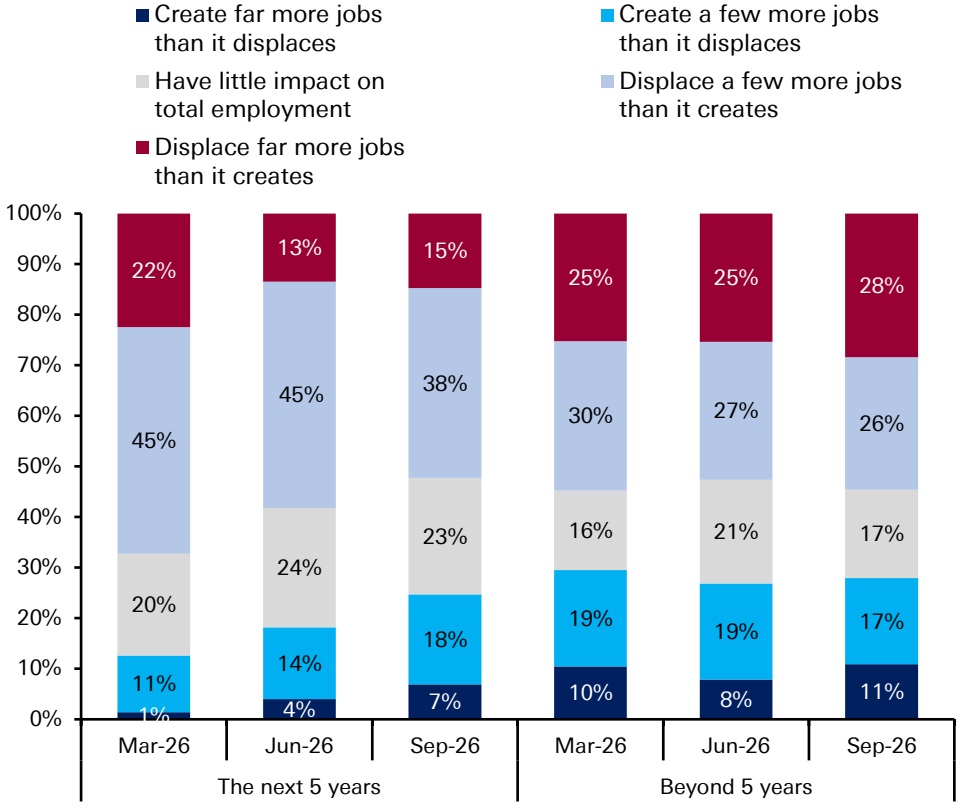



Source: dbDataInsights Survey, Deutsche Bank Research

# Near-term fears of AI-driven job displacement have eased markedly since March, but longer-term expectations remain negative and stable. The share expecting net job displacement over the next 5 years has fallen from 67% in March to 52%, while the longer-term figure remains around 55%. The first chart aggregates. The second is more granular.

## For the following time horizons, which best describes your expectation for AI's impact on employment? Assume a scenario where AI adoption stays on its current trajectory. AI will...



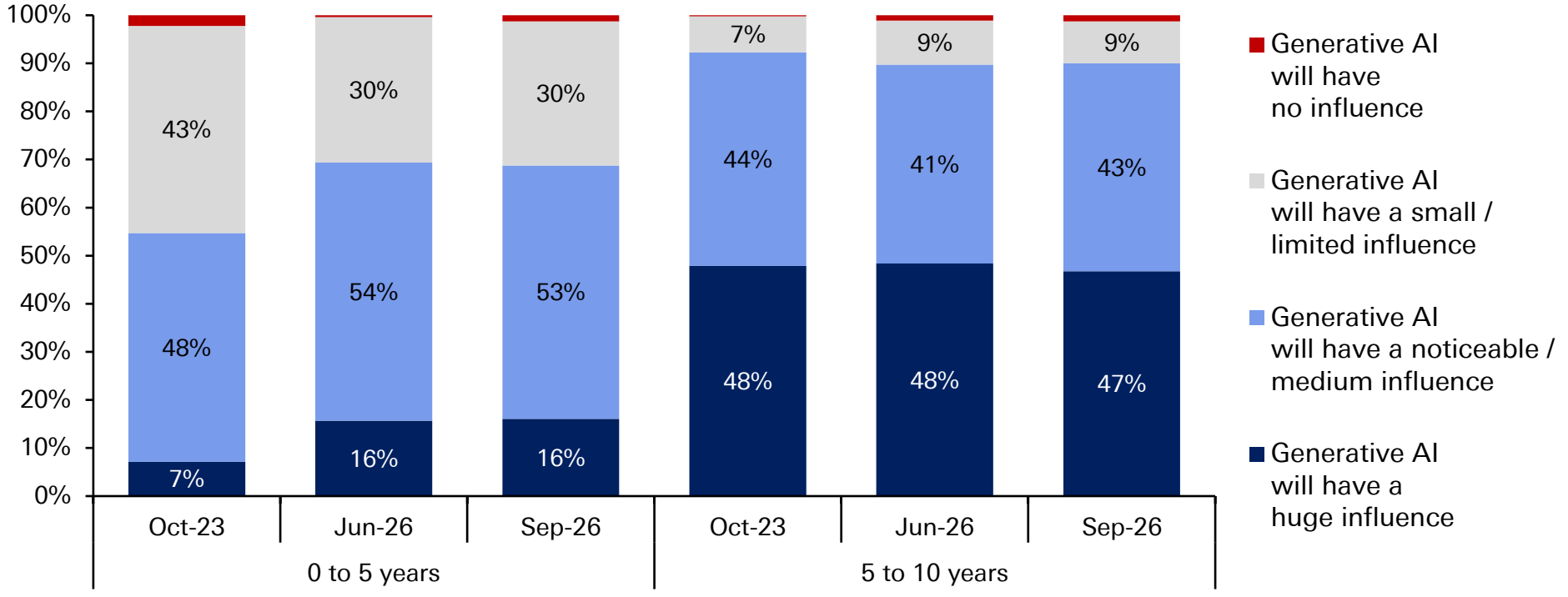

Source: dbDataInsights Survey, Deutsche Bank Research



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# GenAI productivity growth expectations seem to have stabilised, with the strongest impact still expected over the longer term. Less than 1% believe AI will have no positive impact on growth.

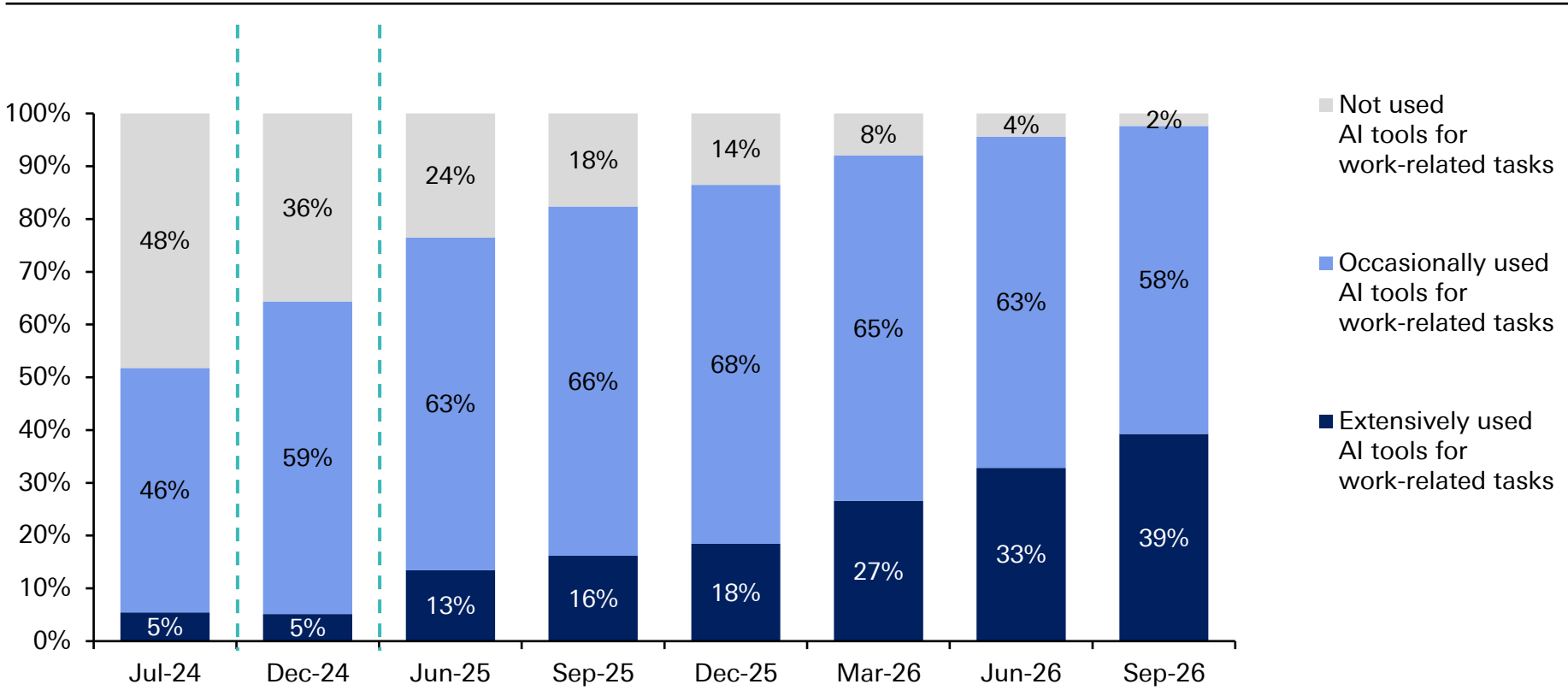
## What influence, if any, do you think future iterations of Generative AI will have on productivity growth within the following time frames...




Source: dbDataInsights Survey, Deutsche Bank Research

# AI adoption at work continues to accelerate over this year alone. Now only 2% have not used AI for work, compared to 8% in March and 14% last December. “Extensively used” has also risen from 18% to 39% since December.

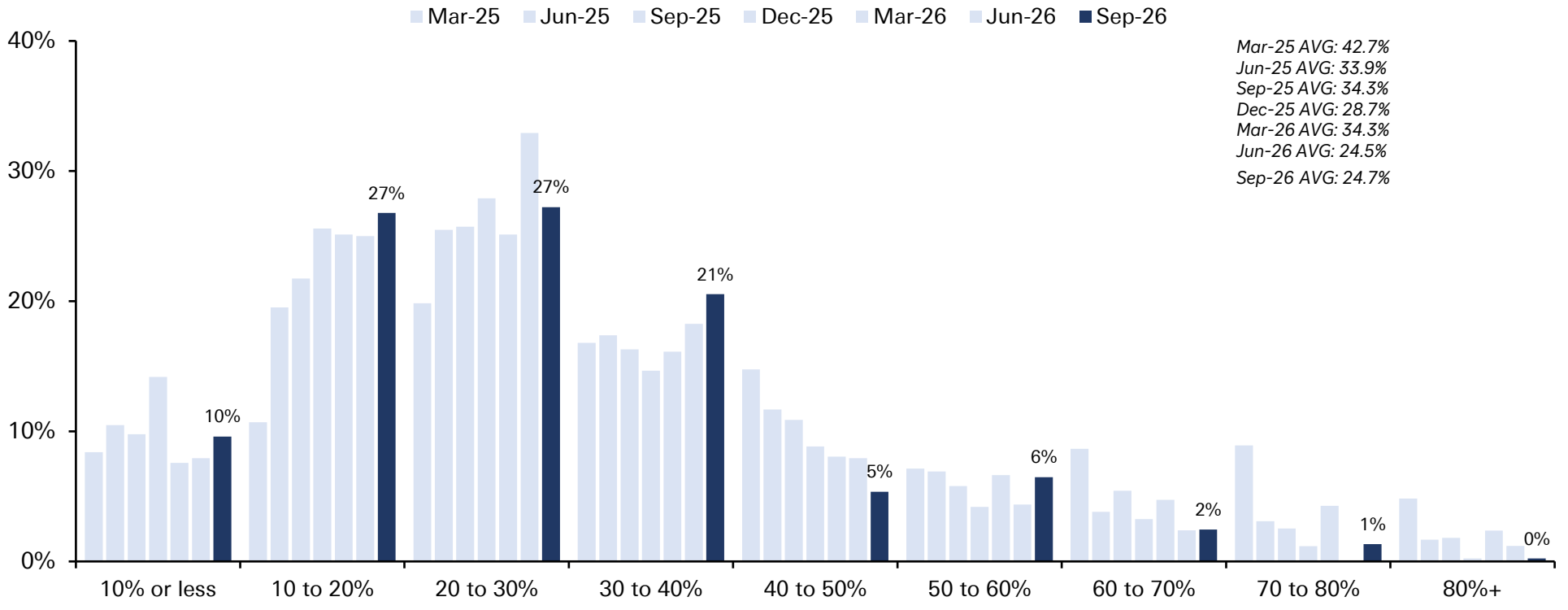
## Which of the following best describes your use of AI tools for work-related tasks within the last three months? I have...




Source: dbDataInsights Survey, Deutsche Bank Research

# Average 12-month US recession-risk expectations have edged up a fraction to 24.7% since June but remain near survey lows with most respondents still concentrated in the 10-30% probability range.

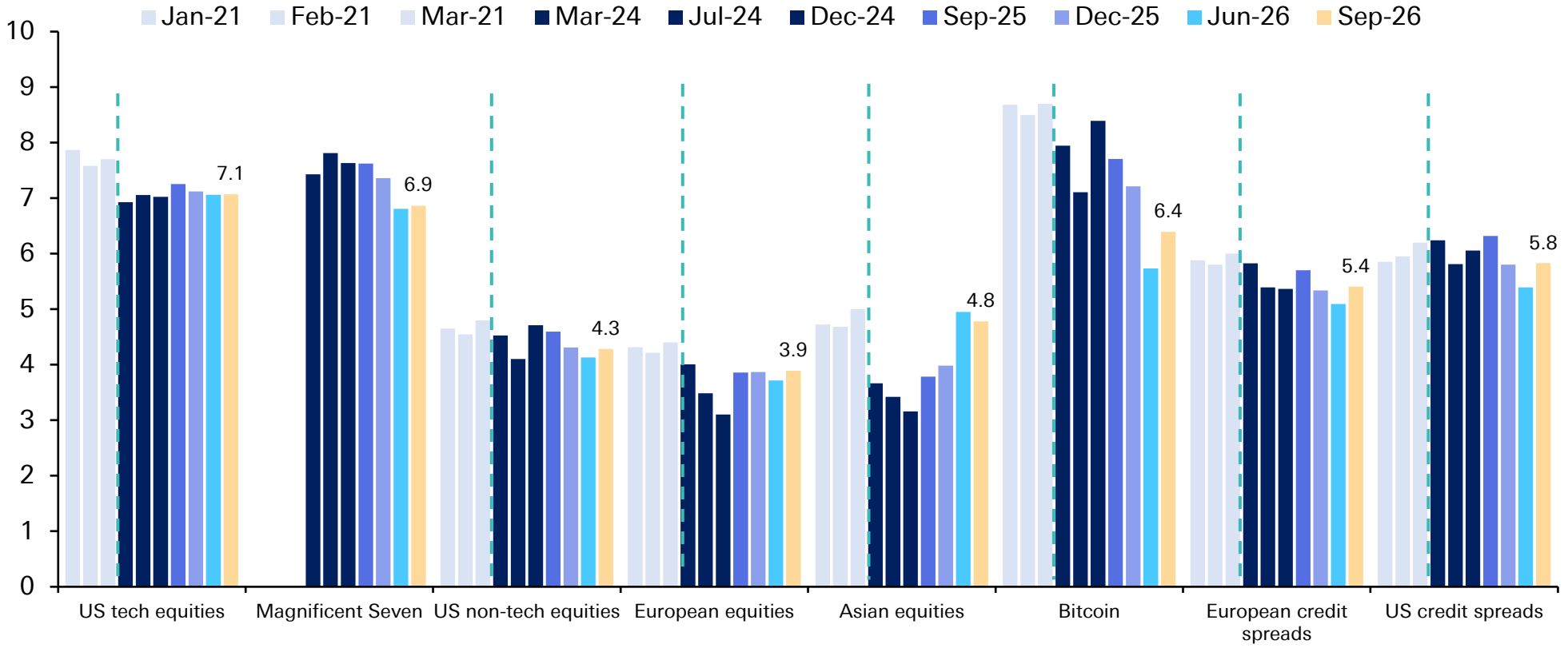
## What do you think the probability of there being a US recession in the next 12 months is?




Source: dbDataInsights Survey, Deutsche Bank Research

Bubble-risk perception has risen in several assets since we last asked this question in June. The Mag 7 has inched up to 6.9 from 6.8 previously, while Bitcoin registered a bigger jump to 6.4 from 5.7. Notably, bubble-risk perceptions for both European and US credit spreads have also risen, with the US back to its December 2025 levels. European and non-US tech equities have the lowest perceived risk.

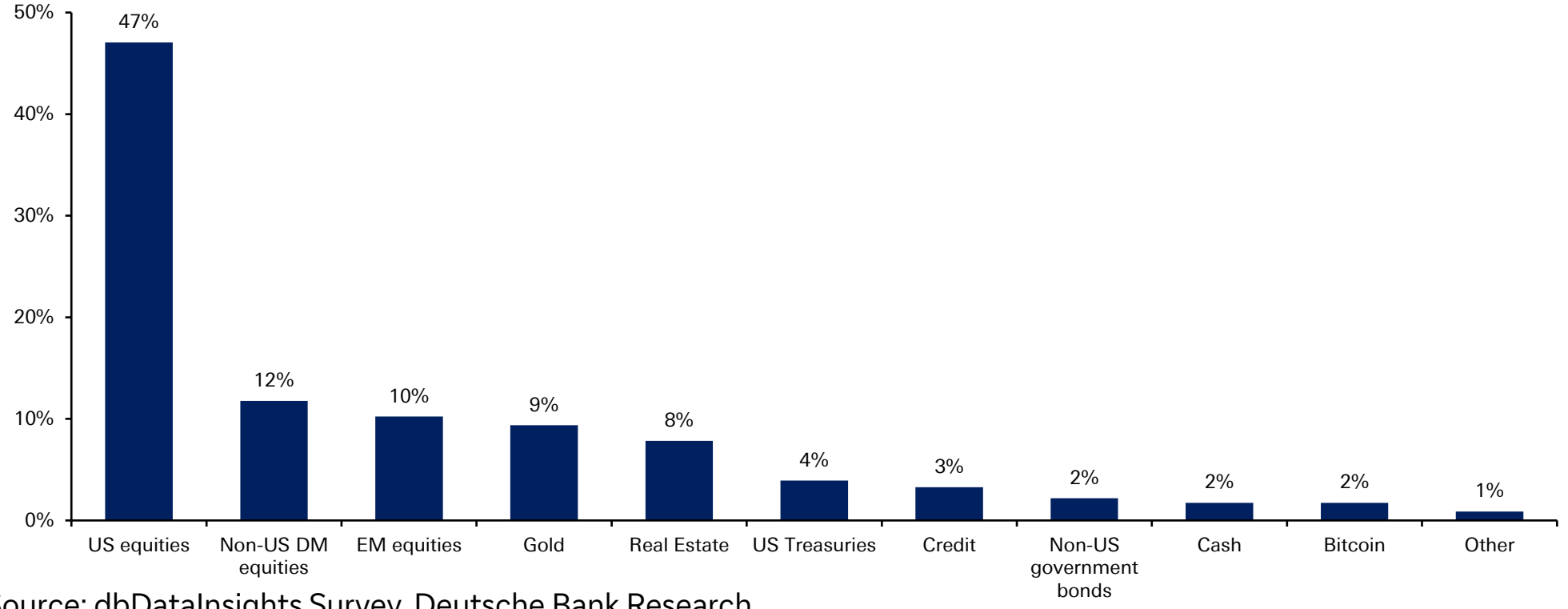
Looking at the following assets and using a scale of 0 to 10 where 0 means 'No Bubble' and 10 means 'Extreme Bubble', please tell us where you think these assets currently are?




Source: dbDataInsights Survey, Deutsche Bank Research.

# Despite the higher perceived bubble risk, nearly half of respondents would choose US equities as their sole asset for the next decade. So confidence in US equities seems to be unrivalled for long-term returns, despite market concerns of elevated valuations.

If you could hold only one of the following assets for the next 10 years, which would you choose.




Source: dbDataInsights Survey, Deutsche Bank Research



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# Appendix 1

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# This Month in Geopolitics: **October 2026**

## Key events, themes and trends to watch in the month ahead

October 2026

Dr. Helen Belopolsky and Miha Hribernik

**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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# **At a Glance:** Geopolitical events to watch in **October 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides. For an overview of key events between October and December 2026, see our latest 90-day Geopolitical Risk Radar.




Source: Deutsche Bank



Deutsche Bank Research Institute

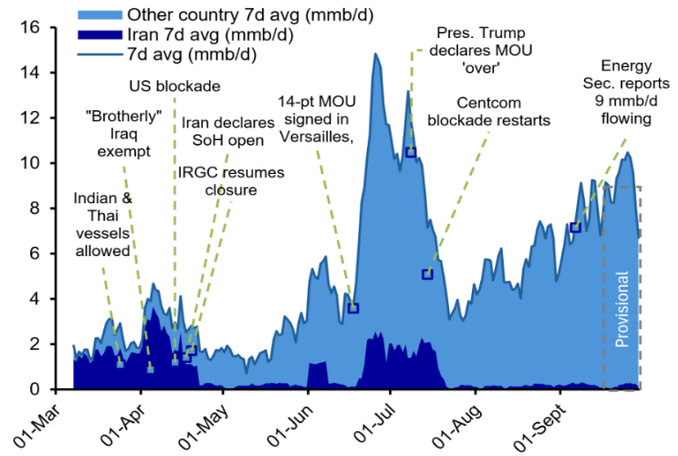
# In the spotlight for October 2026: US-Iran war and EU-China relations

## US-Iran war

**Diplomacy appears firmly back in play.** The past month saw continued fighting and renewed diplomacy. Reporting suggests that the US is prepared to extend troop deployments in the Middle East to 2027. Iran targeted both US forces in the region and some tankers moving through the Strait, but has been unable to prevent a significant rebound in crude flows through Hormuz. On 22 SEP, talks were mediated by Qatar, Pakistan and Egypt, and on 24 SEP, Iran presented the US with a seven-day roadmap to a ceasefire leading to a phased reopening of the Strait of Hormuz and a return to nuclear negotiations, conditional on a US lifting of its blockade of Iran, sanctions relief and broader cessation of hostilities. The Trump Administration rejected the proposal, but negotiations continue.

To watch will be whether October is the month the war transitions into a negotiated settlement, which may hinge on whether Iran is willing to make some concessions on sanctions relief and the nuclear question, and whether Trump is keen to demonstrate that the war has produced a political and economic result in the final weeks before the 3 NOV midterm elections. It is plausible that the current diplomatic channel could produce a temporary de-escalation mechanism as long as there is pathway around the broader nuclear issue.

Strait of Hormuz total liquids exports (mmb/d)




Source: Kpler, Deutsche Bank Research.

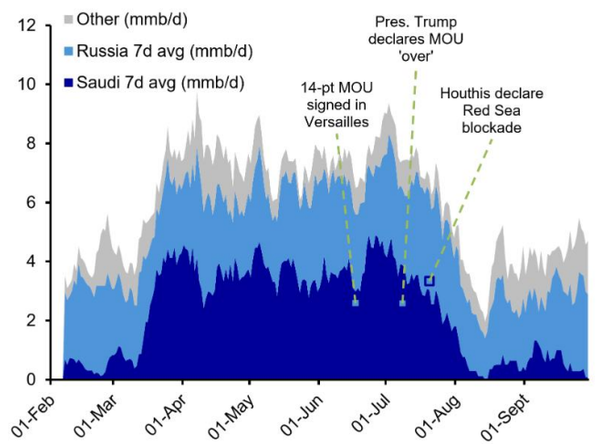
## EU-China trade relations

**Both sides will reconvene for trade talks in Beijing on 8-9 October, which the EU views as a deadline for "tangible results".** Brussels remains focused on its rising trade deficit with China – which has reached a record €1 billion per day – and the impact of Chinese goods on European industry. The bloc has vowed to take measures if talks fail.

While both sides could achieve progress in talks, a lasting breakthrough appears improbable, setting the stage for more trade tensions into 2027. New EU trade investigations, curbs on Chinese imports, and EU-level instruments intended to address alleged Chinese overcapacity, are possible outcomes, to which Beijing has stated it would respond.

China-EU trade talks could also impact China-US relations. Should talks with Brussels fall short of averting increased European trade pressure, we assess that Beijing may further prioritise near-term US stability. See Rules of Engagement: Mapping the Next Phase of China-US Ties for more.

Bab el Mandeb total liquids exports (mmb/d)




Source: Kpler, Deutsche Bank Research. For more, see: Hsueh On Oil: Success of the shuttle system - Deutsche Bank Research

# Key events, issues and themes to watch in **October 2026**

Listed in chronological order

## Latvian elections (3 October)

**Latvia will elect all 100 members of the 15th Saeima.** Latvia, which borders Russia and Belarus, has been among the most vocal supporters of Ukraine. The next government will set the country's course on defence spending, Ukraine aid, sanctions enforcement and border security. The field is highly fragmented with the governing parties lacking a majority in the polls. The fragmentation matters as a weakened and multi-party scramble could mean a slower, more unstable coalition.

## General election in Brazil (4 October)

**Brazil's first electoral round is on 4 OCT, with a runoff on 25 OCT.** This is a critical election as it will decide whether Latin America's biggest economy continues President Lula's state-led social spending or pivots to a more fiscally-orthodox agenda under Bolsonaro. Polls have gone from a Lula lead in the spring to a statistical dead heat between Lula and Bolsonaro. A Brazilian Supreme Court crisis – which has implicated Supreme Court Justices and politicians on both sides – is weighing on the race. Congressional and gubernatorial races will also impact the governability of the country.

This election will also shape whether the country hedges between Washington and Beijing under Lula or tilts toward closer US trade and security alignment under Bolsonaro with potential implications for tariffs, BRICS+ cohesion, and Latin America's broader positioning between the two powers.

## Annual meetings of the International Monetary Fund and World Bank (12-18 October)

**The meetings will take place in Bangkok, bringing finance ministers, central bankers, policy-makers and business leaders together.** The agenda is expected to centre on global growth, financial stability, debt, trade and investment, development finance and the impact of AI on jobs and productivity, with particular attention given to mobilizing private capital. The key watchpoint will be the outlook for global growth and inflation, debt vulnerabilities, the future of the international financial architecture and how policy-makers respond to increasing fragmentation in trade and capital flows.



Source: Waranont (Joe) on Unsplash

# Key events, issues and themes to watch in **October 2026**

Listed in chronological order

## European Council meeting (15-16 October)

**Leaders are expected to focus heavily on Ukraine, European security and defence, competitiveness and the EU's economic position in a more contested global environment.** They will also take forward negotiations on the 2028-2034 EU long-term budget. They set themselves a soft deadline in June to get the budget broadly locked down because missing it pushes ratification into 2027 and risks a funding gap for beneficiaries from JAN 2028. To watch will be a genuine breakthrough on the multiannual framework or more likely what "substantial" progress may look like. Ukraine funding is the other pressure point. This meeting will be important for understanding the direction of European defence spending and industrial policy, the next phase of budget negotiations and the EU response to growing external economic pressure and strategic dependencies.

## Alberta independence referendum (19 October)

**The Canadian province of Alberta will hold a referendum, which includes a question about the province's future within Canada.** This is not an independence vote itself but asking voters to choose between remaining a Canadian province and asking the Alberta government to begin the constitutional process required for a future binding referendum on separation. Current polling suggests "remain" is comfortably ahead. The significance is around potential renewed uncertainty around Canada's federal cohesion, energy policy, fiscal transfers and Alberta's relationship with Ottawa and Washington. Even a modest agreement to begin a process towards secession would likely keep alive equalization grievances as a structural fault line in Canadian federalism feeding into political risk.

## Berlin Global Dialogue (20-22 October)

**The event's theme focuses on "Advancing Alliances in a New Era."** It will tackle how governments and businesses respond to a world where geopolitics, economic security, supply chains and technology are increasingly intertwined. There is likely to be a strong focus on the future of the transatlantic relationship, including how Europe adapts to a more demanding security environment. There will also be discussions of tariffs, supply chain resilience, critical capabilities and the weaponization of economic interdependence. It will align with Chancellor Merz's efforts to ensure Europe actively shapes outcomes around strategic autonomy and reducing China exposure in manufacturing.



Source: Bernd Dittrich on Unsplash

# Key events, issues and themes to watch in **October 2026**

Listed in chronological order

## Summit between the EU and the Gulf Cooperation Council (24 October)

**The second summit will take place in Saudi Arabi, bringing EU and Gulf leaders together.** The agenda is expected to focus on regional security and the Middle East, energy and economic security, trade and investment, infrastructure, technology and broader strategic cooperation. The security dimension will be particularly salient given the current crisis in the region. Economically, the relationship is substantial: EU-GCC trade reached €165.6bn in 2025.

To watch will be movement on the long-stalled EU-GCC free trade talks and visa-waiver requests from Qatar and Saudi Arabia in addition to the extent of the focus on Gaza reconstruction, Iran and regional security get relative to the economic agenda.

GCC states have consistently used these summits to press for clearer EU positioning on Middle East conflicts in exchange for economic cooperation. The summit comes amid at a critical juncture where Europe is recalibrating its external partnerships.

## Fifth Plenum of China's Central Committee (26-29 October)

**The Central Committee of the Communist Party of China will hold its Fifth Plenary Session in Beijing.** This will be one of the most important events on the country's political calendar ahead of the 21st Party Congress in late 2027.

The Plenum will focus on Party governance, and investors will scrutinise signals regarding leadership changes and personnel reshuffles, institutional reforms, and political priorities for the next Congress. While fifth plenums historically focus on economic planning, the development guidelines for the 15th Five-Year Plan (2026–2030) were already adopted during the Fourth Plenum in OCT 2025, allowing the upcoming session to focus on personnel and governance matters.




Source: istockphoto

# Key events, issues and themes to watch in **October 2026**

Listed in chronological order

## Legislative election in Israel (27 October)

**Elections will be held for the 26th Knesset, the first since the October 7th attacks and wars in Gaza, Lebanon and Iran.** This is the first Knesset since 1988 to actually complete a full four-year term without early dissolution. Benjamim Netanyahu remains the dominant single figure but faces a fragmented field: Naftali Bennett's new "Together" party is polling as the main rival bloc, alongside Gadi Eisenkot's Yashar, Benny Gantz's Blue and White, and the further right Ben Gvir and Smotrich factions.

Israel's proportional system means the election is effectively about coalition formation. Key faultlines to watch: the ultra-Orthodox draft-exemption fight, how the war's legacy and hostage/reconstruction issues in Gaza factor into campaign messaging, and whether a Netanyahu-led bloc can reassemble a working majority or whether the Bennett/Eisenkot-style "anyone but Bibi" coalitions gain enough traction. The outcome will shape Israel's posture on critical regional issues, including Gaza reconstruction, Iran policy and West Bank settlement activity.

## EU-Canada Summit (29-30 October)

**This summit is shaping up as a particularly important juncture against a more ambitious push to redefine the relationship.** During Mark Carney's SEP visit to Strasbourg, he called for a comprehensive strategic alliance with Europe. European Commission President von der Leyen went further, proposing that Canada become the EU's first "associate member".

The agenda is expected to centre on trade and economic security, defense, critical minerals, energy, digital cooperation and supply-chain resilience, building on the EU-Canada security and Defense Partnership in the EU's SAFE defense initiative. The summit will be watched for concrete progress on a digital trade agreement and broader economic cooperation.



Source: istockphoto

# Key events, issues and themes to watch in **October 2026**

Listed in chronological order

## G20 Foreign Ministers' Meeting (30-31 October)

**The foreign ministers' meeting comes ahead of the December Leaders' Summit in Miami under the US' chairmanship of the G20 this year.** Trump's G20 year has organized around four expert tracks – trade, innovation, energy and deregulation. The session is framed around economic prosperity, energy security and technological innovation, a much more transactional focus than previous foreign ministerials, which typically focused on geopolitical crisis management.

It will be a test of whether the G20 can sustain cooperation amid intensifying geopolitical fragmentation. In addition, Russia has confirmed its intent to actively participate in the meeting. This would further normalize Russia's multilateral engagement. For Europe and other partners, a key question will be the extent to which the G20 can be a forum for broader international cooperation.



Source: Lance Asper on Unsplash

# Appendix 1

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Deutsche Bank
Research

# Q3 Tech Performance Review:
## AI euphoria meets macro headwinds

October 1, 2026

**Marion Laboure | Camilla Siazon**

**Deutsche Bank Research Institute**

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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China and US agree to establish AI safety channel and continue trade and military talks
Chip Stocks Tumble After AI Leaders Call for Slowdown in Development
South Korea $2 trillion stock rout breaks records as SK Hynix results disappoint
Meta unveils AI personal assistant linked to WhatsApp and Instagram
Deutsche Bank
Research Institute

# SUMMARY: AI euphoria meets macro headwinds

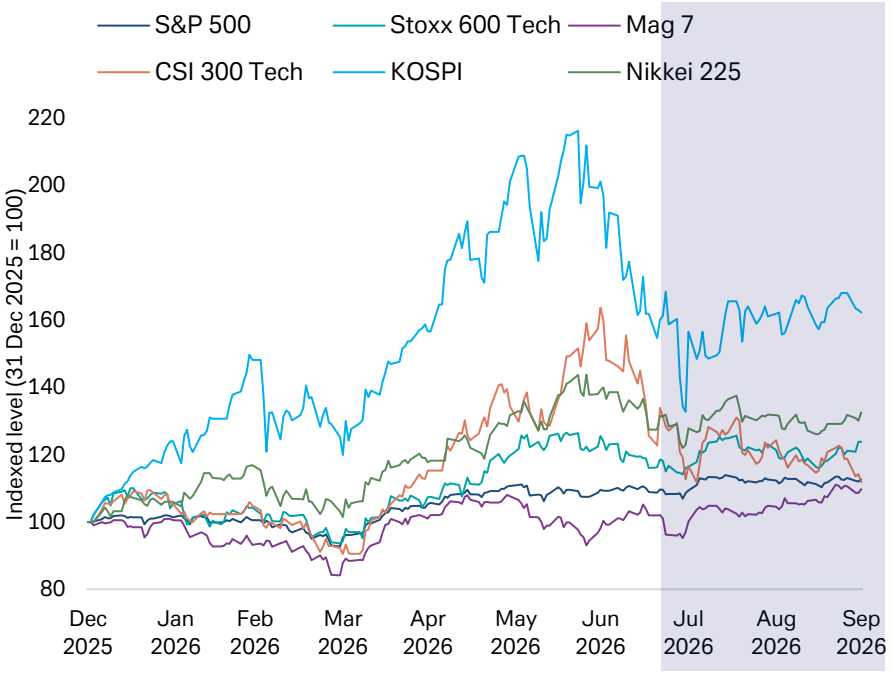
Tech remained a major driver of equity market performance in Q3. The Mag 7 outperformed the broader market while the S&P 500 also posted gains, lifted by mega-cap and tech stocks.

Still, tech's path in Q3 was volatile. July began with a reversal of the AI trade as investors questioned elevated valuations, the sustainability of AI spending and the outlook for semiconductor demand. That led to a large sell-off in chip stocks, with large falls in Q2 outperformers SK Hynix and Samsung contributing to a sharp decline in Korea's KOSPI. The Philly SOX experienced its worst monthly performance since 2008.

By August, however, sentiment had improved. The S&P 500 reached fresh record highs, even as US Treasury yields rose, supported by stronger cloud earnings and signs of AI monetization. September proved more challenging as higher oil prices, inflation concerns and rising bond yields weighed on risk assets, particularly semiconductors. The Fed also raised its policy rate by 25bp on Sep 16.

Yet, despite these macro headwinds, company guidance continued to indicate strong AI infrastructure demand, with Micron's stronger-than-expected revenue guidance on Sep 30 providing further evidence of robust demand for memory chips. Memory supply remains tight, semiconductor capex is accelerating, and the industry's attention has increasingly turned to bottlenecks in optical networking and photonics, as well as concerns around AI regulation and safety. Looking ahead, tech is likely to remain an important driver of equity returns into Q4, although elevated valuations and tighter financial conditions leave the sector sensitive to earnings disappointments.

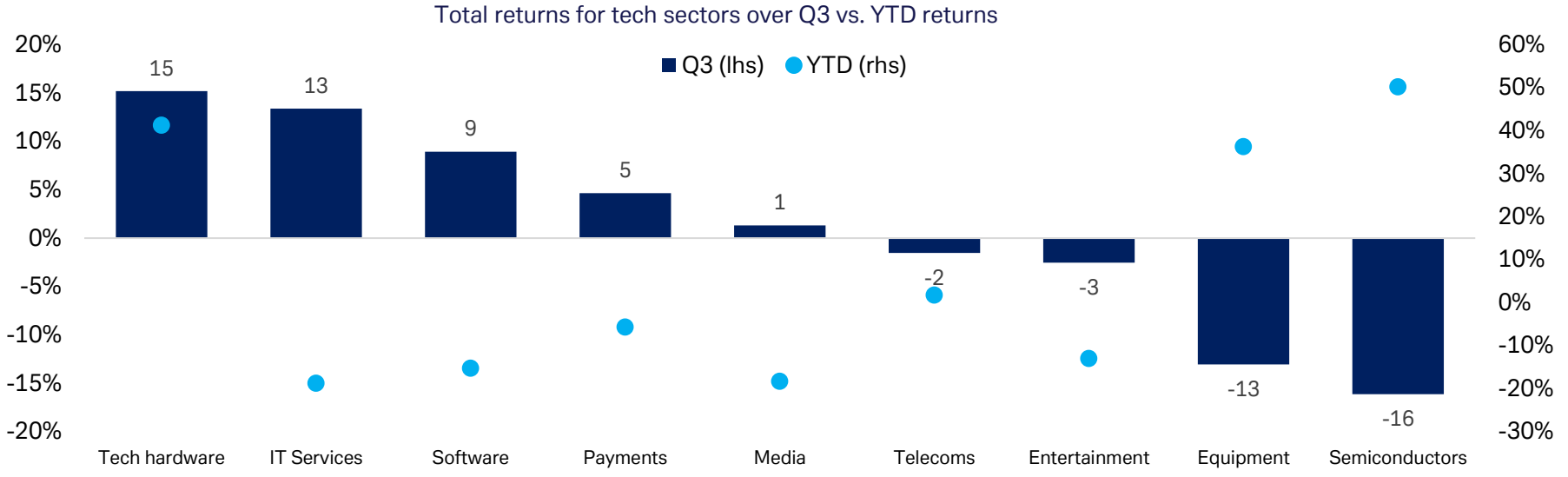
Q3 started with doubts about the AI trade but ended with renewed confidence




Sources: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30. Returns in local currency.

# 1. GLOBAL INDUSTRY BREAKDOWN

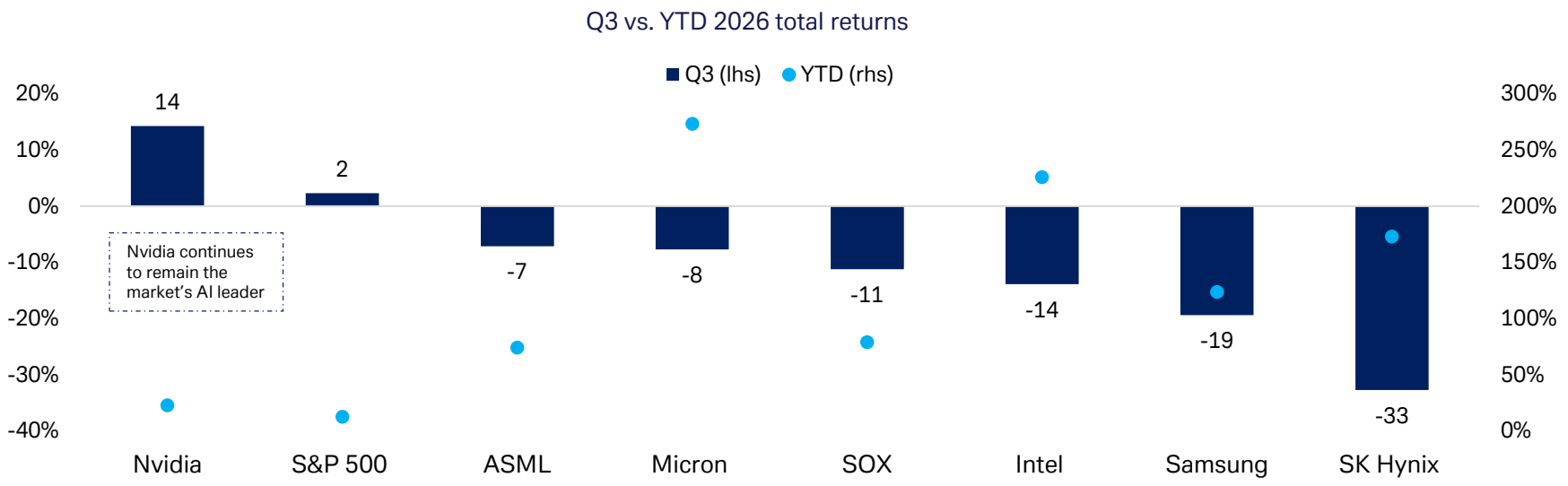
Tech hardware, including Lenovo, Advantech, and HP Inc, rebounded in Q3. This was partly driven by accelerating AI server demand and the ongoing data center build-out, although continuing memory shortages raised costs for hardware manufacturers. Cloud and server providers, like Oracle, Dell and HPE, also highlighted strong AI-related demand. Semis and equipment were the weakest performers. Elsewhere, entertainment and telecoms posted modest declines. Finally, in Q3 investor attention increasingly shifted to AI agents, with Meta's Muse, Microsoft's Copilot and Salesforce's Agentforce among the prominent offerings.




Source: Deutsche Bank Research, Bloomberg Finance LP. Median returns taken. Updated Sep 30.

# 2. SEMICONDUCTORS

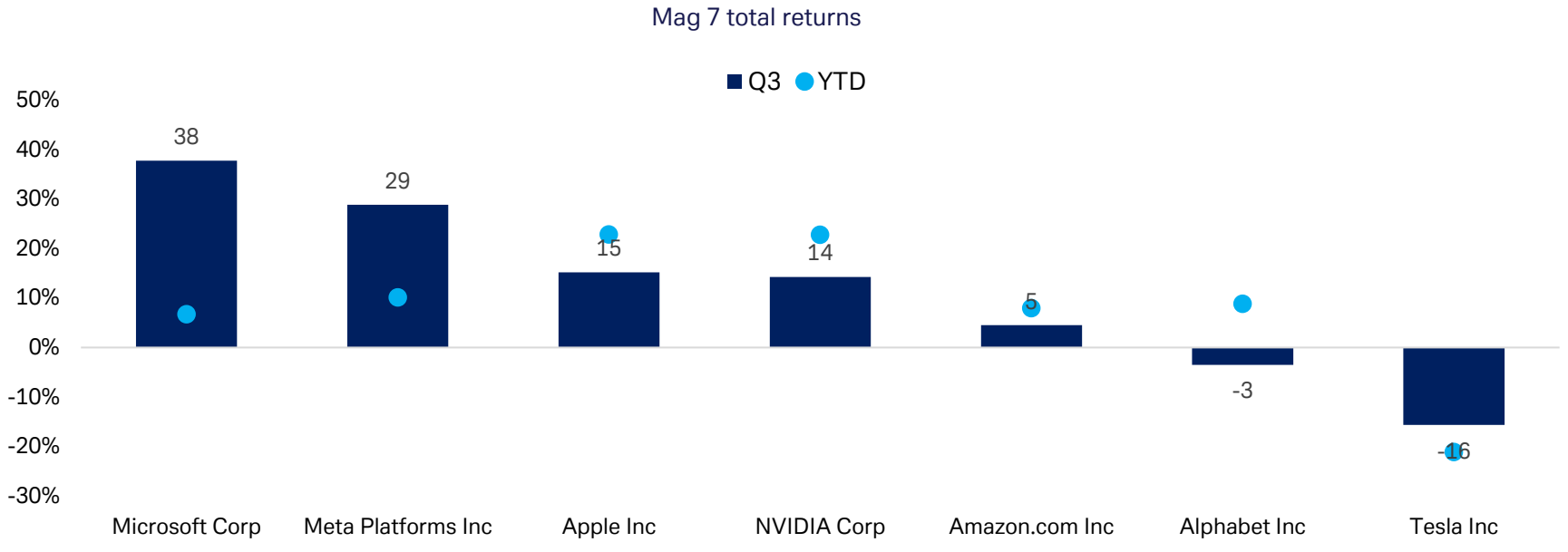
Chip stocks were hurt initially by a reversal of the AI trade and then by rising bond yields. The Philly SOX index fell sharply in Q3, after its worst monthly performance in July (-21%) since 2008. Concerns about higher oil prices and rising rates continued to weigh on the sector in Aug and Sep. SK Hynix was among the weakest performers, amid an earnings disappointment relative to expectations and an unwinding of leveraged positions. Its Nasdaq ADRs began trading in July. Despite these headwinds, the semis outlook remains supported by AI infrastructure demand. Within semis, potential long-term beneficiaries include suppliers of AI infrastructure, memory including high-bandwidth memory (HBM), advanced packaging, photonics and semiconductor equipment.




Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30. Returns in local currency.

# 3. MAGNIFICENT 7

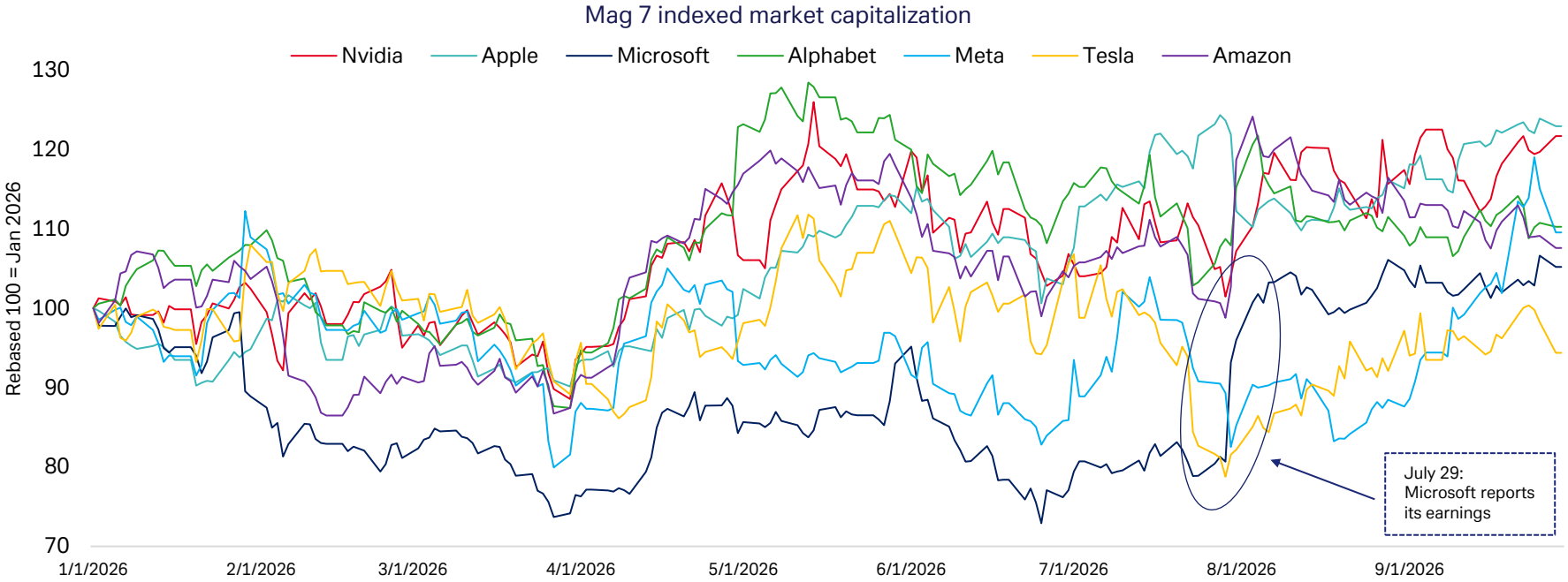
Microsoft led the group in Q3 after its Azure growth materially exceeded expectations, with strong Copilot adoption and guidance for continued positive free cash flow despite substantial capital spending. Meta also became a major focus in Sep following the launch of Muse. Amazon's AWS growth also accelerated, underscoring continued cloud demand. Tesla underperformed and remains the group's worst-performing stock YTD.




Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

# 3.1 MAGNIFICENT 7: Microsoft's earnings rebound

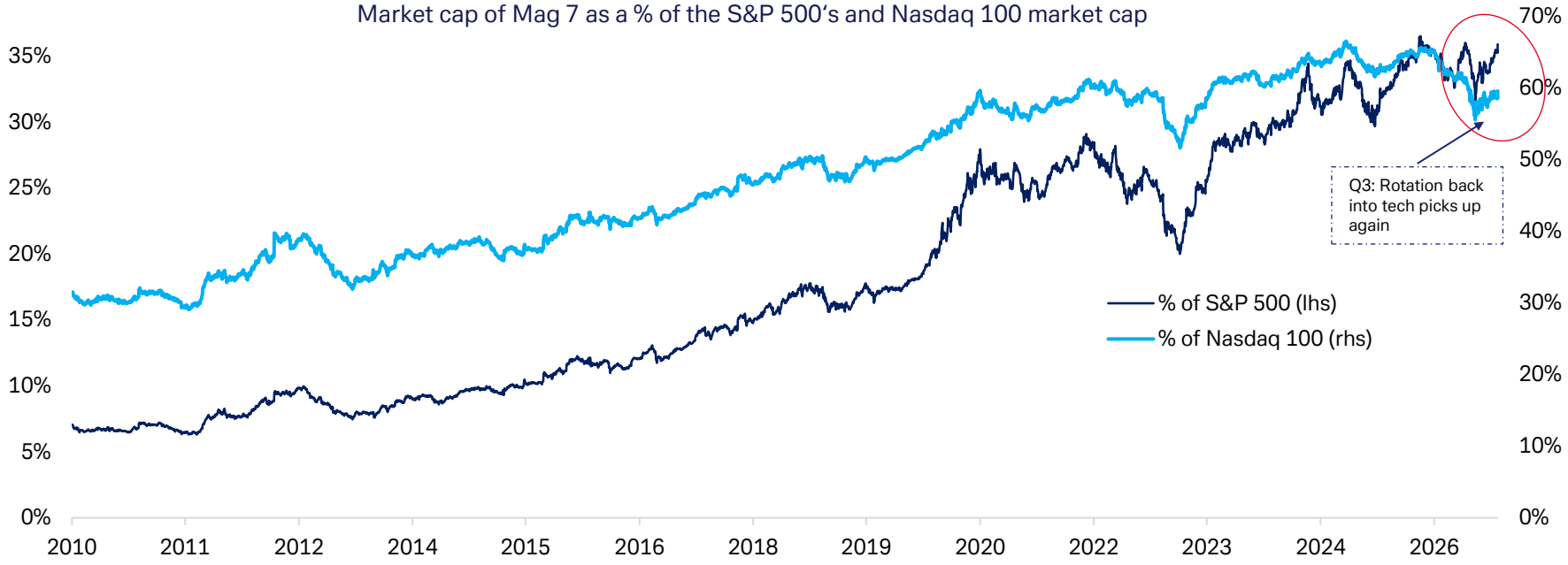
On July 30, Microsoft rose more than 15%, its strongest daily gain since 2008, following results released the previous evening. The move added ~$450bn to its market cap, the largest one-day increase on record at the time. For context, that daily increase is larger than the value of any company in Europe (except ASML).




Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

# 3.2 MAGNIFICENT 7: Market concentration

According to our equity strategists, from end-July to mid-September, tech-exposed stocks within the S&P 500 rose 14% while the remainder declined 3%. This divergence highlights the concentration of market returns in the tech-exposed sector. In late July, our strategists anticipated a rotation back into technology, which remains underway.

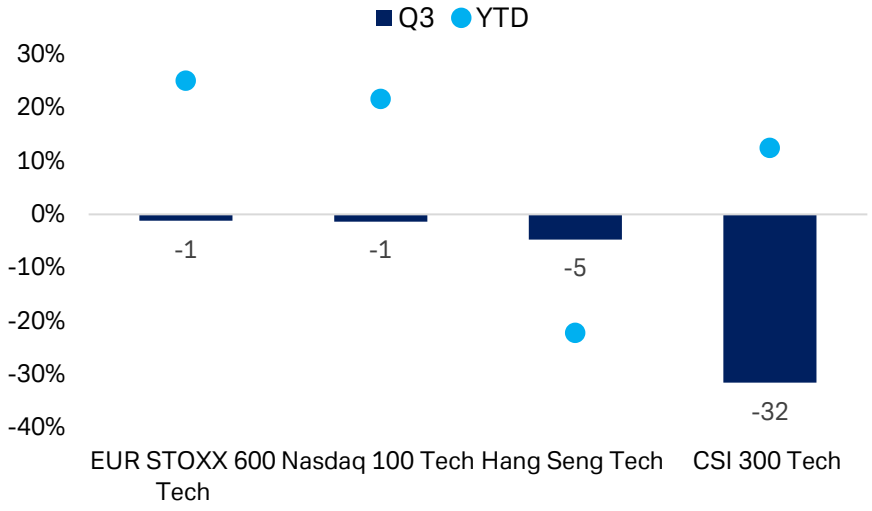



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

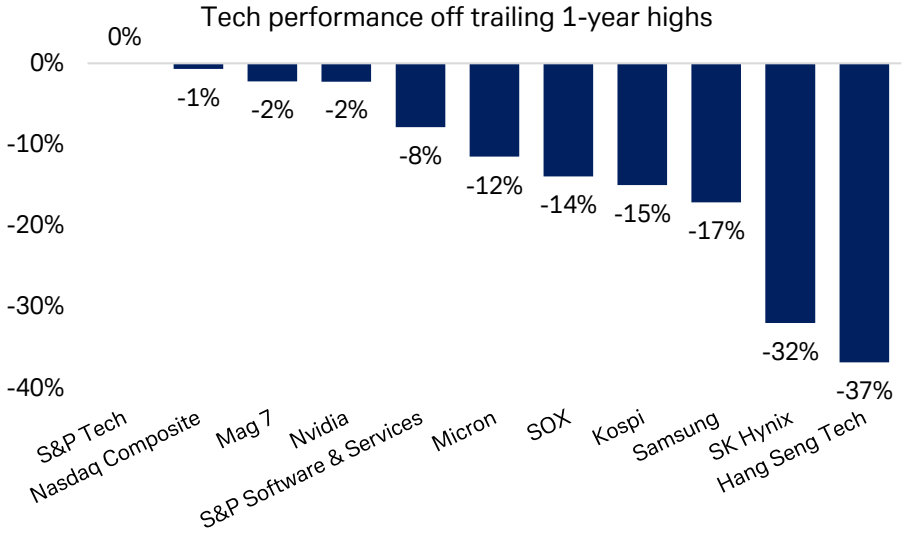
# 4. REGIONAL BREAKDOWN

After outperforming in Q2, mainland Chinese tech stocks posted large declines this quarter, hurt by July's sell-off. The CSI 300 tech series led the declines. Weakness also extended across Asia more broadly, with the KOSPI, Nikkei 225 and Hang Seng Tech down in Q3, driven by a chip sell-off and concerns over further Fed hikes. European tech equities were more resilient, with the STOXX Europe 600 Tech Index down slightly lower in Q3 but still well above its year-end level. Within European tech, Infineon underperformed, while Orange illustrated weakness in telecoms outside the index.

Top global tech indices – Q3 vs. YTD total returns



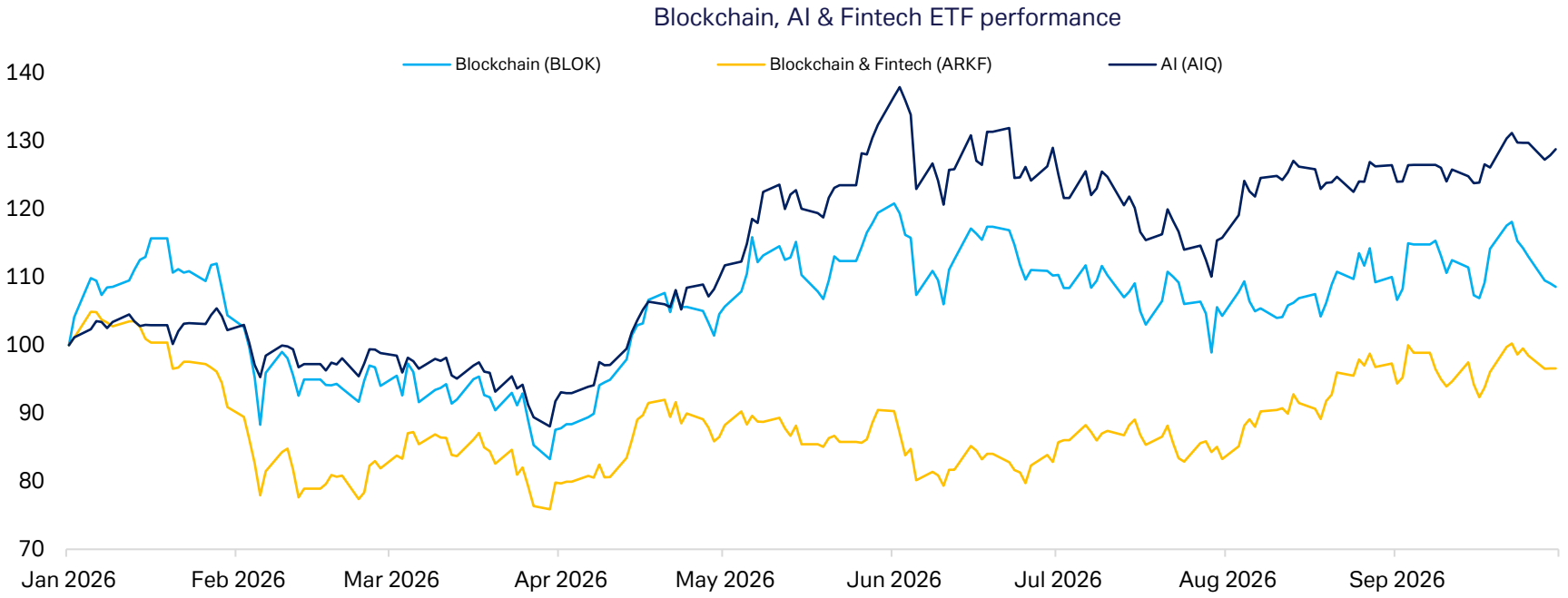

Notably, most of US tech is near recovery from their 1yr highs, unlike those in Asia




Source: Deutsche Bank Research, Bloomberg Finance LP. Data updated Sep 30.

# 5. CRYPTO, AI & FINTECH

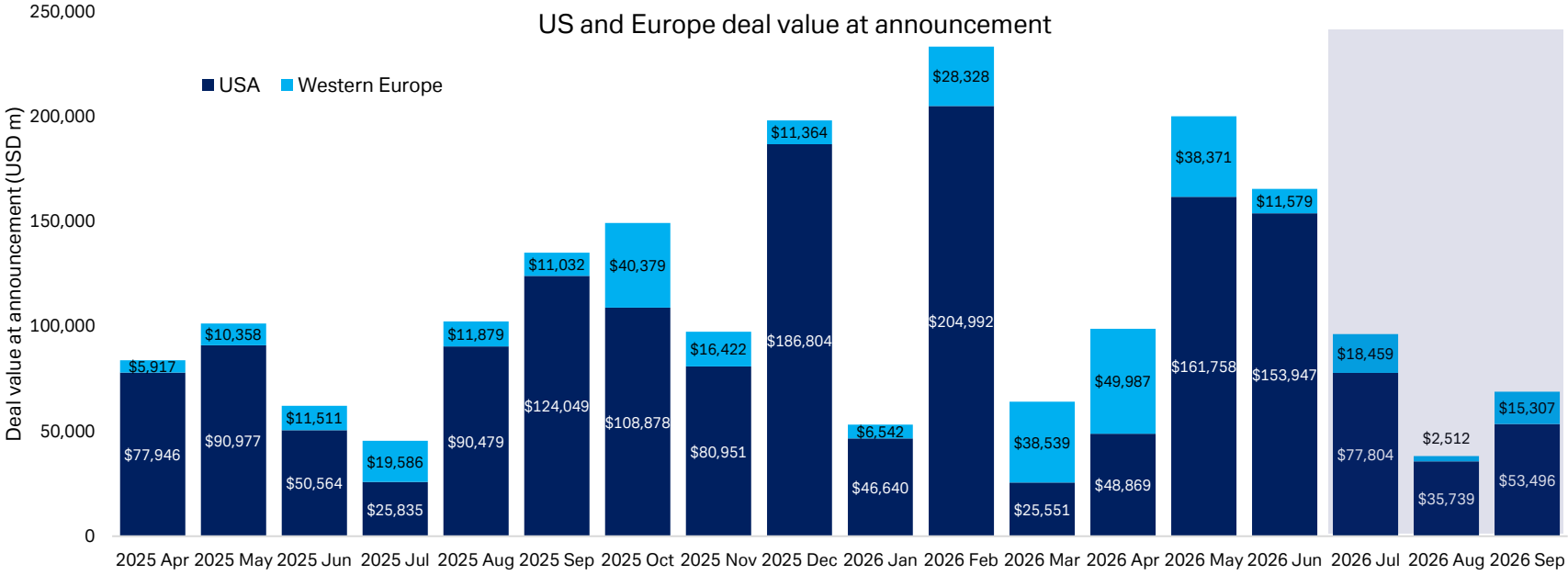
Our thematic indicators also faced macro headwinds in Q3, although the fintech and blockchain ETF shown recovered during the quarter. Bitcoin also recovered strongly during Q3, trading around $83k in late Sep, despite the setback when the Senate failed to advance the CLARITY Act on Sep 15.




Source: Deutsche Bank Research, Bloomberg Finance LP. The following are based on AIQ ETF, BLOK US Equity ETF, and ARKF US Equity ETF. Updated Sep 30.

# 6. M&A

Q3 saw lower announced deal value than Q2, against a backdrop of uncertainty around the upcoming US midterm elections, ongoing geopolitical conflict and higher oil prices. Separately, several companies, including Oura and Holtec, have delayed their IPOs, citing uncertainty. Still, there were some notable deals, including Nvidia's agreement to acquire Hugging Face for ~$12.93bn, announced on Sep 3, and AMD's announced acquisition of World Labs for $8.2bn on Sep 28.




Source: Deutsche Bank Research, Dealogic. Updated Sep 30.

# Appendix 1

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# How Q3's emerging themes shape the coming 12 months

1 Oct 2026

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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# The themes that drove markets in Q3 – based on AI analysis of our daily market commentaries and asset prices

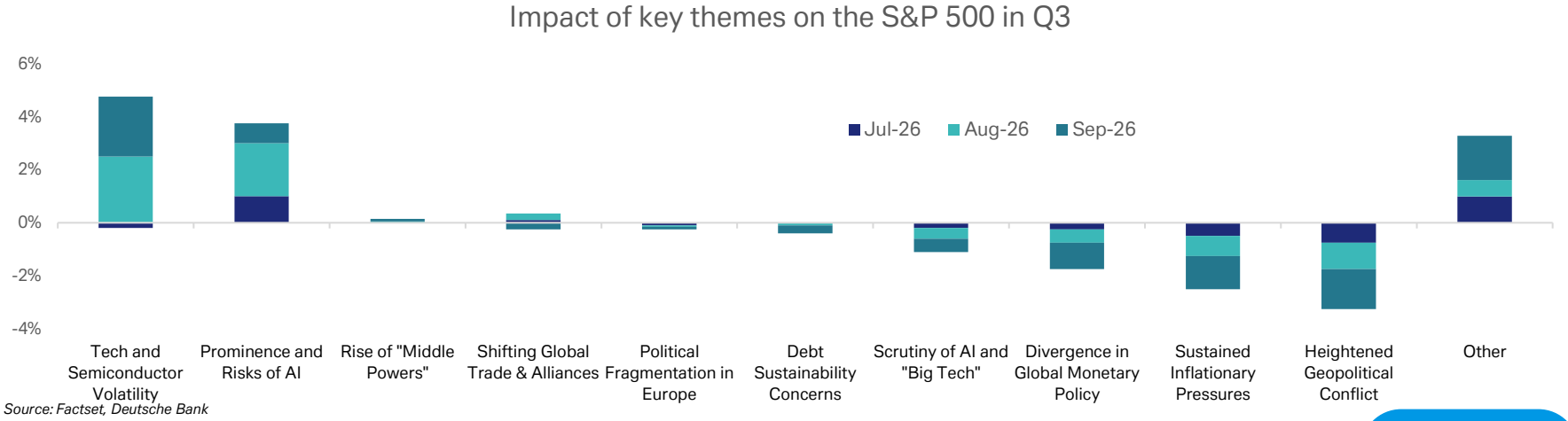


# dbLumina's estimate of how each key theme in Q3 impacted the S&P 500

**Chip volatility**
The extraordinary moves in Korean chipmakers drove material but more muted trading in the US

**AI existential risks**
The AI doom discussion also raised the prominence of AI business models

**Geopolitical conflict**
The market has finally realised there are no easy exit routes with Iran

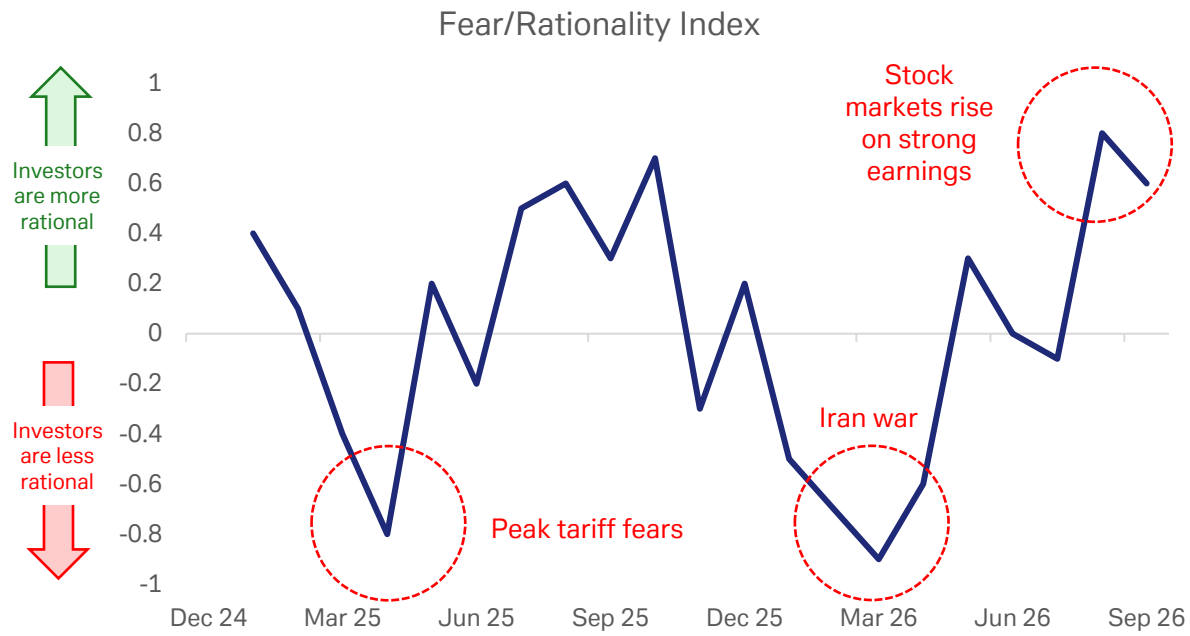



*Source: Factset, Deutsche Bank*

# Are investors rational?

The stock market strength and resilience over the last 3 months appears to be a rational response to strong earnings and only limited signs that higher energy prices are materially affecting broad corporate profits.

## Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions




*Source: Factset, Deutsche Bank*

## What does it mean to be 'rational'?

**Risk-off periods:**
The tariff fears in 2025 and then the Iran war in 2026 were key buying opportunities in equity markets. Yet, the overwhelming fear exhibited by investors in these periods largely meant they sold off risk assets in an emotional and irrational response.

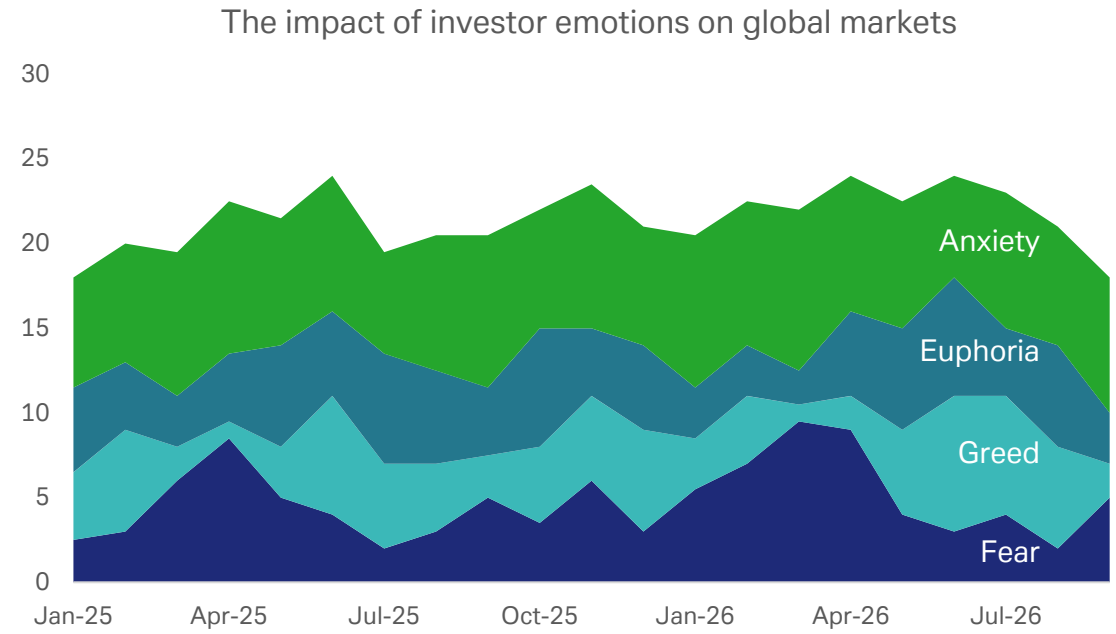
**Risk-on periods:**
It can be rational to buy into a rising market. Our indicator suggests this is what happened in Q3 as earnings growth continued to be strong.

**Fear Indicators:**
Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted.

# Emotions cooled as ‘rational’ investing took over

## Lower realised volatility in markets in Q3 helped clam emotional market responses. That indicates investors are detaching from the volatile macro and geopolitical backdrop

### Emotion dipped markedly in Q3




*Source: Factset, Deutsche Bank*

### Month-by-month

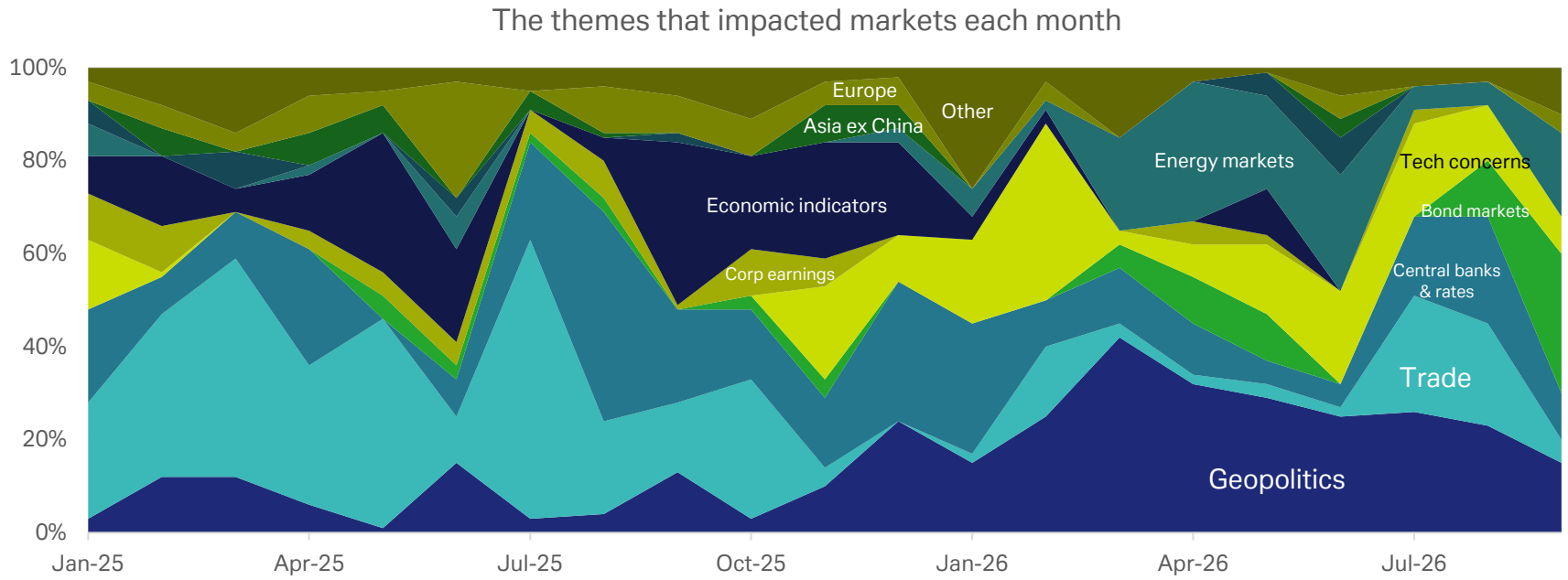
July 2026: A fear-driven sell-off in semiconductor stocks, particularly in Korea, was balanced by optimism from strong earnings reports from major tech firms and a significant drop in oil prices.

August 2026: An easing in inflation pressures lowered expectations of Fed rate hikes and supported a rally in Treasury markets. Strong corporate earnings season and signs of de-escalation in the Middle East

September 2026: Rising energy prices and renewed uncertainty in the Middle East fuelled a sell-off in global sovereign bonds. Concerns about debt sustainability and hawkish central bank commentary weighed on equity markets despite some encouraging economic signals.

# Our AI analysis shows how groups of themes have driven markets over time

Tariffs & 'Liberation day' | Fears of tariff effect on economies | Trade deals propel markets to new highs | Tariff fears subside; focus reverts to 'normal' factors | AI & software worries | Iran war | Tech rebound | Oil jumps + bond sell-off

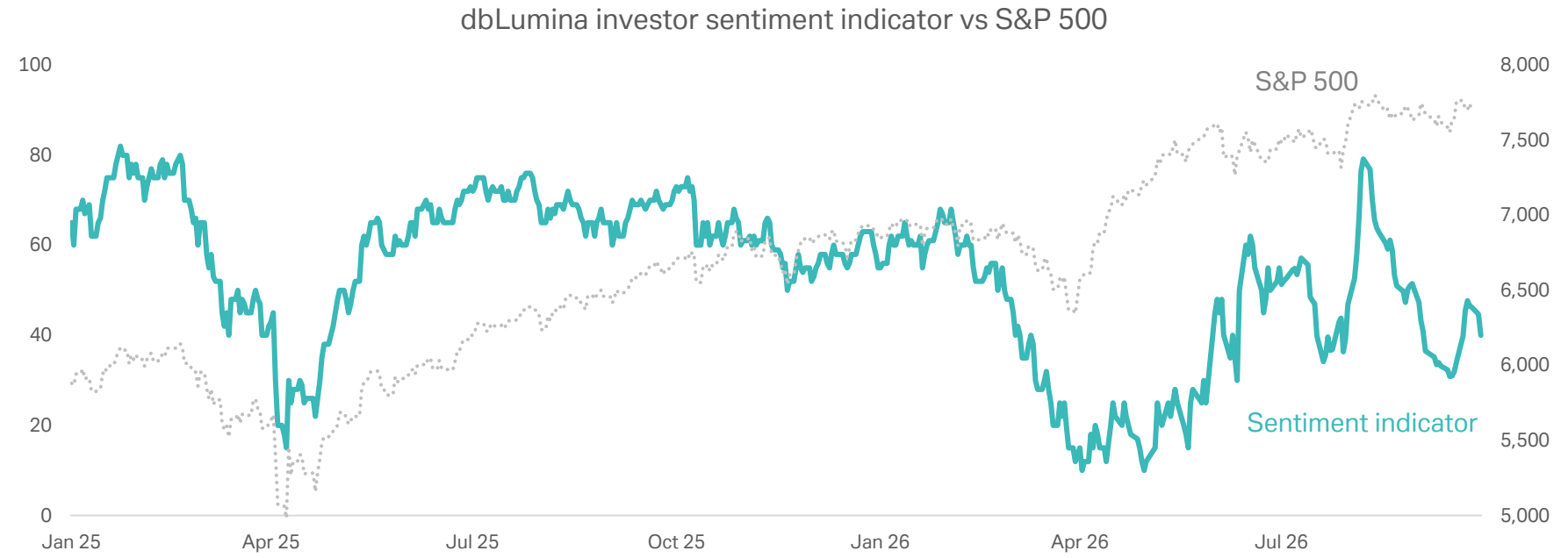



*Source: Deutsche Bank*

# Our gauge of investor sentiment uses AI to interpret macro, economic, and market signals

This is our response to the fact that so many confidence indices today fail to reflect technological and social megatrends that now affect markets. Our sentiment indicator is nowhere near as bad as some others.

**The volatile sentiment in Q3 drove some heady jumps and drops in AI-linked equities and, thus, broader markets. A positive sign is that sentiment has dropped below last year's average even though the S&P 500 is broadly steady**




*Source: Factset, Deutsche Bank*

# Evolving risks: AI can be more objective than a human. Thus, we ask it to analyse our daily market commentaries and asset prices to determine the top broad-based risks in markets

The top-4 are obvious; but sudden market dislocations happen when non-obvious risks arise. Risks 5-10 are acknowledged but simply not priced in



1 Stagflationary shock from energy prices & geopolitics



2 Sovereign debt crisis & contagion



3 Escalation of the US-Iran conflict



4 AI & technology sector dislocation



5 Central bank policy error



6 US-China geopolitical & trade re-escalation



7 Yield curve inversion & recession signal



8 Fracturing of Western alliances (NATO/G7)



9 Cybersecurity & critical infrastructure attacks



10 US political instability & government shutdown

# Emerging risks underestimated by the market – part 1

## 1 Geopolitical fragmentation of global reserve assets

A reallocation of global sovereign reserve assets is accelerating, spurred by geopolitical uncertainty and the fear of trade and financial sanctions. Notably, central banks are repatriating their physical assets, such as the Dutch and French central banks which have recently moved gold reserves out of New York. Furthermore, a World Gold Council survey reveals that 45% of global central banks plan to increase their gold reserves over the next 12 months.

**What the market underestimates:** The market views the recent surges in gold prices as temporary, cyclical safe-haven flows. In reality, sovereign institutions are structurally preparing for a more fragmented global trade system where tariff policies and sanctions may make dollar-clearing assets more difficult.

**Impact on key factors in our megatrend model:**

**Interest rates:** At a time when the "world is awash in debt", long-term interest rates of some countries may be forced structurally higher to attract alternative private capital.

**Inflation:** Unexpected dollar depreciation may increase the cost of imported goods

**Equity Prices:** High-growth US equity valuations, which are highly sensitive to the long-term risk-free rate, may face multiple compression if bond yields are pushed higher.

## 2 The AI “sameness“ trap and the productivity stagnation

The market currently prices in a material, long-term GDP boost from AI-driven productivity gains. However, some research from the IMF suggests that over-reliance on standardised LLMs can produce a high degree of "sameness" and uniformity in cognitive output. Furthermore, escalating intellectual property and copyright litigation threaten to bifurcate the quality of AI models based strictly on developers' licensing budgets.

**What the market underestimates:** Some investors assume that the exponential curve of AI hardware capabilities translates directly into exponential economic output. But there is a risk that if AI tools merely standardise corporate processes rather than innovate them, and legal barriers restrict access to premium data, the productivity boom may be less aggressive than expected.

**Impact on key factors in our megatrend model:**

**GDP growth:** Structural growth ceiling if AI fails to deliver the multi-percentage point boost to global productivity required to offset aging demographics.

**Equity prices:** A downward revaluation of the technology and software sector if massive AI infrastructure expenditures fail to generate revenue according to existing expectations.

# Emerging risks underestimated by the market – part 2

## 3 The fragmentation of the EU's political centre

Advanced economies are facing a dual demographic crisis: rapidly aging populations and underfunded pension systems. This is forcing some older people to remain in or re-enter the workforce out of financial necessity rather than choice.

**What the market underestimates:** Lower unemployment rates are celebrated as a sign of economic strength. However, in structurally ageing societies like Japan and parts of Europe, the growth of the labour force is increasingly driven by part-time, older workers. Furthermore, G20 nations lack adequate, cohesive policy frameworks to address these long-term demographic fiscal drains.

**Impact on key factors in our megatrend model:**

**Inflation:** An older workforce working out of necessity has less discretionary spending power, potentially locking economies into persistent, structural demand-side deflationary forces, offset by supply-side wage pressures in highly skilled sectors.

**GDP growth:** Labour productivity growth that is lower than it might otherwise be as the median age of the workforce climbs, limiting long-term potential GDP.

## 4 Critical mineral isolationism and 'green' supply chain blockades

The geopolitical battleground is focussed on energy blockades (such as the Strait of Hormuz) but underestimates the weaponisation of critical minerals. China currently holds 'escalation dominance' over rare-earth mineral supply chains. The emerging risk is the proactive restriction of these minerals to Western firms, such as recent bans on supplying rare-earth inputs to some US firms.

**What the market underestimates:** The market treats rare-earth supply constraints as temporary trade spat negotiation tactics. In reality, as Western nations aggressively push for reshoring and friend-shoring, China is increasingly using its processing monopoly as a long-term strategic blockade to try to slow Western high-tech, defence and the energy transition.

**Impact on key factors in our megatrend model:**

**Inflation:** Upwards pressure on inflation as the cost of manufacturing electric vehicles, semiconductors, and advanced electronics surges due to supply blockades and the high capital cost of rebuilding localised processing facilities.

**Interest rates:** Central banks may be forced to keep policy rates higher to combat the persistent supply-side inflation generated by supply-chain fragmentation.

# Emerging themes – part 1

## Our dbLumina AI model helps us identify the key emerging themes that have the potential to most impact markets and economies in the short term

### 1 AI's impact on geopolitics and national security

While the previous focus was on the AI hardware squeeze and commercial competition, the narrative is shifting towards AI's role as a critical component of national security and geopolitical leverage.

- **AI access:** We see this in President Trump's framing of AI development as a geopolitical race ("whoever wins AI wins").
- **AI as a bargaining chip:** Several examples over the last 12 months including the 2025 deal whereby AMD agreed to pay 15% of sales from certain AI chips to the US government in exchange for an export license. Even if these deals change, the trajectory is one where access to AI technology is tied to national economic and security interests.
- **Direct military & security applications:** Several examples including, cyberattacks on critical infrastructure (such as a UK power plant), the use of AI in developing "exceptional" phages to fight E.coli, and the growing field of AI security discussions.

**Outlook:** Expect more government intervention, the use of AI technology access as a tool in diplomatic negotiations, and a greater focus on the dual-use nature of AI in both commercial and military contexts. This will likely create regulatory and operational uncertainty for tech firms.

### 2 The fracturing of multilateralism and the rise of "Middle Power Countries"

The trend of reshoring is evolving into a more proactive diplomatic strategy where nations are forming new, often transactional, alliances to navigate a world less dominated by a single superpower.

- **Proactive alliance building**: Canadian Prime Minister Carney has been front and centre making his first international trip to France and the UK in an effort to diversify away from traditional reliance on the US. That follows other key meetings between Japan and India, and more.
- **Bypassing traditional powers:** Greenland signalled its willingness to turn to China for investment if Western countries do not provide support, reinforcing the leverage of some smaller nations.
- **Shifting security arrangements:** The US pushed for a 'reassurance force' in Ukraine, while French President Macron hosted a "coalition of the willing" to discuss Europe's defence without US guarantees. Further examples include the discussion of a Franco-German "nuclear sharing" arrangement.

**Outlook:** The assumptions of a bipolar world order is giving way to a more complex, multi-nodal system. This will create greater emphasis on regional blocs and issue-specific coalitions. The stability of traditional alliances like NATO and the G7 remain deeply uncertain.

# Emerging themes – part 2

These four key themes dbLumina identifies are sometimes ‘known’ by investors, however, they may be underestimated by the market, particularly when there are some potentially series longer-term impacts

## 3 The return of global bond vigilantes

Previous talk of “sovereign fiscal fragility” has now manifested itself into a severe market reaction. Although fears around government debt and deficits pop up from time to time, the magnitude of September’s bond sell-off has been compounded by oil price concerns

- **Accelerating sovereign sell-off:** Concerns about inflation and debt sustainability are not new but the peaks in bond yields are. Yields on 10-year Treasuries, bunds, and gilts all rose sharply, with bunds hitting their highest level since 2009.
- **Fiscal policy as a direct market catalyst:** The market is increasingly sensitive to fiscal policy announcements. For example, in the UK postponed tax rises, led to a spike in gilt yields. Similarly, France's political turmoil and concerns over its budget deficit caused the Franco-German yield spread to widen significantly.
- **The “awash in debt" narrative:** Treasury Secretary Bessent's comment at the G20 that "The world is awash in debt" signalled that the issue is increasingly top-of-mind for policymakers.

**Outlook:** Markets will be increasingly quick to punish any perceived fiscal irresponsibility. Expect higher borrowing costs, currency volatility, and increased risk of financial instability, especially for countries with high debt levels.

## 4 The bifurcation of economies: corporate resilience vs. consumer strain

The strain in the US consumer and labour markets is now diverging sharply from the health of large corporations, reinforcing the K-shaped economy.

- **Strong corporate performance:** Earnings season for Q2 showed remarkably solid global growth, and the resilience of large corporations was a recurring theme that helped support equity markets despite other headwinds.
- **Weak consumer & labour signals:** Economic reports have showed some signs of a weakening labour market, falling consumer confidence, and stress among low-income Americans. The New York Fed's survey also showed households expect a worse financial situation.
- **Policy and market disconnect:** While strong corporate performance depends on a robust economy, the underlying consumer weakness is concerning. Expect the market to continue its twitchy reaction to data, as investors struggle to reconcile these conflicting signals.

**Outlook:** This theme is potentially the most consequential for the domestic US economy. If corporate profits and equity markets "detach" from the real economy as consumers perceive it, it could lead to significant social and political instability, making economic policy even more difficult to navigate.

# The impact of today's themes 1 year ahead:

We asked our AI model, dbLumina, to analyse the impact of the themes for the quarter. The 'positive' and 'negative' indicators are relative to expectations at the beginning of the quarter. For example, a net negative score for the S&P 500 does not mean we expect the index to fall. Rather it means we see the rising probability that returns will be lower than previously expected.

| Theme | GDP Growth | S&P 500 | Interest Rates[1] | Inflation[1] |
|---|---|---|---|---|
| 1. Heightened geopolitical conflict | Negative | Negative | Higher | Higher |
| 2. Sustained inflationary pressures | Negative | Negative | Strongly higher | Strongly higher |
| 3. Divergence in global monetary policy | Mixed | Volatile | Divergent | Divergent |
| 4. Prominence and risks of AI | Slightly positive | Positive | Neutral to higher | Neutral to higher |
| 5. Shifting global trade & alliances | Neutral to negative | Neutral to negative | Neutral | Neutral |
| 6. Rise of 'Middle Power Countries' | Neutral | Neutral | Neutral | Neutral |
| 7. Political fragmentation in Europe | Slightly negative | Negative | Neutral | Neutral |
| 8. Scrutiny of AI and "Big Tech" | Neutral | Negative | Neutral | Neutral |
| 9. Debt sustainability concerns | Slightly negative | Slightly negative | Slightly higher | Slightly higher |
| 10. Tech and semiconductor volatility | Neutral | Volatile | Neutral | Neutral |

| | GDP Growth | S&P 500 | Interest Rates[2] | Inflation[2] |
|---|---|---|---|---|
| Sum of positive/negative indicators (positive is +1, negative is -1) | -4 | -5 | -5 | -5 |

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

*Source: Deutsche Bank*

# The impact of today's themes 3 years ahead:

The medium-term picture is far more optimistic than the short-term picture. The implicit assumption is that the macro and geopolitical volatility that is currently weighing on markets will either resolve itself over the coming 3 years or become normalised enough that the underlying earnings power of the US economy will drive equity markets ahead and, to a lesser extent, economy growth.

| Theme | GDP Growth | S&P 500 | Interest Rates[1] | Inflation[1] |
|---|---|---|---|---|
| 1. Heightened geopolitical conflict | Slightly negative to neutral | Neutral | Stabilising | Moderating |
| 2. Sustained inflationary pressures | Neutral | Neutral to slightly positive | Structurally higher | Stabilised but higher |
| 3. Divergence in global monetary policy | Converging | Positive | Converging | Converging |
| 4. Prominence and risks of AI | Strongly positive | Strongly positive | Uncertain | Lower |
| 5. Shifting global trade & alliances | Slightly positive | Positive | Neutral | Neutral |
| 6. Rise of 'Middle Power Countries' | Slightly positive | Slightly positive | Neutral | Neutral |
| 7. Political fragmentation in Europe | Uncertain | Uncertain | Neutral | Neutral |
| 8. Scrutiny of AI and "Big Tech" | Neutral | Neutral | Neutral | Neutral |
| 9. Debt sustainability concerns | Strongly negative | Strongly negative | Highly volatile | Highly volatile |
| 10. Tech and semiconductor volatility | Neutral | Less volatile | Neutral | Neutral |

| | GDP Growth | S&P 500 | Interest Rates[2] | Inflation[2] |
|---|---|---|---|---|
| Sum of positive/negative indicators (positive is +1, negative is -1) | +1 | +5 | -1 | Nil |

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

*Source: Deutsche Bank*

# Appendix 1



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